

Freshwater Aquarium Fish Deaths Despite Acceptable Water Tests

A Research Compilation on Unexplained Mortality, Water-Change-Associated Losses, and the Limits of Hobbyist Water Testing

Research compilation prepared as NotebookLM source material for FishKeepingLifeCo / Tank Guide Life
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BIBLIOGRAPHY AND SOURCE LIST

- Tier 1–4: Peer-reviewed, veterinary, university extension, and government sources
- Tier 5–6: Aquaculture literature and water utility / municipal sources
- Tier 7–9: Aquarium publications, manufacturer documentation, and experienced hobbyist technical sources
- Tier 10: Forum and community sources (anecdotal; cited only to document what hobbyists report and where disagreements lie)
- Sources consulted and deliberately not used

APPENDIX A. GLOSSARY AND ALTERNATE TERMINOLOGY

APPENDIX B. DOCUMENT INTEGRITY NOTES

- B.1 Known limitations of this compilation
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DOCUMENT PURPOSE AND SCOPE

0.1 What this document is

This is a **research and data-compilation document**. It is not a script, an outline, an article, or a narrative. It exists to serve as a retrievable source corpus for an AI research assistant (NotebookLM) that will later be queried about freshwater aquarium fish mortality occurring despite apparently acceptable water-test results, and about the role water changes and associated maintenance may play in such events.

The document deliberately favors completeness and redundancy over concision. Concepts are restated in multiple sections where they are relevant to more than one line of inquiry, because chunk-based retrieval systems perform better when each section is self-sufficient.

0.2 The central question

If ammonia, nitrite, nitrate, pH, and temperature appear acceptable, what else could cause aquarium fish to become sick, stressed, behave abnormally, or die — and under what circumstances could a water change or associated maintenance contribute to the problem?

0.3 Scope boundaries

- **In scope:** freshwater home aquariums; beginner and intermediate hobbyist scenarios; tropical and temperate community species; common commercial equipment and consumables.
- **Partially in scope:** ornamental pond keeping, shrimp/snail invertebrate keeping, and aquaculture literature — used where the mechanism transfers to home aquaria.
- **Out of scope:** marine/reef systems except where a mechanism is identical; commercial aquaculture economics; veterinary drug dosing protocols.

0.4 Evidence classification key used throughout

Every substantive claim in this document is intended to carry one of the following labels. Where a label is absent, treat the claim as **[GENERALLY SUPPORTED]** unless the surrounding text indicates otherwise.

Label	Meaning
[ESTABLISHED]	Supported by peer-reviewed research, veterinary/aquatic-animal-health references, or government/extension publications. Mechanism is understood and reproducible.
[GENERALLY SUPPORTED]	Well accepted in aquatic animal health and aquaculture, but specific thresholds for ornamental species may be extrapolated rather than directly measured.
[PLAUSIBLE, POORLY DOCUMENTED]	The mechanism is chemically or biologically coherent, but direct evidence in home-aquarium conditions is thin or absent.
[ANECDOTAL]	Reported repeatedly by hobbyists or retailers; not verified experimentally. May still be practically useful.
[DISPUTED]	Credible sources actively disagree. The disagreement is described rather than resolved.
[MISLEADING]	Contains a kernel of truth but is commonly stated in a way that produces wrong conclusions.
[UNSUPPORTED / MYTH]	Contradicted by evidence, or asserts a mechanism that does not occur under realistic aquarium conditions.

0.5 A note on how to read disagreements in this document

Several topics in aquarium keeping have no settled answer. Two broad camps recur in the sources consulted:

1. **The conservative-husbandry camp** — represented by most retailers, many hobbyist publications, and manufacturer literature. Emphasizes small water changes, gentle handling, drip acclimation, tank-water-only filter rinsing, and ammonia-binding conditioners. Errs toward caution.
2. **The mechanism-and-dose camp** — represented by aquaculture literature, some veterinary sources, and technically-oriented hobbyist authors. Emphasizes measured concentrations, exposure times, and LC50/threshold data; argues that several standard hobby cautions describe risks too small to matter at realistic doses.

Both camps are cited. Where they conflict, the conflict is stated explicitly and the reasoning of each side is given. Neither is presented as settled.

0.6 Critical framing statement for the whole document

“Water quality” and “fish health” are not the same variable. Every common hobbyist test kit measures a small number of dissolved nitrogen species plus pH, and sometimes hardness. A fish can die of starvation, internal parasites, aggression, hypoxia, thermal stress, chemical contamination, shipping injury, or old age in water that is chemically flawless. Conversely, water can be seriously harmful in ways that none of the five standard tests detect. A clean test panel eliminates approximately five possible causes out of several dozen. Additionally, and repeatedly under-appreciated: **a test performed after an incident does not describe the water during the incident.** UF/IFAS states this directly in the context of diagnostic submissions — a single water sample provides a one-time evaluation with no insight into past trends, and fish may become sick weeks after a water-quality problem has been corrected [UF-FA16].

SECTION 1. WHAT “PERFECT WATER PARAMETERS” ACTUALLY MEANS

1.1 What hobbyist test kits actually measure

1.1.1 The standard five

Test	What it physically measures	Common units
Ammonia	Total ammonia nitrogen (TAN) — the sum of NH3 (un-ionized ammonia) and NH4+ (ammonium). Most hobby colorimetric kits (e.g., salicylate or Nessler chemistry) report TAN, not the toxic fraction.	mg/L as NH3 or as N — labeling varies by brand
Nitrite	Nitrite ion (NO2-), usually via Griess diazotization	mg/L
Nitrate	Nitrate (NO3-) after cadmium- or zinc-based reduction to nitrite, then Griess reaction	mg/L
pH	Negative log of hydrogen ion activity, by colorimetric indicator dye or electrode	pH units (log scale)
Temperature	Thermal state of the water at one point in the tank at one moment	°F / °C

1.1.2 The extended set some hobbyists own

GH (general hardness), KH (carbonate hardness / alkalinity), phosphate, chlorine/total chlorine, copper, iron, TDS/conductivity meter, dissolved oxygen (rare), ORP (rare).

1.2 The TAN vs. un-ionized ammonia problem — the single most important testing subtlety

[ESTABLISHED] Ammonia in water exists in equilibrium between NH3 (un-ionized, uncharged, lipid-soluble, crosses gill membranes readily, toxic) and NH4+ (ionized ammonium, charged, poorly membrane-permeant, far less toxic). The equilibrium position is governed principally by pH, secondarily by temperature, and marginally by ionic strength [UF-SS538; SRAC-4603].

Key quantitative anchors:

- UF/IFAS EDIS publication FA16 (“Ammonia in Aquatic Systems”) provides the standard conversion table, calculated from Emerson et al. (1975), for converting a measured TAN value into un-ionized ammonia using pH and temperature [UF-FA16].
- Un-ionized ammonia toxicity is generally described as beginning around **0.05 mg/L**, and UF/IFAS notes that as un-ionized ammonia approaches **2.0 mg/L**, fish will begin to die [UF-FA16].
- SRAC 4603 states that in general **less than 10% of total ammonia is in the toxic un-ionized form when pH is below 8.0**, and that this proportion increases dramatically as pH rises [SRAC-4603].
- The European Inland Fisheries Advisory Commission reported ammonia was approximately **10 times more toxic at pH 8.0 than at pH 7.0**; an increase of one pH unit increases un-ionized ammonia roughly tenfold [UF-SS538].
- Russo and Thurston (1991), summarized by UF/IFAS, reported 96-hour LC50 values for un-ionized ammonia ranging from **0.08 to 3.8 mg/L** across fish species — a ~47-fold species range [UF-SS538].
- US EPA has recommended acute ammonia criteria of 2.9 or 5.0 mg N/L and chronic criteria of 0.26 or 1.8 mg N/L, and considers water below 0.020 mg/L un-ionized ammonia safe for fish reproduction [TNC-AMMONIA-REVIEW].

Practical retrieval statement: the same TAN reading of 0.25 mg/L is a near-non-event in a pH 6.5 blackwater tank and a meaningful chronic stressor in a pH 8.4 Rift Lake tank. **A single number without pH and temperature is not interpretable.** This is the most common form of “my parameters looked fine” that was in fact not fine.

1.2.1 Test kits that measure free ammonia instead of TAN

Some products (notably Seachem’s ammonia alert/test chemistry) report **free ammonia (NH3)** rather than TAN. Because the toxicity thresholds published in the research literature are defined in terms of NH3, free-ammonia readings map more directly onto danger levels. This creates a documented source of hobbyist confusion when two kits from two manufacturers report apparently contradictory values for the same water; they are measuring different things [AQSCI-AMMONIA].

1.3 What common test kits do NOT measure

This list is central to the document's core question. None of the following is detected by the standard five tests:

Dissolved gases and respiration - Dissolved oxygen (mg/L or % saturation) - Dissolved carbon dioxide - Total dissolved gas pressure / gas supersaturation - Nitrogen gas supersaturation

Disinfectants and their residues - Free chlorine (unless a chlorine test is separately owned) - Chloramine and its breakdown intermediates - Disinfection byproducts (trihalomethanes, haloacetic acids)

Metals and inorganic contaminants - Copper, zinc, lead, iron, cadmium, nickel (unless separately tested) - Aluminum (relevant in some low-pH source waters)

Organic contaminants - Pesticides and insecticides (pyrethroids, fipronil, imidacloprid, organophosphates) - Surfactants (soap, detergent residue) - Solvents, VOCs, paint and aerosol propellant residues - Petroleum products - Personal-care residues (lotions, sunscreens, fragrances, hand sanitizer) - Household cleaner residues (ammonia-based cleaners, bleach, quaternary ammonium compounds)

Biological - Pathogen load of any kind (bacteria, ciliates, flagellates, monogeneans, microsporidia, viruses) - Bacterial toxins and exotoxins - Dissolved organic carbon (DOC) load / "organics" - Biological oxygen demand (BOD)

Physiological and physical - Osmolality / total ionic strength (TDS meters approximate this, but see 13.6) - Specific ion composition (sodium vs. calcium vs. magnesium vs. potassium) - Water velocity, flow patterns, dead zones - Light intensity and photoperiod - Sound and vibration - Nutritional status of the fish - Physical injury, internal injury, or parasite burden of the fish

Historical - Any parameter's value at any previous point in time

1.4 Limitations of liquid reagent test kits

[ESTABLISHED / GENERALLY SUPPORTED]

1. **Colorimetric subjectivity.** Reading a color against a printed card depends on ambient light temperature, card fading, the reader's color vision, and the background surface. Two people can read the same vial as 0 ppm and 0.25 ppm.
2. **Discrete card steps, not continuous scales.** A card with steps at 0, 0.25, 0.50, 1.0 cannot express 0.15. "Zero" on such a card frequently means "below approximately 0.25 mg/L," not "absent."
3. **Reagent degradation.** Reagents oxidize, evaporate through imperfectly closed caps, and degrade with heat and light exposure. Most manufacturers state multi-year shelf life for unopened liquid reagents, but degradation after opening is not well characterized in consumer documentation.
4. **Reagent separation — the nitrate bottle problem.** Nitrate tests that use a zinc- or cadmium-based reductant contain a suspension that settles. If bottle #2 is not shaken vigorously (API's own instructions specify 30+ seconds of shaking of the bottle and then 1 minute of shaking the vial), **the test reports falsely low or zero nitrate.** This is one of the most frequently reported causes of a hobbyist believing their nitrate is 0 ppm in a tank that has never had a water change [ANECDOTAL, extremely widely reported; manufacturer instruction basis].
5. **Timing sensitivity.** Colors continue to develop. Reading a nitrate test at 2 minutes instead of 5, or reading an ammonia test at 10 minutes instead of 5, can shift the apparent result by a full card step.
6. **Drop-size variance.** Reagent bottles held at an angle rather than fully vertical deliver inconsistent drop volumes, changing reagent-to-sample ratio.
7. **Temperature of the sample.** Some reactions develop more slowly in cold samples.
8. **Cross-reactivity and interference.** Certain medications, dyes, ammonia-binding conditioners, and high organic loads can interfere with colorimetric chemistry. Notably, **some ammonia-binding water conditioners can cause false positives or altered readings on Nessler-type ammonia tests,** and conversely salicylate tests may still read bound ammonia as present. This is a documented source of panic-driven, unnecessary water changes.

1.5 Limitations of test strips

[GENERALLY SUPPORTED]

1. **Broad range bands.** Strip color blocks are often spaced at intervals such as 0 / 20 / 40 / 80 / 160 ppm nitrate. Precision is low by design.
2. **Moisture sensitivity.** Strips absorb ambient humidity through an imperfectly sealed container and degrade rapidly once opened. Shelf life after opening is typically far shorter than for liquid reagents.
3. **The ammonia gap.** Many multi-parameter strips (commonly 5-in-1 or 6-in-1) **do not include ammonia at all.** A hobbyist using only strips may believe they have tested "everything" while never having measured the single most acutely toxic common parameter.
4. **Read-time sensitivity and pad bleed.** Colors on adjacent pads can bleed into each other if the strip is held vertically or shaken; several products include a "bleed pad" specifically to mitigate this.
5. **Disagreement with liquid kits is commonly reported.** Hobbyist forums contain numerous reports of strips and liquid kits differing by an order of magnitude on the same sample [ANECDOTAL]. In most such reports the underlying cause is never resolved; both a degraded strip and an unshaken nitrate reagent are plausible.

[DISPUTED] Whether strips are meaningfully less accurate than liquid kits in practice. Hobby consensus strongly favors liquid kits; some experienced keepers report strips are adequate for trend monitoring on established tanks when stored dry and replaced annually.

No high-quality independent comparative validation study specific to consumer aquarium strips was located during this research.

1.6 User procedure and interpretation errors

- Sampling from the same spot every time (e.g., always at the surface near the filter outflow), missing gradients.
- Sampling immediately after feeding, dosing, or a water change, capturing a transient rather than a steady state.
- Contaminated vials: soap residue, previous reagent residue, hands, or storing vials with caps on while damp.
- Using tap water to “rinse” vials, introducing chlorine/chloramine and its own ammonia.
- Testing the tank but never testing the source water.
- Confusing units: nitrate reported as NO3- vs. NO3-N differs by a factor of ~4.43. A “40 ppm” reading means very different things in the two conventions. The same conversion issue applies to ammonia (NH3 vs. N).
- Believing that “0 nitrate” in a stocked, unplanted, un-water-changed tank is a good result rather than a red flag for a failed test.
- Treating a normal result as proof of tank health rather than as the absence of one specific abnormality.

1.7 Why aquarium conditions change between tests

[ESTABLISHED] Aquaria are dynamic. Within 24 hours, the following can change materially:

- **pH** — diurnal swings driven by photosynthesis (CO2 consumption raises pH during the photoperiod) and respiration (CO2 production lowers pH overnight). In poorly buffered water this swing can exceed 1.0 pH unit. In natural ponds the daytime pH can reach 8.8 [AQSCI-CO2PH].
- **Dissolved oxygen** — lowest just before dawn in planted or heavily stocked systems.
- **Ammonia** — spikes after feeding, after a death, after a filter disturbance; consumed within hours to days by an intact biofilter.
- **Temperature** — heater cycling, room temperature, lighting heat load.
- **Nitrite** — transient during biofilter disruption.

A test performed at 7:00 p.m. describes 7:00 p.m. A fish that died at 5:00 a.m. died under different conditions.

1.8 Why testing after an incident may not reveal the cause

This is one of the most important conceptual points in the entire document.

[ESTABLISHED] Several genuinely lethal conditions leave little or no trace by the time an aquarist tests:

Transient cause	Why it is gone by test time
Chlorine exposure	Free chlorine reacts with organics and dissipates; residual is consumed within minutes to hours in an organic-rich tank
Chloramine exposure	The chlorine moiety is consumed; the released ammonia is oxidized by the biofilter within hours to days
Ammonia spike	Consumed by nitrifying organisms; a functional biofilter can clear a moderate spike within hours
Nitrite spike	Same
Low dissolved oxygen	Re-equilibrates with the atmosphere as soon as surface agitation resumes or biological demand falls
Gas supersaturation	Degasses to equilibrium quickly once at atmospheric pressure
Temperature excursion	Heater recovers, or room warms
pH excursion from CO2	Degasses within hours
Pesticide / aerosol contamination	Adsorbs to surfaces and organics, degrades, or is removed by carbon; not measurable by hobbyist means at any time
Osmotic/ionic shock	Not a persistent water property at all — it is an event, not a state

Retrieval statement: “I tested right after and everything was fine” is fully consistent with the fish having been killed by any of the above.

1.9 Water quality versus fish health

[ESTABLISHED] Aquatic-animal-health references consistently frame water quality as a *predisposing factor* rather than the sole determinant of disease. UF/IFAS notes that fish submitted to diagnostic laboratories are often tested only for infectious agents, and that even a submitted water sample provides only a one-time evaluation offering no insight into past trends; fish may become sick weeks after a water-quality problem has been corrected [UF-FA16].

The standard conceptual model in fish health is a three-way interaction:

Host (species, age, genetics, nutrition, immune status) × Pathogen (presence, virulence, load) × Environment (water quality, temperature, stocking, stability)

Disease occurs when the intersection is unfavorable. Perfect environment does not remove the other two axes.

1.10 Is “perfect parameters” scientifically meaningful?

[MISLEADING as commonly used]

The phrase conflates several distinct claims:

1. “The five things I tested were within a range I read on the internet.”
2. “The water is optimal for these specific species.”
3. “The water is not harming my fish.”
4. “Nothing about my aquarium is harming my fish.”

Only (1) is supported by a test panel. (2) requires species-specific knowledge. (3) requires far more measurements than five. (4) is not a water-chemistry claim at all.

There is also no single “perfect” target. Requirements are species-specific and, importantly, **population-specific**: commercially bred neon tetras raised for generations in moderately hard water do not necessarily require the pH 5.0 blackwater of the wild Rio Negro population. Conversely, wild-caught specimens of the same nominal species may not tolerate the conditions captive-bred conspecifics thrive in.

1.11 Species-specific parameter requirements — selected reference ranges

[GENERALLY SUPPORTED — ranges are hobby- and literature-derived consensus, not universal law]

Group	Typical pH	Typical GH	Typical temp	Notes
Soft-water South American tetras/rasboras (wild)	4.5–6.8	0–4 dGH	74–82°F	Captive-bred stock far more tolerant
Commercially bred community livebearers (guppy, platy, molly)	7.0–8.2	8–20 dGH	74–82°F	Mollies in particular do poorly in very soft water
Rift Lake cichlids (Malawi/Tanganyika)	7.8–8.6	10–20 dGH	76–82°F	High KH needed for pH stability
Discus	5.5–7.0	1–6 dGH	82–86°F	High temp raises O2 demand and lowers O2 solubility
Goldfish (temperate)	7.0–8.4	8–20 dGH	65–72°F	Very high bioload per fish
White Cloud Mountain minnow	6.5–8.0	5–19 dGH	64–72°F	Frequently mis-sold as tropical
Corydoras spp.	6.0–7.6	2–12 dGH	72–78°F	Many species stressed above 80°F
Betta splendens	6.5–7.5	3–12 dGH	78–82°F	Labyrinth organ; surface air access required
Neocaridina shrimp	6.5–8.0	6–12 dGH	65–78°F	Copper-sensitive; GH needed for molting
Caridina (Taiwan bee) shrimp	5.5–6.5	4–6 dGH	68–74°F	Very sensitive to TDS and to sudden change

Interpretive note: “acceptable” ranges published for the hobby generally describe conditions under which the species will survive and often breed. They do not describe a cliff edge on either side. The more defensible generalization is that **stability within a tolerable range matters more than hitting a specific number**, with the important exception of parameters that affect toxicity chemistry (pH’s effect on ammonia and metals) and osmoregulatory load (GH/TDS extremes).

1.12 Acclimation history and previous water conditions

A fish’s tolerance is not a fixed species property; it is modified by the conditions it was raised and held in. Relevant history includes:

- Water chemistry at the breeding farm (often Southeast Asian or Florida farm water — frequently harder and more alkaline than hobbyists assume for “soft water” species)
- Water chemistry at the transshipper and importer
- Water chemistry at the wholesaler
- Water chemistry at the retail store (often a central recirculating system serving all tanks, meaning every species in the store is in identical water)
- Duration at each stage
- Whether the fish was medicated prophylactically at any stage
- How many acclimation events the fish has already survived in the preceding two weeks

[GENERALLY SUPPORTED] A fish that has moved through four water chemistries in ten days has a substantially depleted stress-response reserve compared to the same fish moved once.

1.13 How someone can obtain good readings while a serious problem exists — enumerated pathways

1. The problem is a parameter not on the test panel (oxygen, chlorine, metals, pesticides, organics).
2. The problem was transient and has resolved (Section 1.8).
3. The problem is biological, not chemical (pathogen, parasite, injury, starvation, aggression).
4. The test is being performed incorrectly (unshaken reagent, expired strip, contaminated vial).
5. The test is being read incorrectly (color match, unit confusion, card step resolution).
6. The reading is correct but the interpretation is wrong for the pH/temperature context (TAN vs. NH3).
7. The reading is correct for the sampled location but not for the whole tank (gradients, dead zones, substrate interface).
8. The reading is correct but the value is inappropriate for the species being kept.
9. The value is within a “safe” range but the *rate of change* to reach it was harmful.
10. The parameter is fine but the fish arrived already compromised (Section 6).
11. The parameter is fine now, but the fish sustained irreversible gill or organ damage during an earlier event.
12. Everything is genuinely fine and the fish died of age, congenital defect, or an idiopathic cause.

SECTION 2. WATER CHANGES AS A CAUSAL PATHWAY

2.1 Framing

Water changes are, on balance, the single most reliable husbandry practice in freshwater aquarium keeping. This section is not an argument against them. It is a catalogue of the specific mechanisms by which a water change **can** cause harm, so that a genuine causal link can be distinguished from coincidental timing.

A critical distinction runs through this entire section:

The water change is the most memorable event of the week. It is therefore the event most likely to be blamed, whether or not it was responsible. Section 3 covers the maintenance actions that happen simultaneously and are frequently the true cause.

2.2 Temperature differences and rapid temperature change

[ESTABLISHED — mechanism] / [DISPUTED — magnitude of hobby thresholds]

Mechanism. Fish are ectotherms. Metabolic rate, enzyme kinetics, cardiac output, and immune function all track body temperature, which tracks water temperature within minutes for small fish. A rapid drop reduces metabolic rate and immune competence; a rapid rise raises oxygen demand while simultaneously lowering oxygen solubility. Acute thermal change also triggers a heat-shock protein (HSP) response and a cortisol stress response.

What is well supported: thermal stress is real, is dose-dependent, and produces measurable physiological markers.

What is disputed: the commonly repeated hobby claim that a change of 2°F is dangerous. One technically-oriented review compiles university research in which measurable HSP induction in fish required sudden upward shifts on the order of **14°F (Airaksinen et al. 1998)**, **22°F (Basu et al. 2001)**, and **27°F (Currie et al. 2000)**, and notes that in those studies fish did not die or show severe problems despite the magnitude of the shift [AQSCI-SHIFTS]. That source argues the 2°F threshold is a myth.

Counterpoint: aquaculture and veterinary sources consistently advise minimizing rapid thermal change, and thermal stress is a recognized predisposing factor for outbreaks of *Ichthyophthirius* and *Flavobacterium*. Small, chronically repeated thermal insults may matter more than the acute-mortality literature captures. Additionally, laboratory subjects in HSP studies are typically healthy, well-fed, unstressed animals — not a fish that arrived from a shop three days ago.

Practical synthesis for retrieval: a 3–5°F difference during a routine water change on healthy established fish is unlikely to be independently lethal. A 15–20°F cold-water shock delivered rapidly to a large fraction of the tank volume is a credible acute killer, particularly for small-bodied species, and is a very common beginner error when a hose is run straight from a cold tap in winter.

Aggravating factors: large change percentage; small tank (less thermal inertia); fish already stressed; species with narrow thermal tolerance; simultaneous addition of cold and unconditioned water.

2.3 Chlorine

[ESTABLISHED]

Mechanism. Free chlorine (hypochlorous acid / hypochlorite) is a strong oxidant. At the gill, it damages the lamellar epithelium, causing hyperplasia, lamellar fusion, and loss of gas-exchange surface. It also oxidizes hemoglobin and damages the mucus layer. Death is by asphyxiation with visibly normal water.

Concentrations. Municipal free chlorine residuals commonly run in the **0.2–2.5 mg/L** range at the tap. Patent literature and aquaculture sources describe **chlorine at 0.2–0.3 ppm as rapidly toxic to fishes**, with US EPA recommending an upper limit of

0.003 mg/L for continuous exposure by coldwater and warmwater fishes [USPTO-4666610]. In other words, ordinary tap water is roughly two orders of magnitude above the chronic safety threshold.

Symptom timing. Minutes to a few hours. Gasping, erratic swimming, darting, “flashing,” reddened gills, sudden death. Larger and slower fish sometimes survive an event that kills small tetras outright, because gill surface area to body mass differs.

Why the test panel misses it. Free chlorine is not on the standard five. It is also consumed rapidly by organics in the tank, so a chlorine test done an hour later may read zero.

2.4 Chloramine

[ESTABLISHED — hazard] / [DISPUTED — magnitude of the residual-ammonia hazard]

What chloramine is. Monochloramine (NH_2Cl) is chlorine pre-combined with ammonia, used because it persists further into the distribution system and produces fewer regulated disinfection byproducts than free chlorine. Roughly a third to 40% of US municipalities use chloramine as their secondary disinfectant.

Key facts: - Chloramine **does not dissipate by letting water stand.** Water utilities state this explicitly [ACWD-CHLORAMINE]. This defeats the traditional “age your water overnight” practice, which was developed in the free-chlorine era. - Chloramine is toxic to both freshwater and saltwater fish; koi are as susceptible as other species [ACWD-CHLORAMINE]. - Patent literature cites chloramine toxicity data of approximately **LC0 = 0.1-0.15 mg/L, LC50 = 0.3-0.35 mg/L, LC100 > 0.8 mg/L**, and describes lethal action typically appearing within **5 to 30 minutes** — with the further statement that replacing more than 10% of aquarium water with chloraminated drinking water may prove fatal to all fish [USPTO-4786434]. Patent-application toxicity claims should be treated as motivated but are directionally consistent with utility guidance. - Conventional thiosulfate dechlorinators **break the chlorine-ammonia bond, neutralize the chlorine, and leave ammonia behind** [FRITZ-CHLORAMINE; AQCOOP-COND].

The residual-ammonia dispute — described, not resolved:

Position A (manufacturers, most retailers). The released ammonia is toxic and should be handled with a conditioner formulated to bind or detoxify ammonia in the same step. Toxicity depends heavily on pH; the same ammonia reading represents very different risk at pH 7.0 vs. pH 8.2 [FRITZ-CHLORAMINE]. Aquarium Co-Op similarly describes ammonia-binding conditioners as temporarily converting ammonia to an inert state for up to 24 hours, during which the biofilter consumes it [AQCOOP-COND].

Position B (technically-oriented critics). The arithmetic is unfavorable to the alarm. If municipal chloramine is 1–3 ppm, neutralizing 2 ppm of chloramine (MW ~51.5) releases roughly **0.62 ppm of total ammonia**, of which at pH 7.0 and 25°C only about **0.57% is unionized NH_3** — approximately **0.0035 ppm NH_3** . That is well below the ~0.05 ppm NH_3 alert threshold, and partial water changes dilute it further. On this reading, the ammonia-detoxifying claims are marketing rather than necessity, and getting rid of the chlorine is the part that matters [AQSCI-CHLORAMINE].

Where both sides agree: pH is the deciding variable. Above pH 8.0, the same released ammonia becomes materially more dangerous. Both sides also agree that some utilities have recently been reported as running unusually high chloramine, with hobbyists reporting **4–8 ppm ammonia straight from the tap**, implying chloramine on the order of 12–24 ppm; this is far outside normal and would change the calculation entirely [AQSCI-CHLORAMINE, reporting social-media accounts — ANECDOTAL as to frequency].

Practical retrieval statement: the residual-ammonia risk from chloramine is usually small in soft-to-neutral water with an intact biofilter, and non-trivial in high-pH tanks, very large water changes, tanks with impaired biofilters, or unusually high local chloramine. Testing your own tap water for ammonia resolves the question locally in five minutes.

2.5 Improper dechlorination, forgotten conditioner, incorrect dosage

[ESTABLISHED]

Failure modes, roughly in order of observed frequency:

1. **Forgotten entirely.** Most common in the “just refilling from the hose” scenario, in a hurry, or when a second person performs maintenance.
2. **Dosed for the wrong volume.** Dosing for the replacement volume vs. the full tank volume is a real ambiguity; most product labels permit dosing for the tank volume when adding conditioner directly to the tank. Under-dosing occurs when the hobbyist doses for the bucket but the bucket is refilled multiple times.
3. **Expired or degraded product.** Thiosulfate solutions degrade; a bottle open for years may be substantially weaker. Sulfurous smell in many products is normal, not an indicator of freshness.
4. **Conditioner added to the tank but new water added faster than mixing.** A fish sitting directly in the inflow stream can experience full-strength tap water for seconds to minutes even though the tank average is fine. Small fish near the fill point are disproportionately affected.
5. **Conditioner formulated for chlorine only, in a chloramine municipality.** Utilities note that all methods for removing chloramine will also remove free chlorine, but **the reverse is not true** [GSWSA-FLUSH-FAQ].
6. **Massive overdose.** Thiosulfate is a reducing agent; excess thiosulfate consumes dissolved oxygen. Patent and hobbyist sources both note that inadvertent or purposeful overdosing can measurably reduce dissolved oxygen and cause respiratory stress [USPTO-4666610]. Aquarium Co-Op explicitly recommends adding an air stone for 3–4 hours after very heavy dosing [AQCOOP-COND].

[GENERALLY SUPPORTED]

2.6 Treating water before addition vs. dosing the aquarium

[PLAUSIBLE, POORLY DOCUMENTED — practical difference]

Two approaches:

- **Pre-treat:** condition water in a container, let it mix, then add. Guarantees no fish is ever exposed to untreated water. Requires containers and time.
- **Dose-the-tank:** add conditioner to the aquarium (usually dosed for full tank volume), then add tap water. Convenient for large or python-style refills.

Arguments for pre-treatment: eliminates the localized-exposure failure mode (2.5, item 4); allows temperature matching and CO₂ degassing before addition; allows the water to be tested before it touches fish.

Arguments that dose-the-tank is adequate: thiosulfate reaction with chlorine is rapid (effectively instantaneous on mixing); tank volume dilutes incoming water continuously; the practice is used routinely at scale in retail and by experienced keepers without documented harm.

Where the difference likely matters most: very large percentage changes; small tanks; high chlorine or chloramine municipalities; fast refill rates; fish that will not move away from the inflow (bottom dwellers under a fill stream, fish in a breeder box, sick or weak fish).

2.7 Municipal water treatment changes and seasonal chemistry

[ESTABLISHED] This is an under-recognized cause of “nothing changed on my end” mortality events.

2.7.1 Free chlorine conversion (“chlorine burn”)

Many chloraminating utilities perform an annual temporary switch from chloramine to free chlorine, typically lasting **30 to 60 days**, to oxidize accumulated ammonia and control nitrifying biofilm in the distribution system. A peer-reviewed study of EPA Region 6 systems reports that most systems conduct a free chlorine conversion once per year, that it can be initiated anywhere from winter to fall, and that it can last 30–60 days [EPA-R6-FCC].

Utilities routinely publish notices naming **fish, pond, and aquarium owners** alongside dialysis patients as the two groups requiring precautions [GREENSBORO-FCC; GUC-FCC; LAKEJACKSON-FCC; KILGORE-FCC; PALMBAY-FCC].

The specific hazard windows: - Free chlorine residuals during the conversion may be **higher** than the usual chloramine residual. - During the **transition weeks at each end**, water may contain chlorine, ammonia, and/or a mixture of the two [COCOA-FCC]. - Conversions are frequently combined with **distribution-system flushing** via fire hydrants, which mobilizes sediment, iron, manganese, and biofilm, and can produce discolored water [BRUNSWICK-FCC].

Why hobbyists miss it: the notice arrives as a bill insert or a website post; the water still looks and tastes normal to a human; no household routine changes.

2.7.2 Other municipal and seasonal variables

- **Source switching.** Utilities with multiple sources (surface reservoir vs. wellfield vs. purchased water) may switch seasonally, changing hardness, alkalinity, pH, and TDS substantially — sometimes within a single week.
- **Spring runoff / turbidity events.** Higher organic load leads to higher disinfectant dosing.
- **Corrosion-control chemistry.** Utilities dose orthophosphate, zinc orthophosphate, or lime for lead and copper control. Changes to this program alter phosphate levels (a common unexplained algae trigger) and change metal solubility in household plumbing.
- **pH adjustment.** Utilities may adjust finished-water pH seasonally.
- **Main breaks and repairs.** Following a repair, chlorine is typically boosted locally for disinfection, and disturbed pipe scale enters the water.
- **Hydrant flushing.** Same mechanism, scheduled.

[GENERALLY SUPPORTED] **Practical diagnostic:** if an unexplained mortality event coincides with a water change, the utility’s public notices page and water-quality hotline are legitimate investigative steps that hobbyists almost never take.

2.8 Sudden pH changes

[DISPUTED — see also Section 17]

Mechanism as usually stated. A rapid pH shift disrupts acid-base balance, gill ion transport, and the mucus layer; fish become lethargic, lose appetite, clamp fins, produce excess mucus, and may die within days.

Evidence complications:

1. **pH is a log scale of hydrogen ion activity, not a substance.** Two waters at pH 7.0 can be radically different in ionic composition. The physiological insult attributed to “pH shock” is frequently an ionic/osmotic insult that happens to be accompanied by a pH change.
2. **The cause of the pH change matters more than the magnitude.** A pH swing driven by CO₂ (planted tanks, injected CO₂, degassing tap water) is generally tolerated far better than a numerically identical swing driven by a change in mineral content [ANECDOTAL but mechanistically coherent and widely reported by experienced keepers].
3. **Older field evidence of tolerance.** Wiebe (1931), as summarized in a technical review, moved several fish species between lakes at pH 4.4–6.4 and lakes at pH 8.2–8.7 in both directions, with no mortality and no observed “shock” [AQSCI-SHIFTS, citing Wiebe

1931]. This is old, low-control work, but it is direct evidence against the strongest form of the pH-shock claim.

4. **Counter-evidence.** Veterinary and hobby references describe recognizable pH-shock presentations, and note the specific case of a fish in a transport bag: opening a bag after several hours can produce a rapid pH rise within about 20 minutes as CO₂ escapes, which can seriously stress or kill the fish [AQWIKI-PHSHOCK]. Note that in this specific case the harm is more plausibly attributable to the accompanying jump in un-ionized ammonia than to pH per se — see Section 7.

Synthesis for retrieval: “pH shock” as a stand-alone, magnitude-driven killer is **poorly supported**. pH change as (a) a proxy for ionic change, (b) a modifier of ammonia and metal toxicity, and (c) an indicator of buffer collapse is **well supported**. The claim should not be repeated in its simple form.

2.9 GH, KH, TDS and conductivity differences; osmotic stress

[ESTABLISHED — mechanism] / [DISPUTED — practical thresholds in home aquaria]

Mechanism. Freshwater fish are hyperosmotic to their environment: their body fluids are saltier than the water. They constantly gain water osmotically across the gills and lose ions, and must actively pump ions in (via gill chloride cells) and excrete large volumes of dilute urine. Moving a fish abruptly to water of very different ionic strength changes the osmotic gradient and the workload on this system. Extremely soft, mineral-poor water increases ion loss; very hard water reduces it and can present a different regulatory challenge to soft-water-adapted species.

Where the risk is genuinely high [GENERALLY SUPPORTED]: - Moving fish from hard water into near-zero-TDS RO water without remineralization. - Moving soft-water-adapted wild-caught fish into water above ~15 dGH. - Invertebrates (shrimp) — molting depends on GH; Caridina shrimp are the classic case of genuine TDS sensitivity, and shrimp keepers routinely observe mortality after TDS swings that fish ignore. - Very young fry and very small-bodied fish, which have high surface-area-to-volume ratios.

Where the risk is likely overstated [DISPUTED]: routine 25–50% water changes with the keeper’s own consistent tap water, in a tank whose TDS has drifted upward from evaporation, in robust community fish. Some experienced sources argue “TDS shock” in this scenario is essentially never the cause of death; others report shrimp losses under exactly these conditions.

Key point about direction: the dangerous direction in a neglected tank is usually **from high-TDS old water toward low-TDS fresh water** (the classic Old Tank Syndrome water change), not the reverse.

2.10 Large water changes

[DISPUTED — see also Section 17.1 and 17.2]

The hobby’s inherited “never more than 25%” rule has weak documented origin. Competing positions:

Position A — large changes are fine and often better. Frequent large changes (50–90%) are standard practice in fish breeding, retail systems, discus keeping, and fry rearing. Large changes reduce nitrate, dissolved organics, and hormone accumulation more effectively. The principal requirements are matched temperature and complete dechlorination.

Position B — large changes carry real risk in specific circumstances. The risk is not the volume itself but the **magnitude of the delta** between old and new water. In a well-maintained tank, tank water and treated tap water are nearly identical, and a 90% change moves almost nothing. In a neglected tank they can be radically different, and a 90% change is a massive, instantaneous chemistry change.

Synthesis: the operative variable is **difference**, not **percentage**. This reframing resolves most of the apparent contradiction between the two camps and is the single most useful conceptual correction available in this topic area.

2.11 Neglected aquariums receiving sudden large water changes

[GENERALLY SUPPORTED — this is a real and specific hazard]

The classic dangerous sequence:

1. Tank is not water-changed for months. Nitrate climbs (100–400+ ppm is achievable). Dissolved organics accumulate. Nitrification produces hydrogen ions, consuming carbonate buffer; KH falls. pH drifts downward, sometimes to 6.0 or below. TDS climbs, partly from nitrate and partly from evaporation-driven mineral concentration.
2. Fish physiologically acclimate to these conditions over weeks. Ion regulation, blood pH, and renal function adjust. This is genuine physiological acclimation, not just tolerance.
3. Something prompts intervention (fish look bad, water smells, guests are coming). A 75–90% water change is performed with fresh tap water.
4. The new water is higher pH, higher KH, higher in some minerals, dramatically lower in nitrate and organics, and possibly a different temperature.
5. Fish that were stable in the old conditions are moved across a very large chemistry gap in minutes.

Mechanisms actually implicated in step 5: - The un-ionized ammonia trap. If the tank had accumulated ammonia at low pH (mostly as harmless ammonium), raising the pH by 1.5–2 units converts a large fraction to toxic NH₃ essentially instantly. This is the most defensible and most dangerous mechanism in the whole scenario, and it follows directly from established ammonia chemistry [UF-FA16; SRAC-4603]. - **Acid-base and ionic disruption** as above. - **Loss of a low-pH environment that had been suppressing the biofilter**, if the tank was in a stalled state.

Consensus corrective practice [GENERALLY SUPPORTED]: in a badly neglected tank, perform a **series of small changes** (10–20%) over one to three weeks rather than one large change; test ammonia before changing water; consider that the fish’s current

environment, while bad, is one they are adapted to. A veterinary telehealth source describes a case-style example of a neglected 50-gallon tank with nitrate over 150 ppm and pH at 6.0 recovering over a four-week phased change plan [ASKAVET-OTS — note this is a commercial veterinary blog, not peer-reviewed].

2.12 Old Tank Syndrome — what the evidence supports

[GENERALLY SUPPORTED as a description; DISPUTED as to mechanism and prevalence]

The standard definition. A gradual decline in a chronically under-maintained aquarium characterized by: high nitrate, accumulated organics, depleted carbonate hardness (KH), and depressed pH. Frequently invoked to explain why long-term resident fish appear healthy while every newly added fish dies within 24–48 hours.

What is well supported: - Nitrification is an acidifying process. Oxidation of ammonia to nitrate produces hydrogen ions, which consume carbonate alkalinity. In low-KH source water, pH will fall over time without water changes. This is straightforward chemistry. - Practical Fishkeeping's treatment emphasizes two consequences of carbonate depletion: compromised osmoregulation, and loss of buffering allowing pH to drop or swing, either of which can stress or kill fish [PFK-OTS]. - Nitrifying bacteria performance is pH-dependent. Below roughly pH 6.5, classical ammonia-oxidizing bacteria work poorly, which can cause ammonia to accumulate in an apparently "cycled" tank [AQSCI-OTS]. - The "new fish dies, old fish fine" pattern is genuinely diagnostic-adjacent, because it points to a difference between resident (acclimated) and newcomer (unacclimated) animals rather than to an acutely toxic condition.

What is disputed or overstated: - One technical source argues that hobbyists label **any** ammonia spike as "old tank syndrome" or a "crash," when the vast majority of ammonia spikes are not accompanied by a pH drop and are therefore not OTS at all — the most common actual cause being an unnoticed dead fish decaying [AQSCI-OTS]. - The same source notes OTS-type pH collapse can also occur in a **very new** tank cycled in low-KH water, meaning "old" is a misnomer. - Hobbyist forum threads asking for a single categorical case of OTS-caused fish death produce mostly near-misses and second-hand reports rather than documented mortality [UKAPS-OTS — ANECDOTAL]. - Some popular write-ups assert that nitrates "consume the buffer." This is loose: it is the hydrogen ions co-produced during nitrification, not nitrate itself, that consume alkalinity. Nitrate concentration is a *marker* of accumulated nitrification, not the acidifying agent.

Retrieval statement: OTS is best understood as **carbonate depletion plus organic accumulation in a chronically under-maintained low-KH system**, with the pH drop as the key measurable signature. High nitrate alone is not OTS. An ammonia spike alone is not OTS.

2.13 Water changes after long periods of evaporation and top-offs

[GENERALLY SUPPORTED]

Mechanism. Evaporation removes water and leaves solutes behind. Topping off with tap water adds the tap water's full mineral load again. Repeated over months, GH, KH, TDS, sodium, and any non-volatile contaminant in the source water all **concentrate**. Nitrate and dissolved organics also concentrate.

Consequences: - A tank that started at 8 dGH can reach 15–20 dGH over months of top-offs, with no water change ever performed. - TDS may be double or triple the source water. - The keeper's mental model ("my tank matches my tap water") becomes false without any visible sign. - The eventual water change is then a large ionic step-down, not a step-up — the reverse of the Old Tank Syndrome direction, and the classic scenario for shrimp losses.

Note on interaction with OTS: evaporation concentration (raising minerals) and nitrification acidification (consuming carbonate) act in opposite directions on KH and can partially mask each other. A tank can simultaneously have elevated TDS and depressed KH. This is why TDS alone is not a substitute for KH testing.

2.14 Well water considerations

[GENERALLY SUPPORTED]

Private wells lack the standardization of municipal supply and introduce distinct risks:

- **Dissolved CO₂.** Groundwater is frequently supersaturated with CO₂ under pressure. On drawing, pH reads low; after degassing, pH rises substantially — sometimes by more than a full unit. A hobbyist who tests tap water immediately and matches "pH 6.6" will find their tank at 7.8 within a day. See 2.20.
- **Dissolved iron and manganese.** Ferrous iron in solution oxidizes on aeration to ferric hydroxide, producing rust-colored precipitate. Aeration-driven CO₂ loss raises pH and accelerates this oxidation [ASRS-AERATION].
- **Hydrogen sulfide.** Genuine H₂S is present in some well waters (the "rotten egg" well). Unlike the aquarium-substrate H₂S myth (Section 3.5), this is a real and documented source-water phenomenon.
- **Low dissolved oxygen.** Groundwater can arrive substantially undersaturated in oxygen.
- **Nitrate.** Agricultural areas can have well nitrate at or above the 10 mg/L NO₃-N drinking-water limit — equivalent to ~44 ppm NO₃, meaning fresh water changes may *add* nitrate.
- **Hardness extremes.** Some wells produce very hard water (20–30+ dGH); others are extremely soft.
- **Seasonal variability.** Water table changes alter chemistry.
- **No disinfectant residual.** A benefit for aquarists (no dechlorination needed) but means the water is not sterile.
- **Softener interaction.** See 2.15.

2.15 Household water softeners

[ESTABLISHED — mechanism]

Conventional ion-exchange softeners **exchange calcium and magnesium ions for sodium** (or potassium in KCl systems). Critically:

- **Softened water is not soft in the aquarium sense.** GH (calcium + magnesium) falls; **TDS and conductivity remain essentially unchanged**, because ions were swapped, not removed. Sodium content rises substantially.
- **KH is largely unaffected.** Bicarbonate remains; it is now sodium bicarbonate rather than calcium bicarbonate. The water may therefore have low GH and normal-to-high KH — a combination that is uncommon in nature and unhelpful for most fish and plants.
- Consequences for livestock: shrimp cannot molt properly without calcium and magnesium; livebearers and Rift Lake cichlids kept in softened water frequently show poor health; plants may show calcium/magnesium deficiency.
- **Regeneration cycles** can push a slug of high-salinity brine into household lines. A water change performed shortly after a regeneration cycle can deliver anomalously high sodium/chloride.
- Many homes have an **unsoftened outdoor spigot or a bypass tap** — often the correct water source for aquarium use.

Practical diagnostic: softened water typically reads **low GH, moderate KH, and normal-to-high TDS**. That combination is a strong fingerprint.

2.16 Reverse osmosis water and remineralization errors

[GENERALLY SUPPORTED]

RO water properties: near-zero TDS, near-zero GH, near-zero KH, essentially no buffering capacity. pH readings on pure RO water are meaningless and unstable — the meter has almost no ions to measure and the water's pH is dominated by dissolved atmospheric CO₂.

Failure modes:

1. **Using straight RO water for a water change.** Rapidly dilutes tank minerals and buffering. Repeated over time, drives the tank toward zero KH and pH instability. **[GENERALLY SUPPORTED — a genuine cause of loss]**
2. **Remineralizing with the wrong product.** GH-only remineralizers (calcium/magnesium salts) add hardness but no buffering; the resulting water is pH-unstable. GH+KH products add both. Rift Lake buffers add a great deal of both. Using a GH-only product where KH was needed produces a tank prone to pH collapse.
3. **Dosing errors.** Remineralizer dosing is by volume and requires a TDS meter or GH/KH test to verify. Eyeballing produces batch-to-batch variation, meaning every water change is a small ionic step in a random direction.
4. **Exhausted RO membrane or DI resin.** A failing membrane silently passes progressively more TDS, including whatever the failing prefilter is no longer removing. Without a TDS meter on the product line this is invisible.
5. **Failed carbon prefilter with chloramine.** Chloramine damages thin-film composite RO membranes and requires adequate carbon contact time to remove. A spent carbon prefilter can allow chloramine through.
6. **Assuming RO water is safe by definition.** RO removes ions; it does not remove all organics or all dissolved gases.

2.17 Extremely low-mineral water

[GENERALLY SUPPORTED]

Beyond the osmoregulatory load, very low mineral content has systemic effects:

- **No buffering.** pH becomes hostage to CO₂, organic acids, and nitrification. Large swings become possible from small inputs.
- **Nitrification impairment.** Carbonate is a carbon source for autotrophic nitrifiers and buffers the acid they produce. At very low KH, biofilter performance can degrade. Some sources note that soft-water systems may be dominated by different nitrifying organisms (ammonia-oxidizing archaea, *Nitrospira*, *Nitrotoga*) that tolerate low pH better than the classical *Nitrosomonas/Nitrobacter* model, which complicates simple predictions [GLASSBOX-OTS — hobby-technical source].
- **Increased toxicity of metals.** Copper, zinc, aluminum, and cadmium are all more toxic in soft water (Section 8.2).
- **Molting failure in shrimp and snail shell erosion.**

2.18 Source-water contamination, heavy metals, plumbing, and hot-water heaters

[GENERALLY SUPPORTED]

- **Copper from plumbing.** Copper leaches from copper pipe, especially in **new installations, acidic water, soft water, and after prolonged stagnation**. The US EPA action level for copper in drinking water is **1.3 ppm** — a concentration far above the threshold of concern for sensitive invertebrates.
- **Lead.** From legacy lead service lines, lead solder in pre-1986 plumbing, and brass fixtures. Mobilized by disturbance, by water chemistry changes (the Flint and Washington DC cases both involved disinfectant/corrosion-control changes), and by stagnation.
- **Zinc.** From galvanized pipe and some fittings.
- **Iron and manganese.** More of a well-water and main-flushing issue.
- **Stagnation effect.** Water that has sat in household pipes overnight or over a vacation accumulates more leached metal. **Running the tap for 1-2 minutes before drawing aquarium water is a documented mitigation** for metals, and is standard advice in drinking-water guidance.
- **Hot water heaters.** Multiple concerns: (a) tank-type heaters contain a **sacrificial anode rod**, typically magnesium or aluminum,

which corrodes by design and adds those metals to the water; (b) the tank accumulates sediment, scale, and biofilm; (c) hot water is more aggressive toward plumbing metals, mobilizing more copper and lead; (d) hot water systems are the classic *Legionella* habitat (a human, not fish, concern); (e) hot water is degassed relative to cold, and hot-then-cooled water has a different dissolved gas profile. **[GENERALLY SUPPORTED]** The standard hobby advice — mix hot and cold at the tap for temperature matching, accepting a small hot-water fraction — trades a modest metals/scale risk against a thermal-shock risk. For sensitive invertebrates and soft-water systems, heating cold water separately is the conservative option.

- **New plumbing, water heaters, or fixtures.** Flux, solder, cutting oil, pipe dope, and PEX-related off-flavor compounds can all appear transiently after plumbing work. A “we had a plumber in last week” answer is diagnostically significant and rarely volunteered.
- **Water filters and softener bypasses.** Point-of-use carbon filters degrade; an exhausted carbon block passes chlorine/chloramine while the user believes it is being removed.

2.19 Sudden dissolved gas changes and gas supersaturation

[ESTABLISHED — phenomenon] / [DISPUTED — frequency in home aquaria]

Mechanism. Gas solubility rises with pressure and falls with temperature. Water that is saturated with air at low temperature and/or high pressure becomes **supersaturated** when it is warmed or brought to atmospheric pressure. Fish breathing supersaturated water can develop **gas bubble disease (GBD)** — emboli forming in blood and tissues, analogous to decompression sickness in divers [WIKIVET-GBD].

Documented causes in captive systems: - Air leaks on the suction side of pumps, cavitating pumps, and valve/pipe leaks that entrain air under pressure [TOLEDO-GBD; WIKIVET-GBD] - Sudden temperature gradients [WIKIVET-GBD] - Excessive aeration combined with high temperature [PETPLACE-SUPERSAT] - Heavy algal blooms producing oxygen faster than it can diffuse out (mainly ponds) [WIKIVET-GBD] - Air transport of fish [WIKIVET-GBD]

Clinical signs: bubbles under the skin and in the fins; bubbles in the eye (exophthalmia/“popeye”-like presentations); abnormal buoyancy; acute mortality. A veterinary account describes emboli in gill lamellae, opercula, around the mouth, and “silver lines” of gas visible between fin rays in acute cases [TOLEDO-GBD].

The home-aquarium water-change scenario. In principle: cold water saturated with air enters the home, is heated (in the heater or by mixing) while still under line pressure, and is then released to atmospheric pressure in the aquarium — producing supersaturated water. The visible sign is the familiar **cloud of fine bubbles coating glass, decor, and fish after a large water change**.

How much of a risk is this actually? - One technical review acknowledges the mechanism is theoretically possible from a tap during a large water change, but argues that aeration on essentially any aquarium resolves the excess long before gas bubble disease develops, and concludes it is not something home aquarists need to worry about [AQSCI-GBD]. - Another technical source, from a group that measures total dissolved gas directly, states that conventional aquarium pumps rarely generate supersaturation sufficient to produce classical GBD by themselves, and that risk arises only when several factors coincide. It makes the important clarification that **the visible bubbles in an aquarium are usually a consequence of degassing, not the cause of the disease** — the dangerous condition is invisible, clear, supersaturated water [FOSSA-GBD]. - Conversely, at least one retail operation reported having to install a large aerating/degassing device because supersaturated incoming water was forming bubbles on fish and killing them [AQSCI-GBD, reporting a commercial account — ANECDOTAL].

Retrieval statement: GBD is a real, veterinary-recognized condition; the “bubbles on the glass after a water change” phenomenon is real and is evidence of supersaturation; but progression to clinically significant GBD in a normally aerated home aquarium appears **uncommon**. Highest-risk configurations: very large water changes, direct-from-tap refill through a hose, cold source water heated in-line, minimal surface agitation, and small tanks. Pre-mixing and aerating replacement water in a bucket eliminates the mechanism entirely.

2.20 Replacement-water CO₂ and its effect on pH

[ESTABLISHED]

Mechanism. Water under pressure in a distribution system or aquifer holds more CO₂ than it can hold at atmospheric pressure. Dissolved CO₂ forms carbonic acid species and lowers pH. On standing or aeration, CO₂ outgasses toward atmospheric equilibrium and **pH rises**.

Magnitudes. A worked example from a hobby forum, using measured KH: tap at KH 270 ppm and pH 7.2 implies ~30 ppm CO₂; after aeration to atmospheric equilibrium, the same water reads **pH 8.0** [AQADVICE-DEGAS]. A technical source gives the general case: low-CO₂ tap water can come out at pH 7.8 and drop to 6.8 as it absorbs atmospheric CO₂, while high-CO₂ tap water can come out at 6.2 and stabilize at 7.8 [AQSCI-CO₂PH].

Consequences for the aquarist: 1. **The pH you measure at the tap is not the pH the water will have in your tank.** Testing tap water requires letting a sample sit (or aerating it) for 12–24 hours first. Failing to do this is an extremely common error and leads people to “correct” a problem that does not exist. 2. **A large water change with high-CO₂ water produces a transient pH depression followed by a rise over hours** as the tank degasses. Fish experience a pH excursion in two directions. 3. In heavily planted or CO₂-injected tanks, this interacts with the diurnal photosynthetic pH swing. 4. **Direction matters for ammonia.** If the tank contains any ammonia, the post-water-change pH *rise* increases the toxic NH₃ fraction.

Related important correction: surface agitation causes CO₂ to **leave** the water (outgas) toward atmospheric equilibrium; it does not add CO₂ to the tank. Aeration raises pH in CO₂-rich water and does not lower it. **[ESTABLISHED — this is a frequently reversed claim in hobby writing.]**

2.21 Dissolved oxygen differences in replacement water

[GENERALLY SUPPORTED]

- Cold tap water is typically near or above saturation for its temperature; when warmed in the tank it becomes supersaturated (see 2.19).
- Hot tap water is typically **degassed** — it has lost oxygen in the heater. Filling largely with hot water introduces oxygen-poor water.
- Well water may arrive substantially undersaturated.
- Heavy dechlorinator overdose consumes oxygen (2.5, item 6).
- A large water change in a tank with high biological oxygen demand can transiently disturb the DO balance while the new water equilibrates.

None of this is measurable with the standard five tests.

2.22 Very cold or very warm replacement water

Covered mechanistically in 2.2. Additional consequences worth separating:

- **Very cold water** carries more dissolved gas (supersaturation risk on warming), can cause acute thermal stress, and slows the biofilter transiently.
- **Very warm water** carries less oxygen, accelerates metabolism and thus oxygen demand, and increases the un-ionized ammonia fraction at any given pH and TAN.
- **Directional asymmetry [GENERALLY SUPPORTED]:** rapid warming is generally considered more dangerous than rapid cooling of equal magnitude, because it raises oxygen demand at the same moment it lowers oxygen availability. Rapid cooling is the more common accident.

SECTION 3. MAINTENANCE PERFORMED DURING A WATER CHANGE — THE CONFOUNDED VARIABLE

3.1 Why this section matters more than Section 2

A “water change” as performed by most hobbyists is not one action. It is a bundle:

siphon substrate → remove water → possibly clean filter → possibly clean glass → possibly move or scrub decor → possibly trim or uproot plants → possibly unplug equipment → refill → possibly dose fertilizer, conditioner, buffer, or medication → possibly feed → restart equipment

Any of ten or more distinct interventions occurred within a thirty-minute window. When fish die afterward, “the water change” is blamed because it is the label for the whole bundle. **Isolating which component was responsible is the central diagnostic task**, and it is the reason a timeline reconstruction (Section 16.6) is more valuable than another round of testing.

3.2 Deep substrate disturbance and gravel vacuuming

[GENERALLY SUPPORTED]

What actually happens. Substrate accumulates detritus, uneaten food, feces, decomposing plant matter, and a large heterotrophic microbial population. Disturbing it releases: - Fine particulates into the water column (mechanical gill irritation; increased turbidity) - Dissolved organics and ammonia from the interstitial water, which can be considerably more concentrated than the water column - A pulse of oxygen demand as newly suspended organics are aerobically decomposed - Bacteria into suspension, sometimes producing a visible bacterial bloom over the next 24–72 hours

The oxygen-demand mechanism is the most underrated item in this entire document. Stirring a large mass of accumulated organics into an oxygenated water column creates an immediate biological oxygen demand spike. In a warm, heavily stocked tank with modest surface agitation, this can measurably depress dissolved oxygen for hours. Symptoms — surface gasping, rapid opercular movement, lethargy — are identical to ammonia poisoning, and the aquarist tests for ammonia, finds none, and concludes the water is fine.

First-time deep-clean risk. A tank that has never had its substrate vacuumed, then receives a thorough deep vacuum, releases far more accumulated material than a routinely maintained tank. The risk scales with the interval since the last cleaning, not with the vigor of the current cleaning.

3.3 Disturbing accumulated organic material and dead zones

Beyond open substrate: under and behind rocks, driftwood, and ornaments; inside hollow decorations; under filter intakes; in undergravel filter plenums; inside canister filter media baskets; inside sponge filters that have not been squeezed in a year. All of these can accumulate anaerobic-adjacent (hypoxic) sludge. Moving a large ornament that has sat undisturbed for a year can release a visible black cloud with an obvious smell.

3.4 Anaerobic and hypoxic zones — what is actually true

[DISPUTED — one of the most contested topics found in this research]

Position A — hypoxic zones exist and matter, but true anaerobic H₂S production does not occur meaningfully in aquarium substrates. One detailed technical treatment argues that anoxic conditions sufficient to produce poisonous quantities of hydrogen sulfide are effectively impossible in an aquarium substrate, and that hobbyists routinely misattribute deaths to substrate H₂S. The same source is explicit that **low-oxygen (hypoxic) conditions absolutely can exist** — behind a stopped filter, in a deep pocket of an ornament where uneaten food accumulates away from flow, or in a poorly aerated, overstocked, overfed tank — and that under those conditions organic bacterial toxins along with ammonia and nitrite can be produced and can kill fish, but that this is very rare [AQSCI-ANAEROBIC].

Position B — H₂S does form in substrates but never reaches harmful concentrations in a well-agitated system. Another source concedes that H₂S does develop in substrates when multiple conditions are met, but states it never reaches a concentration sufficient to cause health issues in closed systems with good surface agitation or aeration, because H₂S is readily driven out of solution by agitation [PAKIS-SUBSTRATE].

Position C — H₂S is a real risk in specific failure modes. A third treatment argues sulfate reduction requires total deprivation of both oxygen and nitrate, and identifies specific realistic failure modes: an undergravel filter plate with accumulated mulm when the filter stops (power failure or mechanical failure), and canister/FSB filters that cease operation with organic load inside [AQPONDANSWERS-H₂S].

What all three positions agree on: 1. The classic hobby claim that a deep sand bed spontaneously produces enough H₂S to kill a tank is **not supported**. 2. Stopped-flow situations with accumulated organics are the genuine risk configuration. 3. Surface agitation strips H₂S efficiently. 4. Hypoxic pockets producing ammonia, nitrite, and bacterial metabolites are more plausible than sulfide as a cause of harm.

Retrieval statement: substrate hydrogen sulfide should be classified as **[PRIMARYLY MYTHOLOGICAL as usually invoked]** but **[PLAUSIBLE in the specific case of a filter or plenum that has been stopped with organics inside and is then restarted]**. The stopped-canister-restart scenario is discussed further in 3.9.

3.5 Cleaning filters and replacing filter media

[ESTABLISHED — biofilter location] In a mature freshwater aquarium, nitrifying biofilm is distributed across all wetted surfaces — filter media, substrate, glass, decor, plant surfaces, and hardscape. The filter typically holds the largest share because it combines high surface area with high flow (oxygen and substrate delivery), but it is not the sole location.

Failure modes ranked by realistic severity:

1. **Wholesale media replacement.** Discarding all cartridges/sponges/ceramic media and installing new. This is the single most damaging routine maintenance error in the hobby, and it is **actively encouraged by the design of cartridge-based filters and by the marketing of monthly cartridge replacement**. Result: substantial loss of nitrifying capacity, ammonia and nitrite spikes over the following days. **[ESTABLISHED]**
2. **Cleaning all biological media at once, thoroughly.** Even without discarding, aggressive scrubbing removes biofilm. Standard mitigation: clean half the media per session, alternate sessions. **[GENERALLY SUPPORTED]**
3. **Allowing media to dry out.** Nitrifying bacteria in an unmoistened medium lose viability progressively. Leaving media on a counter while attending to something else is a common accident. **[GENERALLY SUPPORTED]**
4. **Filter off too long.** See 3.6.
5. **Rinsing biological media under chlorinated tap water.** See 3.5.1 — **[DISPUTED]**.

3.5.1 The tap-water rinse controversy

Standard hobby position: never rinse biological media in tap water; chlorine and chloramine kill nitrifying bacteria; use removed tank water.

Challenge to that position: disinfection is a function of **concentration × contact time (CT)**, not an instantaneous event. One technical treatment argues that at typical tap residuals (0.25–1.0 ppm) the contact time needed is on the order of an hour or more, whereas a real-world rinse lasts 15–60 seconds; it cites Cunliffe (1991), “Bacterial Nitrification in Chloraminated Water Supplies,” reporting that chloramine at typical tap concentrations took about an hour and a half to kill even small quantities of nitrifying bacteria. It further notes that bacteria in biofilm are physically protected by the extracellular polymeric matrix, and reports hobbyist and commercial accounts of routinely rinsing sponges in chlorinated tap water for years without ammonia spikes [AQSCI-TAPRINSE].

Another hobby-technical write-up reaches the same conclusion, adding that laboratory-cultured bacteria are typically **less** resistant to disinfectants than established natural populations, and that concentration matters more than exposure time for inactivation [AQUASCAPEGUIDE-TAPRINSE].

Contrary claims: numerous consumer-facing articles assert dramatic figures — e.g., that 0.2 ppm chlorine destroys up to 70% of surface-adhered *Nitrospira* within 30 seconds, or that even 10 seconds of tap contact destroys surface biofilm irreversibly. **These specific figures could not be traced to any primary source during this research**, appear on low-quality or AI-generated content farms, and should be treated as **[UNSUPPORTED]** unless a citation surfaces.

Retrieval statement: the balance of traceable evidence favors “a brief tap-water rinse of biological media is unlikely to crash a mature biofilter,” while the balance of hobby practice favors caution. Because tank water is free and already in a bucket, the practical cost of the conservative approach is near zero — which is why the myth persists harmlessly. The claim should be classified **[DISPUTED / probably overstated]** rather than repeated as fact.

The genuinely risky version of this action is different: soaking media in tap water for an extended period, using hot water, using soap or vinegar or bleach, or using water that is very different in temperature from the tank.

3.6 Filter remaining off too long, and forgetting to restart

[GENERALLY SUPPORTED / ESTABLISHED for the canister case]

Nitrifiers are aerobic. Without flow, oxygen inside dense media is consumed within hours. Short interruptions (under an hour) are generally inconsequential. Longer interruptions progressively kill the aerobic population.

The canister filter restart hazard [GENERALLY SUPPORTED]. A sealed canister full of organic-rich media that has been switched off for many hours goes hypoxic. Heterotrophic and facultative anaerobic metabolism produces reduced compounds, organic acids, and foul-smelling metabolites; the water inside becomes oxygen-depleted and organically loaded. **Restarting the canister injects that entire volume into the aquarium in seconds.** This is the most defensible “sludge injection” scenario in home aquaria, and is a recognized hazard among experienced keepers. Correct practice: if a canister has been off for more than a few hours, open it, discard the internal water, rinse the media, and restart.

Forgotten filters. The most common variant is unplugging a filter to do maintenance and not plugging it back in. Consequences accrue over 12–72 hours: falling oxygen (no surface agitation), rising ammonia (biofilter dying and no mechanical circulation), and a fish population showing hypoxia signs by morning. This is a leading candidate whenever “everything was fine, then next morning they were all at the surface.”

3.7 Heaters: forgetting, damage, and exposure during draining

[ESTABLISHED]

1. **Heater unplugged and not restarted.** Slow temperature decline over hours; in a cool room, a 78°F tropical tank can reach the low 60s within a day. Fish become lethargic, stop eating, and become susceptible to *Ichthyophthirius*.
2. **Heater left running while exposed to air during draining.** Glass heaters heat rapidly without water to conduct heat away. Two outcomes: thermal shock cracking the glass tube when water returns (immediate failure, and potentially an electrical hazard), or the heating element damaging itself. Older glass heaters are more vulnerable than modern titanium or shatterproof designs. Standard practice: unplug the heater before draining and wait several minutes after refilling before restoring power.
3. **Thermostat failure in the “on” position.** The most lethal single equipment failure in the hobby. A stuck-on heater in a small tank can raise temperature to lethal levels within hours. Frequently discovered as “all fish dead, water 95°F, everything else normal.”
4. **Thermostat failure in the “off” position.** Slower, less lethal, more insidious.
5. **Heater displaced during maintenance** so its thermostat sits in a dead zone or against the glass, producing a false reading and a gradient.
6. **Heater sized wrong.** Over-powered heaters overshoot; under-powered heaters cannot hold temperature in a cold room.

3.8 Equipment malfunction after maintenance, altered flow, reduced surface agitation

[GENERALLY SUPPORTED]

- **Impeller not reseated** after cleaning → filter runs but moves little water, or runs dry and overheats.
- **O-ring pinched or not lubricated** on a canister → leak, air ingress on the suction side, loss of prime, and potentially air entrainment (see 2.19).
- **Air lock** after refilling → filter humming without flow.
- **Spray bar or outflow repositioned** → surface agitation reduced. **This is the specific, frequently-missed cause of post-maintenance oxygen problems.** Lowering the water level, redirecting an outflow below the surface, or removing an air stone during cleaning all cut gas exchange without any visible sign.
- **Water level left low or high** relative to a HOB filter outflow, changing the amount of splash and therefore gas exchange.
- **Air pump check valve reinstalled backwards**, or air line pinched behind the tank.
- **Powerhead or wavemaker left unplugged.**
- **Intake sponge or prefilter reinstalled clogged**, reducing flow immediately.
- **CO2 system disturbed** — a bumped needle valve or a solenoid left in manual can dump CO2 overnight, driving pH down and displacing oxygen.

3.9 Moving decorations, trapping and injuring fish

[GENERALLY SUPPORTED]

- Fish crushed or pinned under repositioned rocks and wood. Bottom dwellers, loaches, and plecos actively seek out and hide inside caves and under objects.
- Fish trapped inside hollow ornaments during draining and left in a few centimeters of water.
- Fish siphoned into a gravel vacuum (fry and very small fish routinely) or trapped against the intake.
- Fish netted and injured — net damage to the mucus layer and to fins provides entry points for opportunistic bacteria such as *Flavobacterium* and *Aeromonas*. **Handling injury frequently manifests 2–5 days later as a bacterial lesion**, by which time the connection to maintenance is no longer obvious.

- Fish jumping out during maintenance, when the lid is off and the water is being disturbed. Frequently discovered hours later behind the stand.
- Territorial reorganization: rearranging hardscape resets established territories and can precipitate a burst of aggression over the following 24–72 hours (Section 10).

3.10 Removing too much plant material at once

[PLAUSIBLE, POORLY DOCUMENTED]

Several mechanisms have been proposed; the evidence in home aquaria is thin but the chemistry is coherent:

- **Loss of nitrogen uptake capacity.** In a heavily planted tank where plants remove a meaningful fraction of ammonia directly, removing most of the biomass removes that sink. Combined with an unchanged feeding regime, ammonia and nitrate can rise.
- **Loss of oxygen production** during the photoperiod.
- **Loss of biofilm surface.** Plant leaves carry nitrifying and heterotrophic biofilm.
- **Decomposition of cut and damaged tissue.** Uprooting disturbs substrate (3.2) and leaves severed root mass to decompose.
- **Loss of cover** → behavioral stress in species that require it (Section 10.7).

Uprooting sword plants, crypts, and other heavily rooted species is simultaneously a substrate-disturbance event and should be treated as such.

3.11 Disturbing established biological surfaces

Scraping all glass surfaces, scrubbing all hardscape, replacing all substrate, or bleaching decor removes established biofilm — both nitrifying and the general heterotrophic community that competes with opportunistic pathogens. This is rarely acutely lethal on its own but contributes to the cumulative disturbance load of a “big clean.”

A useful integrating concept for retrieval: the “spring cleaning” event. Deep gravel vacuum + full filter clean + media replacement + decor scrub + large water change + rearranged hardscape, all in one session. Each component is individually survivable; performed simultaneously, they remove most of the system’s biological buffering and impose several stressors at once. **The most defensible general advice found across sources is to separate maintenance actions in time** — filter one week, substrate another — rather than to avoid any of them.

3.12 Accidental introduction of cleaning chemicals, soaps, aerosols, and lotions

[ESTABLISHED as a hazard class; dose-dependent]

Realistic vectors:

Vector	Notes
Bucket previously used for household cleaning	Surfactant residue persists through rinsing; the classic case. Aquarium-only, labeled buckets are the standard mitigation
Hands with soap, hand sanitizer, or lotion residue	Alcohol-based sanitizer evaporates quickly but leaves emollients; lotions leave lipids and fragrances. Rinsing hands in plain water before entering the tank is standard practice
Sponges or scrubbers previously used with detergent	Same as buckets
Glass cleaner sprayed at the aquarium hood/glass	Aerosol drift into an open-top tank. Ammonia-based glass cleaners are the worst case. Correct method: spray onto the cloth, away from the tank
Air fresheners, aerosol deodorants, hairspray, dry shampoo, perfume	Enter through surface gas exchange. A tank in a bathroom or bedroom has continuous low-level exposure
Insecticide sprays, foggers, “bug bombs”	Extremely high risk — see Section 8.6
Flea/tick spot-on treatments on pets in the room, and hands after application	Fipronil, permethrin, imidacloprid — see Section 8.6
Paint, varnish, stain, solvent fumes	Room renovation is a recognized cause of whole-tank losses
Nicotine and vape aerosol	Reported anecdotally; nicotine is an insecticide-class alkaloid
Scented candles, incense, wax melts	Anecdotal
Oven self-clean cycle / non-stick pan overheating	Releases PTFE decomposition products; documented lethal to birds, plausible for other air-exchanging systems; not documented for fish
Cleaning products used on the aquarium stand or floor beneath	Ammonia and bleach fumes

Important calibration: trace exposure is not automatically lethal. The dose that matters depends on the compound (pyrethroids are toxic at nanomolar concentrations; a fragrance may be inert at any realistic dose), the tank volume, and the surface exchange rate. The correct framing is **hazard class × dose × volume**, not “any chemical near a tank kills fish.”

3.13 Which maintenance risks are well documented and which are exaggerated

Risk	Assessment
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Evidence strength		
Wholesale filter media replacement causing ammonia/nitrite	[ESTABLISHED]	Well documented. Real.
Filter off many hours, then restarted with contents intact	[GENERALLY SUPPORTED]	Real, especially canisters.
Substrate disturbance causing oxygen demand and bloom	[GENERALLY SUPPORTED]	Real and under-recognized.
Heater left on while exposed to air	[ESTABLISHED]	Real.
Reduced surface agitation after maintenance	[GENERALLY SUPPORTED]	Real and under-recognized.
Soap/detergent from a shared bucket	[GENERALLY SUPPORTED]	Real; frequency unknown.
Insecticide/aerosol exposure	[ESTABLISHED as toxic]	Real, dose-dependent, potentially catastrophic.
Net/handling injury leading to bacterial infection days later	[GENERALLY SUPPORTED]	Real; commonly missed because of the delay.
Rinsing filter media briefly in tap water crashing a cycle	[DISPUTED]	Probably overstated.
Substrate hydrogen sulfide killing fish	[PRIMARILY MYTH as usually stated]	Overstated; specific stopped-filter scenarios excepted.
Gravel vacuuming “destroying the beneficial bacteria”	[MISLEADING]	Substrate holds biofilm, but vacuuming the top layer removes detritus, not most of the colony.
Large water changes killing beneficial bacteria	[UNSUPPORTED]	See 17.2.

SECTION 4. OXYGEN PROBLEMS

4.1 Why oxygen is the leading hidden variable

[ESTABLISHED] In aquaculture, dissolved oxygen is described as the most important chemical parameter in determining water quality, and low DO is responsible for more fish kills than temperature, alkalinity, or salinity. UF/IFAS ranks ammonia as the most important water-quality parameter affecting fish **after oxygen** [UF-FA16].

Home aquarists essentially never measure it. There is no dissolved-oxygen pad on any common test strip and no DO test in any common master kit. **Oxygen is therefore the single largest measurement gap between what kills fish and what hobbyists test.**

4.2 Reference concentrations

[ESTABLISHED / GENERALLY SUPPORTED — figures are from aquaculture and fisheries sources; ornamental species vary]

Threshold	Value	Source basis
Healthy target, warmwater species	~5 mg/L or above	University of Kentucky aquaculture extension states warmwater fish require 5 mg/L to maintain good health [UKY-POND]
Healthy target, coldwater species	~7 mg/L	Same source [UKY-POND]
General culture guidance	Should not fall below 50% of saturation	Global Seafood Alliance [GSA-DO-REQ]
Stressful	Below ~3 mg/L for warmwater species	GSA: below 3 mg/L is stressful to shrimp and most warmwater fishes, reducing survival and production [GSA-DO-DYNAMICS]
Prolonged exposure lethal	Below 3 mg/L (warmwater); below 5 mg/L (coldwater)	UK extension [UKY-POND]
Acutely lethal	1.0–1.5 mg/L for a few hours	GSA [GSA-DO-DYNAMICS]
Minimum survivable (short term)	~1.0 mg/L warmwater; 2.5–3.5 mg/L coldwater	GSA [GSA-DO-REQ]
Research hypoxia definition	1–2 mg/L sustained for a few hours adversely affects growth and can cause death	Abdel-Tawwab et al. 2019 review [ABDELTAWWAB2019]; Diaz & Rosenberg consider 2 mg/L hypoxic for vertebrates

Saturation reference points (freshwater, sea level, approximate):

Temperature	Approx. DO at 100% saturation
15°C / 59°F	~10.1 mg/L
20°C / 68°F	~9.1 mg/L
25°C / 77°F	~8.3 mg/L
28°C / 82°F	~7.8 mg/L

30°C / 86°F	~7.6 mg/L
32°C / 90°F	~7.3 mg/L

Key implication: a discus tank at 86°F holds roughly 25% less oxygen at saturation than a coldwater tank at 59°F, while the fish’s metabolic oxygen demand is substantially *higher*. Temperature squeezes oxygen from both ends simultaneously.

4.3 Causes of low dissolved oxygen in home aquaria

Reduced supply: 1. **Reduced surface agitation.** Gas exchange in an aquarium occurs almost entirely at the air–water interface. A tank with a still surface, a surface film, or a floating-plant mat has drastically reduced exchange. This is the most common cause and is invisible. 2. **Surface biofilm/protein film.** Physically impedes exchange. 3. **Filter failure or reduced flow.** Removes both circulation and surface disturbance. 4. **Power outage.** Everything stops at once. 5. **Air pump failure, kinked airline, clogged air stone, backwards check valve.** 6. **Low water level relative to an HOB outflow** (less splash) or a spray bar redirected below the surface. 7. **Lid sealed with condensation** (minor; reduces exchange somewhat). 8. **High altitude.** Lower atmospheric pressure lowers saturation concentration. 9. **Degassed hot tap water used for a refill.**

Increased demand: 10. **High temperature.** Raises metabolic demand, lowers solubility. 11. **Overstocking.** 12. **Heavy feeding and organic decomposition.** Bacterial respiration during decomposition is a primary driver of oxygen depletion; expert guidance in aquaculture notes decomposition can drive DO below 3 mg/L in high-load systems. 13. **Bacterial bloom.** A cloudy-water bacterial bloom is a large, rapidly respiring biomass. Blooms following overfeeding, a dead fish, or a filter disruption can cause serious oxygen depletion, and the bloom is often blamed for “looking bad” rather than for consuming oxygen. 14. **Dead and decomposing organisms** — fish, snails, shrimp — especially unnoticed ones behind equipment. 15. **Plant respiration at night.** Plants consume oxygen continuously and produce it only under light. In a densely planted, low-flow tank, the pre-dawn hours are the daily oxygen minimum. **[ESTABLISHED]** 16. **CO2 injection.** Two distinct mechanisms: (a) CO2 does not displace oxygen in the way often described, but elevated dissolved CO2 impairs fish respiration by reducing the blood’s ability to offload CO2 and shifting hemoglobin oxygen affinity (Bohr/Root effects) — fish can suffocate in oxygen-adequate water; (b) many CO2 systems reduce surface agitation deliberately to conserve CO2, which reduces oxygen input. Overnight CO2 accumulation with the solenoid stuck open is a recognized killer. 17. **Algae die-off** after an algaecide treatment or a blackout, producing a large decomposing biomass. 18. **Substrate disturbance** creating an oxygen-demand pulse (3.2). 19. **Heavy dechlorinator overdose** consuming oxygen chemically (2.5). 20. **Medications** — see 4.4.

4.4 Medications that affect oxygen availability or respiration

[GENERALLY SUPPORTED]

- **Methylene blue, malachite green, formalin.** Formalin in particular consumes oxygen as it oxidizes, and standard veterinary guidance for formalin treatment includes providing supplemental aeration. Formalin also removes oxygen from the water directly.
- **Potassium permanganate.** A strong oxidant; consumes oxygen demand in the water and damages gill tissue at excessive doses.
- **Copper-based treatments.** Damage gill epithelium, reducing effective gas-exchange surface.
- **Antibiotics.** Can suppress the nitrifying population (Section 12.6), producing a delayed ammonia problem; some also cause bacterial die-off and a subsequent bloom.
- **Salt at therapeutic concentrations.** Slightly reduces oxygen solubility.
- **Any medication instructing removal of activated carbon** implicitly removes a mechanical/chemical polishing stage.
- **Dye-based and organic medications** raise oxygen demand as they degrade.

Cross-cutting rule: the standard instruction “increase aeration during treatment” exists precisely because most fish medications either reduce oxygen availability, damage gills, or both.

4.5 Species-specific oxygen requirements

- **High demand / low tolerance:** hillstream loaches, *Danio/Devario* from fast water, most rheophilic species, large-bodied active fish, coldwater species kept warm, goldfish at elevated temperature.
- **Moderate:** most community tetras, rasboras, barbs, livebearers.
- **Tolerant, with caveats:** anabantoids (bettas, gouramis) and *Corydoras* can breathe atmospheric air, but this is an adaptation to low-oxygen conditions, not immunity — persistent surfacing indicates a problem. Note that **a betta in low-oxygen water may show no distress at all**, which means a betta’s apparent health is not evidence that oxygen is adequate for other fish.
- **Loaches (*Pangio*, *Botia*) and some catfish** also gulp air facultatively.

4.6 Clinical presentation of hypoxia

- Congregating at the surface, gulping at the air–water interface (“piping”)
- Congregating directly at the filter outflow or air stone
- Rapid, exaggerated opercular (gill cover) movement
- Flared opercula; mouth held open
- Lethargy alternating with bursts of agitation
- Loss of appetite
- Loss of equilibrium in late stages

- **Larger fish and higher-demand species affected first**; this is a useful differentiator
- Dead fish frequently show **arched backs, flared opercula, and open mouths**

4.7 Why hypoxia is confused with ammonia/nitrite poisoning and with disease

This is one of the most important diagnostic points in the document.

Sign	Hypoxia	Ammonia toxicity	Nitrite toxicity	Gill parasites/columnaris
Surface gasping	Yes — primary sign	Yes	Yes	Yes
Rapid gill movement	Yes	Yes	Yes	Yes
Reddened gills	Sometimes	Yes	Sometimes	Yes
Brown gills	No	No	Yes (methemoglobin)	No
Lethargy	Yes	Yes	Yes	Yes
Improves immediately with aeration	Yes	Partially	Partially	No
Largest fish affected first	Typically yes	Not specifically	Not specifically	Not specifically
Worst before dawn	Typically yes	Not specifically	Not specifically	No
Ammonia test positive	No	Yes	No	No
Flashing/scratching	No	Sometimes	No	Yes
Visible lesions	No	No	No	Often

The decisive practical test: adding vigorous aeration produces near-immediate improvement in hypoxia and little immediate change in the other categories. Because aeration is harmless in essentially every scenario, **it is the correct first action in almost any aquarium emergency** (Section 16.3).

“Brown blood disease.” Nitrite oxidizes hemoglobin to methemoglobin, which cannot carry oxygen. The fish experiences functional hypoxia in fully oxygenated water, and gills and blood appear brown. Chloride is competitively protective at the gill: aquaculture guidance recommends adding chloride at approximately **10 mg/L chloride for each 1 mg/L nitrite** to reduce nitrite’s toxic effect [THEFISHSITE-AMMONIA, summarizing UF/IFAS]. Nitrite is described as toxic at levels as low as **0.10 mg/L** in the same source.

SECTION 5. DISEASE, PARASITES AND PATHOGENS

5.1 Why excellent water parameters cannot rule out disease

[ESTABLISHED] Water chemistry testing detects dissolved inorganic nitrogen species and acidity. It does not detect organisms. A tank can carry a heavy *Ichthyophthirius* infestation, a monogenean burden, a systemic *Aeromonas* infection, or an active *Flavobacterium* outbreak with an entirely unremarkable test panel.

The relationship between water quality and disease is **predisposing, not determining**:

- Poor water quality increases susceptibility by elevating cortisol and suppressing immune function.
- Good water quality reduces susceptibility but does not confer immunity.
- Many important pathogens are **opportunistic** — present in most systems at low levels, causing disease only when host defenses are compromised.
- A subset are **primary pathogens** capable of killing healthy fish in optimal water. *Ichthyophthirius multifiliis* is the canonical example: a veterinary source describes Ich as a motile ciliate that actively feeds on skin and gill tissue, making it a primary pathogen that can kill healthy fish without help from poor environmental conditions or other pathogens [FISHKEEPCHILL-EPISTYLIS].

5.2 Ichthyophthirius multifiliis (Ich, white spot disease)

[ESTABLISHED]

- A relatively large single-celled ciliated protozoan. UF/IFAS describes it as a major problem for freshwater aquarists and commercial producers worldwide, with all freshwater fish species considered susceptible and the parasite found worldwide in both cultured and wild fish [UF-FA006].
- **Life cycle is temperature-dependent** and has a critical implication: the parasite is only vulnerable to treatment during its free-swimming theront stage. The encysted trophont on the fish and the tomont in the substrate are protected. This is why treatment must continue for a full life cycle after visible spots disappear, and why raising temperature (within species tolerance) accelerates the cycle and shortens treatment.
- Visible white spots are the mature trophonts under the epithelium — spots of roughly uniform size, well-defined, distinctly white, and only slightly raised.
- **Gill-only infestations occur** and present with respiratory signs and no visible body spots — a significant cause of “mysterious” mortality.

- Mortality is usually from osmoregulatory failure and secondary infection through the damaged epithelium, plus gill damage.

5.3 Epistylis and the Ich misdiagnosis problem

[ESTABLISHED — the distinction; widely misdiagnosed in practice]

- *Epistylis* is a **sessile, stalked ciliate** that attaches to the fish and filter-feeds on bacteria in the water column — it does not feed on the fish. UF/IFAS covers it under sessile ciliates [UF-FA107].
- A veterinarian interviewed for a hobby publication contrasts the two directly: *Epistylis* does not move; it attaches with a stalk and waves cilia to draw in bacteria as food. Ich, by contrast, is motile and feeds on the fish's skin and gills, causing far more damage. *Epistylis* is described as an **opportunistic** organism that takes advantage of poor environmental conditions — water with a lot of organic debris — and of wounds [FISHKEEPCHILL-EPISTYLIS].
- **Visual differentiation [GENERALLY SUPPORTED]:** *Epistylis* tends to appear fuzzy, translucent, tufted, and to protrude noticeably from the fish, often at wound sites or fin rays; Ich appears as discrete, clearly defined, opaque white dots that only slightly protrude and are distributed more evenly [AQSCI-EPISTYLIS].
- **The clinical consequence of getting it wrong:** *Epistylis* is typically associated with an underlying bacterial component and with high organic load; treating it with antiparasitics targeted at Ich does not resolve it and delays correct management. Reducing organic load, improving water quality, and addressing the bacterial component are the relevant interventions.
- **Do not repeat as fact:** the widely circulated claim that “*Epistylis* is a bacterial infection” — it is a ciliate protozoan, though it is frequently found alongside bacterial infection.

5.4 Velvet (Oodinium / Piscinoodinium / Amyloodinium)

[GENERALLY SUPPORTED]

- A parasitic dinoflagellate. Freshwater form: *Piscinoodinium*. Marine form: *Amyloodinium*.
- Presents as an extremely fine gold, rust, or dusty coating, best seen with a flashlight held at an angle in a darkened room. Much finer than Ich.
- Attacks gills early and heavily; respiratory signs frequently precede visible dusting.
- Highly lethal and fast-moving in small fish; a common cause of rapid unexplained losses in newly acquired bettas, killifish, and small anabantoids.
- Frequently missed entirely because the dusting is difficult to see on light-colored fish and under normal lighting.

5.5 Monogenean flukes (gill flukes, skin flukes)

[ESTABLISHED]

- Multicellular flatworm ectoparasites; *Dactylogyrus* (predominantly gills) and *Gyrodactylus* (predominantly skin) are the classic genera. UF/IFAS provides a pictorial guide [UF-FA111].
- **Reproductive strategy matters clinically:** *Gyrodactylus* is viviparous, bearing live young that are themselves already carrying an embryo — populations can explode extremely rapidly. *Dactylogyrus* is oviparous, with egg stages resistant to many treatments, requiring repeat dosing.
- Signs: flashing/scratching against surfaces, clamped fins, excess mucus, rapid gill movement, reddened or pale gills, gasping, weight loss despite eating. **Often no visible external lesion at all.**
- **Requires microscopy for confident diagnosis** — a skin scrape or gill clip examined at 40–100x. This is the single most valuable diagnostic technique unavailable to most hobbyists.
- Flukes are an extremely common finding on newly imported and farm-raised fish and are a leading candidate for “new fish looked fine, then declined over 1–2 weeks.”

5.6 Internal parasites

[GENERALLY SUPPORTED]

Group	Notes
Nematodes (<i>Camallanus</i> , <i>Capillaria</i>)	<i>Camallanus</i> is visible as red worm-like protrusions from the vent in advanced cases. Common in wild-caught and some farmed livebearers. Causes wasting.
Cestodes (tapeworms)	Wasting despite good appetite
Trematodes (digenean)	Often encysted; frequently incidental
Flagellates (<i>Hexamita</i> , <i>Spironucleus</i>)	Associated with “hole in the head” / head-and-lateral-line erosion in cichlids and discus; white stringy feces; wasting
Microsporidians (<i>Pleistophora hypessobryconis</i> , “neon tetra disease”)	Merck notes it attacks skeletal muscle of neon tetras, angelfish, and other freshwater aquarium fish, with muscle damage causing abnormal movement, and that microscopic examination of diseased tissue is needed for confirmation [MERCK-OWNER]. No effective treatment; euthanasia and removal are standard.
Coccidia	Intestinal; wasting

Diagnostic hallmark of internal parasitism: progressive weight loss over weeks with a normal or even good appetite, a “pinched” or concave belly, white stringy feces, and normal water parameters. This presentation is very often mistakenly attributed to water quality.

5.7 Bacterial infections

[ESTABLISHED]

5.7.1 Columnaris (*Flavobacterium columnare* and related species)

- Merck describes the genus grouping for columnaris, coldwater disease, and bacterial gill disease as having been substantially revised into the genus *Flavobacterium* based on genomic studies; these are gram-negative rod or filamentous bacteria with a distinctive gliding motion, producing skin or gill lesions with slimy or cotton-like surface exudate over surface necrosis, ulceration, and marginal hemorrhage [MERCK-AQUA].
- *F. columnare* is **most common in warmwater species**. Merck notes there are several strains ranging from low to high virulence, that as-yet-unidentified water-quality conditions predispose susceptible fish to infection with the virulent strain, and that **mortality can be acute and may be difficult to control by the time the condition is seen** [MERCK-BACT].
- Presentations: mouth rot / “cottonmouth”; saddleback lesion behind the dorsal fin; pale or grey patches; gill necrosis. Merck’s owner-facing article lists it as saddleback or cottonmouth disease and notes early treatment with potassium permanganate can help, with antibiotics generally needed if not caught early [MERCK-OWNER].
- **It is a bacterium, not a fungus**, despite the cottony appearance. Antifungal-only treatment is ineffective. This is one of the most consequential misidentifications in the hobby.
- **Temperature relationship:** the American Fisheries Society Fish Health Section blue book states severity and occurrence are generally greater at warmer water temperatures (>20°C), though disease can occur in salmonids at 12–15°C [FHS-COLUMNARIS]. An experimental study in golden shiners found significantly higher mortality in high-temperature and columnaris-challenge treatments, and describes *F. cova* (formerly *F. columnare* genomovar II) as generally a secondary or opportunistic pathogen causing heavy mortality in stressed fish, though it can act as a primary pathogen [PMC-GOLDENSHINER].
- **Acute forms can kill within 24–72 hours**, which makes columnaris a leading candidate for “one fish was fine yesterday and dead this morning with a white patch.”
- **Common precipitating events:** netting/handling injury, shipping abrasion, aggression wounds, elevated temperature, high organic load, and — relevantly for this document — **maintenance-associated handling**.

5.7.2 *Flavobacterium psychrophilum* (bacterial coldwater disease)

Merck: most severe at 4–10°C, with signs not expected above 18°C; primarily a salmonid pathogen, though warmwater fish exposed to cold temperatures may rarely be affected [MERCK-AQUA]. Relevant to unheated tanks and to heater failures.

5.7.3 *Flavobacterium branchiophilum* (bacterial gill disease)

Gill-specific; produces respiratory distress with no external body lesions. Another “gasping with clean parameters” candidate.

5.7.4 Motile *Aeromonas* septicemia and *Pseudomonas*

- The classic “hemorrhagic septicemia” presentation: reddened patches, red streaking in fins, ulcers, popeye, dropsy-like fluid accumulation, and internal organ involvement.
- Merck notes that internal organs can bleed and break down, that preventive steps include minimizing stress and crowding, that laboratory tests are needed to determine effective antibiotics, and that **rapidly changing antibiotics or “shotgunning” fish with many different drugs is dangerous and can create resistant strains** [MERCK-OWNER].
- Almost always secondary to stress, injury, or another disease.

5.7.5 *Mycobacterium* spp. (fish tuberculosis / mycobacteriosis)

- **[GENERALLY SUPPORTED]** Chronic, slowly progressive, frequently subclinical for months. Presentations: progressive emaciation, spinal deformity, chronic non-healing ulcers, exophthalmia, granulomas in internal organs.
- Not curable in practice; management is depopulation and disinfection.
- Notable as an explanation for **long-running, low-level, “one fish every few weeks” mortality in a tank with permanently good parameters** — one of the most frustrating patterns in the hobby, and one where water chemistry investigation is guaranteed to be fruitless.
- **Zoonotic:** *Mycobacterium marinum* causes granulomatous skin lesions in humans (“fish tank granuloma”), typically entering via cuts on hands and forearms. Gloves are advised when working in a tank with open cuts.

5.7.6 *Edwardsiella*

Merck notes *Edwardsiella tarda* and other *Edwardsiella* species cause intestinal disease and skin ulceration in many fish species and can also infect reptiles, amphibians, birds, and mammals including people [MERCK-OWNER].

5.8 Fungal infections

[GENERALLY SUPPORTED]

- True fungal/oomycete infection (*Saprolegnia*, *Achlya*) is almost always **secondary** — colonizing already-damaged tissue, dead eggs, or immunocompromised fish.

- Presents as cotton-wool tufts, typically at wound sites, and typically radiating outward with visible filaments.
- **Confusion with columnaris is routine and consequential.** General rules of thumb: true fungus tends to be more three-dimensional, more clearly filamentous, and localized to an injury; columnaris tends to be flatter, greyer, and to spread across the body with the saddleback or mouth distribution. Definitive differentiation requires microscopy.
- The correct question with a fungal diagnosis is always “**what damaged this fish first?**”

5.9 Viral disease

[GENERALLY SUPPORTED — less commonly identified in home aquaria because diagnosis requires laboratory capability]

- Merck describes carp pox (cyprinid herpesvirus-1) as one of the oldest recognized fish diseases, primarily affecting koi, producing smooth raised skin abnormalities with a possibly milky appearance that do not usually cause problems but can become sites of secondary bacterial infection [MERCK-OWNER].
- Lymphocystis (iridovirus) produces cauliflower-like nodules; usually self-limiting.
- Koi herpesvirus (KHV/CyHV-3) — highly lethal in koi and common carp within a temperature window.
- Megalocytivirus / iridovirus infections — implicated in “dwarf gourami disease” and in unexplained losses of gouramis and some cichlids. **[PLAUSIBLE, POORLY DOCUMENTED in the hobby context but well recognized in ornamental fish health literature]**
- Practically: viral disease is a legitimate residual explanation for otherwise inexplicable losses in specific species groups, and there is essentially no home treatment.

5.10 Gill disease as a category

Because the gill is simultaneously the respiratory organ, the primary ion-regulatory organ, and a major nitrogen-excretion route, **anything that damages gills produces a syndrome that mimics both hypoxia and water-quality poisoning:**

Causes of gill pathology: monogenean flukes, Ich (gill form), velvet, *Flavobacterium branchiophilum*, *Epistylis* on gill tissue, ammonia, nitrite, chlorine/chloramine, copper, low pH, particulates and turbidity, gas supersaturation emboli, and physical trauma.

Practical consequence: “gasping at the surface with clean parameters” has at least a dozen distinct causes across four different mechanism classes. It is a symptom, not a diagnosis.

5.11 Secondary infections

The dominant real-world pattern in home aquaria is a **two-stage event**:

1. A primary insult — injury, parasite, water-quality excursion, thermal stress, aggression, handling, shipping — damages the mucus layer, epithelium, or gills, and elevates cortisol.
2. Opportunistic bacteria (*Aeromonas*, *Pseudomonas*, *Flavobacterium*) and oomycetes colonize the damaged tissue.

The keeper observes only stage 2, days later, in water that tests perfectly, and concludes the tank has “a disease.” The correct diagnostic question is what caused stage 1, and the answer is frequently in the maintenance log.

5.12 Disease introduced by new fish, asymptomatic carriers, and incubation

[ESTABLISHED]

- Many fish pathogens persist in carrier animals showing no clinical signs. Merck’s guidance on koi explicitly emphasizes quarantine.
- **Incubation and latency windows commonly cited [GENERALLY SUPPORTED]:**

Agent	Typical time from exposure to visible disease
Ich	3–7 days (temperature-dependent)
Velvet	2–7 days
Monogeneans	Days to 2–3 weeks (population build-up)
Columnaris (acute)	24 hours to a few days after stress
<i>Aeromonas</i> septicemia	2–7 days after the predisposing insult
Internal nematodes	Weeks to months
Mycobacteriosis	Months to over a year
Viral (species-dependent)	Days to weeks, often temperature-gated

- **This explains the single most common pattern reported by beginners:** new fish added; everything fine for 4–10 days; then losses begin, sometimes among the *established* fish rather than the new arrivals. The delay defeats the intuitive association with the purchase.
- **Quarantine.** Standard recommendation across veterinary and hobby sources is **2–4 weeks** in a separate system with separate equipment. A vet-reviewed source specifies quarantining new arrivals 2–4 weeks and cleaning nets and tools between tanks [VETVERIFIED-COLUMNARIS]. Quarantine is the single highest-value practice available to hobbyists and the one most consistently

skipped.

- **Fomite transmission.** Shared nets, siphons, buckets, algae scrapers, and wet hands transfer pathogens between tanks without any fish moving. A fishroom with one bucket is functionally one aquarium.

5.13 Stress-related susceptibility — the physiological link

[ESTABLISHED]

The mechanism connecting husbandry to disease is well characterized:

1. A stressor (handling, transport, crowding, poor water quality, thermal change, aggression) activates the hypothalamic-pituitary-interrenal axis.
2. Plasma **cortisol** and **glucose** rise; lactate may rise.
3. Sustained cortisol elevation is **immunosuppressive**, reducing lymphocyte function and antibody response.
4. Mucus production and composition change, degrading the primary physical/chemical barrier.
5. Opportunistic pathogens that were previously controlled establish infection.

Transport research documents this chain directly. Studies of live fish transport report significantly elevated plasma cortisol and glucose, with high-density transport elevating ammonia nitrogen and temperature, reducing pH, causing mortality at the highest densities, producing progressive gill and liver structural damage, elevated plasma reactive oxygen species and cortisol, and **decreased IgG and antioxidant capacity** [RG-TRANSPORT-DENSITY]. Ammonia and nitrite themselves act as stressors that stimulate corticosteroid release, with elevated circulating corticosteroids linked to impaired immune function and decreased disease resistance [RG-TAMBAQUI].

Retrieval statement: “stress kills fish” is not a vague hobby platitude. It is a specific, measurable endocrine-immune pathway, and it is the mechanism that connects Sections 2, 3, 6, 7, 9, and 10 of this document to Section 5.

5.14 Why fish appear healthy at purchase and decline days later

Consolidated list of non-exclusive explanations:

1. **Incubation period** — infection acquired before or at purchase, not yet visible (5.12).
2. **Cortisol-mediated immunosuppression** from the cumulative transport chain (5.13, Section 6).
3. **Cortisol’s masking effect** — acute stress response can suppress outward signs of illness temporarily; the animal appears alert and active during the acute phase and crashes during the exhaustion phase.
4. **Delayed manifestation of gill or organ damage** from an ammonia, pH, or hypoxia exposure during shipping.
5. **Starvation** — fish are commonly fasted before shipping and may have gone days to weeks with minimal food through the supply chain.
6. **Osmoregulatory debt** accumulated across multiple water chemistries (Section 7).
7. **The new environment’s own stressors** — aggression from established residents, unfamiliar flow, inadequate cover, competition for food.
8. **Prey-species stoicism** — fish evolved to conceal weakness; visible illness in a fish generally indicates advanced disease. The window between “looks fine” and “visibly dying” is genuinely short in many species.
9. **Retail medication** — fish held on prophylactic antibiotics or antiparasitics at the store may be suppressing an infection that resurfaces once medication stops.

5.15 Differentiating water-quality emergencies from infectious disease

Use these as probability shifters, not rules.

Observation	Shifts toward water quality / environment	Shifts toward infectious disease
Onset	Sudden, simultaneous	Progressive, sequential
Distribution	All or most fish affected together	Spreads fish to fish over days
Species pattern	Sensitive species first, but broadly	Often species- or size-specific
Visible lesions	Usually absent	Often present
Response to water change + aeration	Rapid improvement	Little or none
Correlation with an event	Strong (maintenance, refill, outage)	Weaker; correlates with new additions
Feeding behavior	Stops abruptly, all fish	Individual fish stop at different times
Flashing/scratching	Possible (irritants)	Common (ectoparasites)
Timeline to first death	Minutes to hours	Days

Critical caveat, repeated deliberately: these two categories are not mutually exclusive and in practice frequently co-occur, because water-quality stress precipitates opportunistic disease. Finding a plausible water-quality explanation does not rule out disease, and vice versa.

5.16 The hard limit on hobbyist diagnosis

[ESTABLISHED] Definitive diagnosis of most fish disease requires:

- Wet-mount microscopy of skin scrapes and gill clips from a freshly moribund or freshly euthanized fish (parasites, *Flavobacterium* “haystacks”)
- Bacterial culture and antibiotic sensitivity testing (Merck stresses this for choosing antibiotics [MERCK-OWNER])
- Histopathology (mycobacteriosis, viral inclusions, microsporidia)
- PCR/molecular testing for specific agents

Merck notes columnaris can be presumptively diagnosed from visualization of typical organisms on wet mounts of infected skin or gill tissue, and confirmed by isolation on Ordal’s or other cytophaga media [MERCK-AQUA]. None of this is available from a photograph on a forum.

The corollary for this research document: any statement of the form “based on this description, your fish has X” is a hypothesis with a low prior. Photographic and behavioral differentiation lists should be framed as narrowing tools, never as diagnoses.

SECTION 6. LIVESTOCK SOURCE AND PRIOR HISTORY

6.1 The supply chain a typical aquarium fish has already survived

For a farm-raised tropical fish reaching a US hobbyist, the typical chain is:

Breeding farm (Southeast Asia, Florida, Eastern Europe, or Sri Lanka) → **conditioning/fasting period** → **bagging and boxing** → **domestic road transport to exporter** → **air freight, often 24-48 hours with customs delays** → **importer/transshipper holding system, frequently 1-7 days** → **road transport to wholesaler or direct to store** → **retail holding system, 1 day to several weeks** → **bagging** → **car journey** → **hobbyist aquarium**

That is typically **five to eight distinct water chemistries, four to six netting events, and two to four periods of confinement in a sealed bag** — often compressed into 7-21 days.

[GENERALLY SUPPORTED] By the time a fish reaches a home aquarium it has usually undergone more physiological stress in two weeks than it will experience in the following two years, assuming reasonable care.

6.2 Shipping stress: what the research documents

[ESTABLISHED]

Measured effects of live fish transport in the research literature:

- **Water quality inside the container degrades progressively.** Ammonia nitrogen rises, dissolved oxygen falls, pH drops (from respiratory CO₂), and temperature drifts. Reviews report that concentrations of ammonia, nitrite, and CO₂ in transport water can increase substantially over a haul [RG-TRANSPORT-METHODS; RG-TAMBAQUI].
- **Stress markers rise.** Transported fish typically exhibit significantly elevated plasma cortisol and glucose; lactate and malondialdehyde rise; superoxide dismutase and other antioxidant enzyme activities shift [RG-TRANSPORT-METHODS].
- **Density is a key variable.** Mortality rate increases significantly with density and transport duration; high-density transport has been shown to elevate ammonia and temperature, reduce pH, cause mortality in the highest-density group, and produce progressive gill and liver structural damage with decreased IgG [RG-TRANSPORT-DENSITY].
- **Recovery is measurable but not instantaneous.** In one 4-hour bagged-transport study of juvenile piau (*Leporinus friderici*), cortisol rose within 4 hours and returned to control levels about 2 hours after arrival [ACADEMIA-ORNAMENTAL-TRANSPORT]. Cortisol normalization, however, is not the same as immune recovery, which takes substantially longer.
- **Duration limits.** One study of hobbyist-style transport notes that build-up of ammonia and CO₂ limits the duration fish can be kept in sealed oxygenated bags to approximately **24 to 36 hours** [ACADEMIA-ORNAMENTAL-TRANSPORT].

6.3 The critical bag chemistry mechanism (ammonium/ammonia and pH)

[ESTABLISHED — this is one of the most important mechanisms in the document]

Inside a sealed transport bag over hours:

1. The fish excretes ammonia across its gills. **Total ammonia nitrogen rises.**
2. The fish also excretes CO₂. In a sealed bag with limited gas exchange, **dissolved CO₂ accumulates.**
3. Accumulated CO₂ **drives pH down**, often substantially — reports of bag pH falling to 7.2 or well below in a few hours are common, and one study of *Leporinus* transport noted that ammonia levels increased throughout transportation but the low pH kept NH₃ at safe levels for the fish [ACADEMIA-ORNAMENTAL-TRANSPORT].
4. Low pH shifts the ammonia equilibrium overwhelmingly toward **ammonium (NH₄⁺)**, which is comparatively non-toxic. The fish therefore survives a TAN concentration that would be lethal at higher pH.
5. **When the bag is opened, CO₂ outgasses within minutes and pH rises rapidly.** The equilibrium shifts back toward **un-ionized NH₃**, and the fish is suddenly exposed to genuinely toxic ammonia in water it had been tolerating.

This single mechanism underlies the entire drip-acclimation controversy (Section 7.6) and explains why a shipped fish can look fine in the bag and die within hours of a “careful” acclimation.

Contrasting evidence on which factor dominates: a study of marine ornamental *Abudefduf saxatilis* transported for 24/48/72 hours found pH significantly correlated with mortality, especially in the first 24 hours, with hypoxia contributing at 24–48 hours, but **ammonia build-up was not a problem even after 72 hours** [ZOOLOGIA-TRANSPORT]. This suggests the acidification and oxygen trajectory may matter more than ammonia accumulation per se in at least some species — a useful corrective to the assumption that bag ammonia is always the primary killer.

6.4 Retail holding conditions

[GENERALLY SUPPORTED / ANECDOTAL as to prevalence]

Factors that affect a fish’s condition at the moment of purchase:

- **Centralized recirculating systems.** Most chain retailers run many display tanks on one filtration loop. Consequences: (a) every tank has identical water chemistry regardless of species; (b) **a pathogen in any tank is in every tank**; (c) medication of one tank medicates all of them; (d) the store’s water chemistry may differ dramatically from the customer’s.
- **UV sterilization and continuous prophylactic medication** in some systems suppresses visible disease without eliminating carriage.
- **High turnover and short holding times.** A fish purchased the day a shipment arrives has had no opportunity to recover from transport and no opportunity for latent disease to become visible. [GENERALLY SUPPORTED — a widely repeated recommendation among experienced keepers is to ask when the fish arrived and, if recent, to wait a week before buying.]
- **Crowding and mixed-species holding** promotes aggression and pathogen exchange.
- **Feeding regime.** Retail fish are frequently underfed to keep water clean.
- **Staff knowledge** varies enormously; misidentified species and inappropriate stocking advice are common.

6.5 Recently imported and wild-caught fish

- Wild-caught specimens (many plecos, *Corydoras*, apistogramma, wild bettas, many catfish, most freshwater invertebrates from Southeast Asia) carry **higher parasite burdens** and are adapted to specific water chemistry.
- Some are collected using methods that cause injury or, historically, chemical stunning.
- Wild fish frequently refuse prepared foods initially and may starve in a community tank while apparently “settling in.”
- Import stress in wild fish is layered on top of capture and holding stress at origin.

6.6 Starvation and dehydration in the supply chain

[GENERALLY SUPPORTED]

- Fish are deliberately **fasted 24–72 hours before shipping** to reduce ammonia and fecal load in the bag. This is correct practice.
- However, the fast is often longer in reality: fast at the farm + transit + possible fast at the transshipper + limited feeding at retail. A small tetra can go a week or more with minimal intake.
- Small-bodied fish have very limited energy reserves; a neon tetra has days, not weeks.
- The presentation on arrival is a fish that looks thin, has a slightly sunken belly, eats eagerly for a day or two, and then dies — or a fish that never resumes feeding at all.
- **“Dehydration”** in the human sense is not the right frame for a freshwater fish (which gains water osmotically), but **osmoregulatory exhaustion** is: sustained ion loss without adequate energy to fuel active ion uptake. This is why adding a modest concentration of salt or an electrolyte product to quarantine water is a defensible practice for freshly arrived freshwater fish — it reduces the osmotic gradient and the metabolic cost of ion regulation. [GENERALLY SUPPORTED]

6.7 Handling stress and multiple acclimation events

Each netting event involves: chase and capture (exhaustive anaerobic exercise, lactate accumulation), air exposure, physical contact with net mesh (mucus layer abrasion, scale and fin damage), and confinement. The cumulative cortisol load of four to six such events within two weeks is substantial and additive.

Practical implication: the water change performed three days after adding new fish is being performed on animals whose stress-response systems are already at or beyond capacity. The same disturbance that established fish ignore can be decisive for a recent arrival.

6.8 Genetics, inbreeding, congenital problems

[GENERALLY SUPPORTED / ANECDOTAL as to magnitude]

- Mass-production breeding for volume and color, with limited attention to genetic diversity, is widely believed to have reduced hardiness in several popular lines. The **fancy guppy** is the most frequently cited example, with reports of dramatically reduced longevity and disease resistance compared to feeder or wild-type strains. [ANECDOTAL but near-universal among experienced keepers]
- **Dwarf gourami** losses are frequently attributed to an iridovirus endemic in commercial breeding stock. [PLAUSIBLE, POORLY

DOCUMENTED at hobby level; the virus is recognized in ornamental fish health literature]

- **Selectively bred body-form morphs** — balloon mollies, balloon rams, bubble-eye goldfish, veiltail and long-fin morphs — have deliberately altered anatomy (spinal compression, swim bladder distortion, impaired swimming) that reduces functional reserve.
- **Hormone-treated fish.** Some ornamental strains (certain color-enhanced or sex-reversed lines) receive hormone treatment as juveniles, with reduced longevity reported. **[ANECDOTAL]**
- **Dyed and tattooed fish** (“painted glassfish,” dyed tetras) suffer high mortality from the injection or dipping process itself. **[GENERALLY SUPPORTED]**

6.9 Age

Chronically under-appreciated. Typical lifespans under good conditions:

Species	Typical lifespan
Neon tetra	3-5 years
Guppy (fancy)	1-2 years
Guppy (wild-type/feeder)	2-3 years
Betta splendens	2-4 years (often purchased at 6-12 months)
Zebra danio	3-5 years
Corydoras	5-15 years
Common pleco	10-20 years
Angelfish	8-10 years
Goldfish	10-25+ years
Dwarf gourami	3-4 years
Kuhli loach	8-10 years

Key point: many fish are sold at or near adulthood. A betta purchased in a cup is typically 6-12 months old already. **A one-year-old fancy guppy dying is not a mystery.** “Old age” is a legitimate diagnosis and is systematically under-considered in troubleshooting.

6.10 Previous disease exposure and medication history

- A fish that survived a *Flavobacterium* outbreak at the wholesaler may carry gill scarring that permanently reduces respiratory reserve — invisible externally, but manifesting as intolerance of warm water or low oxygen.
- A fish that has been repeatedly exposed to subtherapeutic antibiotics may carry resistant bacteria, making treatment failure more likely.
- Copper-treated fish (common in marine and some freshwater retail systems) may have residual gill and kidney damage.
- Prophylactic treatments at the store may still be present in the bag water and enter the home tank — including copper, which is lethal to invertebrates at concentrations harmless to fish. **This is a specific and concrete argument against adding bag water to an aquarium containing shrimp or snails.**

6.11 Fish that were already dying before purchase

The uncomfortable but well-supported conclusion of this section: **a meaningful fraction of “unexplained” home-aquarium deaths in the first two weeks after purchase involve animals whose trajectory was already established before they entered the hobbyist’s water.** UF/IFAS’s observation that fish may become sick weeks after a water-quality problem has been corrected [UF-FA16] applies directly here — the damaging exposure occurred in the supply chain, and the clinical outcome occurs in the customer’s tank.

This does not absolve home husbandry, and it must not be used to dismiss legitimate tank problems. But any troubleshooting framework that considers only the hobbyist’s aquarium will systematically misattribute causes for new arrivals.

SECTION 7. ACCLIMATION AND OSMOTIC STRESS

7.1 Osmoregulation in freshwater fish — the underlying physiology

[ESTABLISHED]

A freshwater fish is a concentrated solution in a dilute medium. Its body fluids typically run around 280-320 mOsm/kg while the surrounding water may be under 10 mOsm/kg. Consequently:

- **Water enters continuously** by osmosis, mainly across the gills (the thinnest, largest-area interface).
- **Ions leak out continuously** down their concentration gradients.
- The fish compensates by (a) **not drinking**, (b) producing large volumes of **very dilute urine** to void the excess water while reclaiming ions in the kidney tubules, and (c) **actively transporting ions inward** at specialized gill ionocytes (chloride cells), using

ATP-dependent pumps.

This is metabolically expensive and runs continuously. Anything that increases the gradient (moving to softer water), damages the gill (parasites, ammonia, chlorine, copper), or reduces available energy (starvation, hypoxia, high cortisol) degrades the fish's ability to sustain it.

The two failure directions: - **Toward much softer/lower-ionic water:** steeper gradient, faster ion loss, higher pumping cost. Classic failure mode is a fish that becomes lethargic, loses color, and dies over hours to days without obvious lesions. - **Toward much harder/higher-ionic water:** reduced osmotic water influx and altered ion balance; for a soft-water-adapted animal, calcium and magnesium regulation and renal load change. Generally slower and less acutely lethal than the reverse direction in fish, though it is significant for shrimp and for some wild-caught blackwater species.

Calcium's special role [ESTABLISHED]. Environmental calcium reduces gill permeability. Fish in calcium-poor water lose ions faster, and calcium supplementation reduces osmoregulatory stress. This is one reason "GH near zero" is more physiologically demanding than the pH number alone suggests, and one reason RO water without remineralization is genuinely hazardous.

7.2 Why pH differences are usually a proxy, not the mechanism

Restating the point from 2.8 in the acclimation context: two waters at the same pH can have wildly different mineral content, and two waters at different pH can have nearly identical ionic strength. The dangerous acclimation transitions in practice are:

1. **High TAN + low pH → high pH** (the transport bag opening; the neglected tank water change). **Mechanism: un-ionized ammonia.** [ESTABLISHED]
2. **High GH/TDS → very low GH/TDS.** **Mechanism: osmoregulatory.** [ESTABLISHED]
3. **Very low KH water experiencing rapid CO₂ or acid loading.** **Mechanism: acid-base disturbance in a system with no buffer.** [GENERALLY SUPPORTED]
4. **pH change with everything else constant, moderate magnitude.** **Mechanism: weak.** [DISPUTED — see 2.8]

7.3 GH, KH, conductivity, and TDS in acclimation

See Section 13 for full definitions. For acclimation purposes:

- **GH** (calcium + magnesium) is the parameter most directly tied to gill permeability and osmoregulatory cost, and to invertebrate molting.
- **KH** (carbonate/bicarbonate alkalinity) governs pH stability. A fish moving from KH 8 water to KH 1 water is moving into a system where pH can now swing.
- **Conductivity/TDS** is the best single proxy for total ionic strength and therefore for osmotic difference — but it does not identify *which* ions (13.6).
- **Sodium chloride does not register on GH tests.** Softened water and salted water can have low GH and high TDS. This mismatch is a common source of confusion.

7.4 Species differences in tolerance

Category	Examples	Acclimation sensitivity
Highly euryhaline / tolerant	Guppy, molly, platy, zebra danio, goldfish, most farm-bred community fish	Low sensitivity; tolerate wide swings
Moderately tolerant	Most tetras (captive-bred), rasboras, barbs, common corydoras, angelfish	Moderate
Sensitive	Wild-caught apistogramma, wild bettas, chocolate gourami, many <i>Corydoras</i> from soft blackwater, cardinal tetras (wild), rummynose tetra	High — rummynose in particular are widely used as a "canary" species
Very sensitive	Caridina (Taiwan bee, crystal) shrimp, some <i>Sulawesi</i> shrimp, discus (wild), <i>Hypancistrus</i>	Very high; TDS matters directly
Special case	Neocaridina shrimp	Tolerate a wide GH range but are very sensitive to copper and to rapid TDS change

7.5 Acclimation methods and what each actually addresses

Method	Procedure	Addresses	Does not address
Float only	Float sealed bag 15–20 min, net fish out, discard bag water	Temperature	Ionic difference, pH
"Plop and drop"	Net the fish directly from the bag into the tank, discarding bag water immediately (sometimes after a brief float)	Minimizes ammonia exposure time; minimizes handling duration	Temperature (unless floated), ionic difference
Cup / batch addition	Add small volumes of tank water to the bag or container at intervals over	Temperature, gradual ionic transition	Prolongs ammonia exposure after opening

	30–60 min		
Drip acclimation	Siphon tank water into the container at a slow drip over 1–3+ hours	Gradual ionic transition; standard for invertebrates and marine	Prolongs ammonia exposure substantially; oxygen depletion in the container; temperature drift downward
Drip + immediate ammonia binder	Dose an ammonia-binding conditioner into the container at the moment the bag is opened, then drip	Attempts to address both	Depends on the binder working as claimed — itself disputed

7.6 The drip acclimation controversy — presented as a genuine open dispute

[DISPUTED — one of the most active disagreements in the hobby, with essentially no controlled experimental resolution located during this research]

Position A — drip acclimation is standard and correct. Slow, gradual transition minimizes osmotic and pH shock. This is the near-universal recommendation of retailers, most hobby literature, and most online guides, and it is unambiguously correct practice for **invertebrates**, which are genuinely sensitive to salinity/TDS change.

Position B — drip acclimating shipped fish is actively harmful. The argument, stated by a number of online vendors and experienced keepers: the moment the bag is opened, CO₂ escapes and pH rises within minutes, converting relatively harmless ammonium into toxic un-ionized ammonia. Extending the acclimation to an hour or more therefore extends exposure to genuinely toxic ammonia in order to avoid an osmotic transition that most fish tolerate easily. Under this view, “plop and drop” — temperature-float, then transfer the fish out of the bag water quickly — is safer for **shipped** fish [FISHLORE-DRIP; REEF2REEF-DRIP; MEDIUM-ACCLIMATION].

Where the two positions actually converge: - Both agree the bag chemistry mechanism (6.3) is real. - Both agree invertebrates and very TDS-sensitive species warrant slow acclimation. - Both agree bag water should not enter the display tank. - Both agree that **the longer a fish has been in the bag, the stronger the case against a prolonged drip**, because bag TAN is higher and bag pH is lower. - Both agree that when moving fish between two tanks with similar chemistry, acclimation method barely matters.

What is missing: no controlled study comparing acclimation methods for shipped ornamental freshwater fish with mortality and cortisol endpoints was located during this research. Both positions rest on mechanism plus accumulated anecdote. **Hobby forums on this topic are heavily anecdotal on both sides and several contain factually incorrect chemistry.** For example, one forum poster asserts that an ammonia reducer “needs oxygen to decompose ammonia” [HUMBLEFISH-ACCLIMATION] — this is not how thiosulfate or aldehyde-based binders work, and illustrates why forum consensus is not evidence.

A defensible synthesis for retrieval: - Short local transport (30 minutes from the store, bag not opened, pH not yet driven far down): acclimation method is low-stakes; a temperature float plus modest water exchange is fine. - Long shipping (overnight or longer): minimize time between opening the bag and getting the fish into clean, well-oxygenated water. If a slow transition is needed for a genuinely sensitive species, dose an ammonia binder immediately on opening and keep the container aerated. - Invertebrates: drip acclimate. - Always discard bag water.

7.7 Situations where prolonged drip acclimation is plausibly counterproductive

Enumerated for retrieval:

1. Bag has been sealed for many hours (ammonia accumulated, pH depressed).
2. Bag water pH is more than ~0.5 units below tank pH and the fish load in the bag was high.
3. The acclimation container is not aerated — dissolved oxygen in a small still container with a stressed fish falls quickly.
4. The container is out of the water and cooling, so the fish experiences a slow downward temperature drift for an hour.
5. Ambient room temperature is much cooler or warmer than either water.
6. The fish is already showing signs of distress in the bag (lying on its side, gasping, pale).
7. The two water chemistries are already similar, making the entire procedure pointless.
8. Very small fish with minimal energy reserves.

7.8 Distinguishing evidence-based acclimation guidance from tradition

Claim	Classification	Notes
Match temperature before adding fish	[ESTABLISHED]	Uncontroversial; low cost
Never add bag water to the display tank	[GENERALLY SUPPORTED]	Pathogen, medication (esp. copper), and ammonia rationale
Drip acclimate invertebrates	[GENERALLY SUPPORTED]	Molting and TDS sensitivity are real
Drip acclimate all fish for at least an hour	[DISPUTED]	Mechanistically counterproductive for long-shipped fish
pH must be matched to within 0.2 units	[UNSUPPORTED as stated]	See 2.8; the underlying concern is ionic and ammonia chemistry
Turn tank lights off when introducing new fish	[GENERALLY SUPPORTED]	Reduces visual stress and aggression during the first hours; low cost

Feed the tank before adding new fish (to reduce aggression)	[ANECDOTAL]	Plausible and harmless
Rearrange decor when adding fish to a territorial species tank	[GENERALLY SUPPORTED]	Resets established territories; standard cichlid practice
Add fish “in the evening”	[ANECDOTAL]	Consistent with the lights-off rationale
Quarantine new arrivals 2–4 weeks	[ESTABLISHED as best practice]	Strongest recommendation in this section

SECTION 8. TOXINS AND CONTAMINANTS NOT DETECTED BY NORMAL TEST KITS

8.1 Framing: hazard, dose, and volume

The correct model for every item in this section is:

Risk = intrinsic toxicity of the compound × amount that actually entered the water × (1 / tank volume) × exposure duration × species sensitivity × modifying water chemistry

Two errors are common and opposite: - **Over-attribution:** assuming any household chemical near an aquarium is lethal. Most are not, at realistic exposures. - **Under-attribution:** assuming that because the water “looks clean” and tests fine, no chemical exposure occurred. Some compounds in this section are lethal at concentrations that are literally invisible and unmeasurable by any consumer method.

8.2 Heavy metals

8.2.1 Copper

[ESTABLISHED]

Mechanism. Copper damages gill epithelium (impairing gas exchange and ion regulation), disrupts the olfactory system, and inhibits enzymes. It is also the active ingredient in many antiparasitic and algicidal products, which is why it is present in aquarium contexts at all.

Toxicity is strongly modified by water chemistry. Australian/New Zealand water-quality guidance summarizes the key relationships: copper uptake and toxicity generally **decrease with increasing water hardness and alkalinity**, following an exponential inverse relationship, and dissolved organic matter in most fresh waters is generally sufficient to complex copper and remove toxicity — **but often not in very soft waters**. Copper complexing increases at higher pH, though the relationship to toxicity is complex, and copper adsorbs strongly to suspended material [ANZG-COPPER].

Quantitative anchors: - USEPA acute toxicity data for freshwater species in 41 genera at a hardness of 50 mg/L ranged from **17 µg/L to 10,000 µg/L** — a nearly 600-fold species range [ANZG-COPPER]. - A worked hardness comparison: the 96-hour EC50 for a chironomid was **17 µg Cu/L in soft water** (hardness 43 mg/L CaCO₃) versus **98 µg Cu/L in hard water** (hardness 172 mg/L CaCO₃) — nearly a sixfold difference from hardness alone [ANZG-COPPER, citing Gauss et al. 1985]. - The **US EPA action level for copper in drinking water is 1.3 mg/L (1300 µg/L)**, which is far above concentrations relevant to sensitive aquatic organisms. - Hobby sources commonly cite invertebrate sensitivity in the range of **0.02–0.1 mg/L**, with nitrifying bacteria also affected by copper exposure [FISHHANDBOOK-METALS — hobby-technical source; treat specific figures as approximate].

Realistic sources in the home: - Copper plumbing, especially new installations, acidic or soft water, and after stagnation. Aquarium-focused guidance recommends flushing new plumbing thoroughly before drawing aquarium water [IERE-COPPER]. - Copper-containing medications used previously in the same tank (copper adsorbs to silicone, substrate, and decor and can leach back for a long time — a critical point for anyone converting a former marine or previously copper-treated tank to a shrimp tank). - Some plant fertilizers and trace-element mixes contain copper, usually chelated, which reduces bioavailability substantially. - Algaecides. - Brass fittings and some ornaments.

Note: boiling does not remove copper; it slightly concentrates it as water evaporates [IERE-COPPER].

8.2.2 Zinc, lead, cadmium, nickel, aluminum

[GENERALLY SUPPORTED] All follow the same broad pattern: **more toxic in soft, low-alkalinity, low-DOC water**; all are undetected by standard aquarium test kits; all can enter from plumbing, fittings, ornaments not intended for aquarium use, and source water. Lead is a particular concern in older housing stock and is mobilized by changes in disinfectant or corrosion-control chemistry at the utility level. Aluminum becomes more bioavailable and more toxic at low pH, which is relevant to very soft blackwater setups.

8.2.3 Iron

Generally low toxicity to fish at aquarium-relevant concentrations, but ferric hydroxide precipitate can physically coat gills, and very high well-water iron is worth addressing. Iron is also relevant because it **binds sulfide**, which is one reason sulfide accumulation in iron-containing substrates is limited.

8.3 Chlorine and chloramine

Covered in detail in Sections 2.3–2.4. Restated here for retrieval completeness: **neither is on a standard five-parameter test panel**, both are lethal at concentrations present in ordinary municipal tap water, and free chlorine dissipates quickly enough that post-incident testing frequently reads zero.

8.4 Surfactants: soap, detergent, hand sanitizer, lotion

[GENERALLY SUPPORTED — mechanism established; aquarium-specific dose data sparse]

Mechanism. Surfactants are amphipathic molecules that disrupt lipid membranes. At the gill, they destroy the mucus layer and damage the lamellar epithelium directly. They also lower surface tension, which impairs gas exchange at the surface and can affect labyrinth fish and surface-feeding species. Anionic surfactants (most soaps and detergents) are the most toxic class to fish.

Realistic entry routes: shared buckets, sponges, and scrubbers; hands; a decoration washed in the sink; a net rinsed under a soapy tap; an aquarium ornament cleaned in a dishwasher (rinse-aid residue).

Practical calibration: a trace of hand soap residue rinsed from clean hands into a 55-gallon tank is very unlikely to cause an incident. A bucket that was used to mop a floor with a cup of cleaner and then rinsed once is a credible whole-tank kill in a small tank. **Volume ratio dominates.**

8.5 Aerosols, fragrance, and airborne exposure

[PLAUSIBLE, POORLY DOCUMENTED for most fragrance products; ESTABLISHED for insecticidal aerosols]

Mechanism. An aquarium is an open gas-exchange system. Airborne compounds partition into water at the surface according to their Henry's law constants; low-volatility, water-soluble compounds partition efficiently. A filter with strong surface agitation is, in effect, an air scrubber.

Highest concern: insecticide sprays and foggers (8.6), paint and solvent vapors, oven cleaner, and ammonia-based cleaners used in the same room with poor ventilation.

Lower but non-zero concern: air fresheners, perfume, hairspray, dry shampoo, scented candles, essential oil diffusers. **Evidence here is largely anecdotal.** Reports of tank losses following heavy diffuser use exist but are uncontrolled.

Practical mitigation (all low-cost): cover the tank during any aerosol use in the room; turn off air pumps (which actively draw room air into the water) during spraying and for some time after; ventilate; run fresh activated carbon afterward.

8.6 Insecticides — the highest-consequence household category

[ESTABLISHED — extremely high intrinsic toxicity]

8.6.1 Pyrethroids

A published review in *Environmental Toxicology and Chemistry* states plainly that **pyrethroid insecticides are extremely toxic to fish, with 96-hour LC50 values in the nanomolar per liter range**, and that acute lethality symptoms implicate effects on the nervous system, respiratory surfaces, and renal ion regulation [BRADBURY-COATS-1989]. Nanomolar concentrations correspond to fractions of a microgram per liter — orders of magnitude below anything a hobbyist can measure.

A patent comparison provides a concrete anchor: **cypermethrin displayed 100% kill of test fish at a concentration of 0.01 ppm (10 µg/L)** [USPTO-4883789].

Common household pyrethroids: permethrin, cypermethrin, deltamethrin, bifenthrin, tetramethrin, cyphenothrin, prallethrin, esfenvalerate. These appear in household bug sprays, ant and roach sprays, mosquito foggers, wasp sprays, garden insecticides, lawn treatments, and many pet flea products.

8.6.2 Fipronil

The National Pesticide Information Center states fipronil is **highly toxic to both saltwater and freshwater fish and to aquatic invertebrates**, and that two fipronil metabolites tested in freshwater fish and invertebrates were **more toxic than fipronil itself** [NPIC-FIPRONIL]. Regulatory review material notes LC50s to many aquatic invertebrates in the low µg/L range, with the most sensitive reported (mysid shrimp) at an acute EC50 of **140 ng/L** and full life-cycle effects on survival, reproduction, and growth at **less than 5 ng/L** [CAWB-FIPRONIL].

Fipronil is the active ingredient in a number of widely used **spot-on flea and tick treatments for dogs and cats**, as well as ant and roach gels and baits and termite products [NPIC-FIPRONIL].

8.6.3 The realistic household exposure pathways

1. **Indoor pest control spray or fogger used in the room containing the aquarium.** This is the classic whole-tank kill. Foggers are designed to deposit residue on every surface in a room — including the water surface of an open-top aquarium.
2. **Professional pest control treatment of the home**, including perimeter spraying near an intake vent.
3. **Applying a spot-on flea treatment to a pet, then putting hands in the aquarium.** Residue transfers readily from the applicator and from the animal's coat for days.
4. **Pet bedding, collars (some contain permethrin or flumethrin), or a treated animal resting near the tank.**

5. **Lawn or garden treatment**, with residue tracked indoors or carried on hands.
6. **Mosquito control** in the neighborhood, or a plug-in/coil mosquito repellent in the room (these commonly volatilize prallethrin or allethrin).
7. **Treated houseplants** placed near or in the aquarium, or plant leaves/roots entering the water. Systemic insecticides (imidacloprid, dinotefuran) applied to houseplants persist in plant tissue.
8. **Aquatic plants from a nursery** treated with systemic pesticides — a well-known and specific hazard to **invertebrates**, causing shrimp die-offs in tanks where fish are unaffected. Many aquarium plant vendors now explicitly label plants as “shrimp safe” for this reason. **[GENERALLY SUPPORTED — widely reported and mechanistically well founded]**

8.6.4 Why this category deserves special weight in the core question

- Lethal concentrations are far below any hobbyist detection limit.
- The compounds are not on any test panel.
- Exposure leaves no visible trace in the water.
- Symptoms (erratic swimming, spiraling, twitching, loss of equilibrium, gasping, then death) are neurologic and are frequently mistaken for ammonia poisoning or “shock.”
- **Invertebrates die first and at much lower concentrations than fish.** A tank in which all the shrimp and snails die while the fish are merely lethargic is a strong signal for an insecticide or copper exposure rather than a nitrogen-cycle problem.
- Activated carbon and large water changes are the appropriate response, and the exposure route must be identified or it will recur.

8.7 Medication overdoses and interactions

[GENERALLY SUPPORTED]

- **Dosing by tank rating rather than actual water volume.** A “20 gallon” aquarium with substrate, hardscape, and a below-rim water level contains meaningfully less than 20 gallons. Overdose is systematic, not occasional.
- **Stacking medications.** Combining a formalin/malachite green product with an antibiotic, or two antiparasitics, can push oxygen demand and gill damage past tolerance. Merck warns specifically against rapidly changing antibiotics or “shotgunning” fish with many different drugs [MERCK-OWNER].
- **Repeating doses without water changes between them**, causing accumulation.
- **Not removing activated carbon**, then removing it partway through, producing an unpredictable dose curve.
- **Treating in a planted tank.** Copper and some antiparasitics damage plants; dying plants then add organic load.
- **Treating invertebrates with fish medications.** Copper and many antiparasitics are lethal to shrimp and snails at fish-therapeutic doses.
- **Salt.** See 8.9.
- **Temperature interaction.** Many treatments are more toxic at higher temperature, exactly when hobbyists are also raising temperature to accelerate Ich life cycles.

8.8 Fertilizer overdoses and liquid carbon / algaecide misuse

[GENERALLY SUPPORTED]

- **Macronutrient overdose (N, P, K):** most commonly a nitrate problem, or in the case of potassium, generally well tolerated. Rarely acutely lethal.
- **Micronutrient/trace overdose:** the real risk, because trace mixes contain **copper**. Chelation reduces bioavailability, but repeated heavy dosing in a shrimp tank is a documented cause of invertebrate losses. **[GENERALLY SUPPORTED]**
- **Excess iron dosing:** can precipitate and coat surfaces.
- **“Liquid carbon” products (glutaraldehyde-based, e.g., polycycloglutaracetal):** these are biocides. At label dose they are generally tolerated by fish; at elevated dose they damage sensitive plants (*Vallisneria*, mosses), can harm invertebrates, and can cause fish distress. **Spot-dosing directly onto algae with a syringe near fish or shrimp creates a locally very high concentration.** Glutaraldehyde also exerts an oxygen demand. **[GENERALLY SUPPORTED]**
- **Algaecides generally:** in addition to direct toxicity, a successful algaecide treatment produces a large mass of dead algae that decomposes, consuming oxygen and releasing ammonia. **The die-off can be more dangerous than the chemical.** [GENERALLY SUPPORTED]
- **Hydrogen peroxide dosing:** used for algae; overdose damages gills and plants; oxygen release can be violent locally.

8.9 Salt overdoses

[GENERALLY SUPPORTED]

- Sodium chloride at low concentration (roughly 1–3 g/L) is a legitimate supportive therapy: it reduces osmoregulatory workload, is competitively protective against nitrite toxicity (the chloride effect), and has some antiparasitic action.
- **Common errors:** dosing by “tablespoons per gallon” without a defined teaspoon/tablespoon mass; re-dosing the full tank volume at every water change (salt does not evaporate, so re-dosing only for the replaced volume is correct); and dosing salt-intolerant species.
- **Salt-intolerant or salt-sensitive livestock:** most *Corydoras* and other scaleless catfish, many loaches, and most live plants and invertebrates. Anacharis, *vallisneria*, and many stem plants decline at therapeutic salt levels.

- Salt raises TDS and conductivity but **does not register on GH tests**, creating an interpretation trap.

8.10 Contaminated buckets, equipment, and previously-exposed items

- Buckets, hoses, nets, siphons, scrapers, and airline used for any non-aquarium purpose.
- Equipment previously used in a copper-treated tank.
- Equipment previously used in a tank that had a disease outbreak (fomite transmission — Section 5.12).
- Silicone from non-aquarium tubes containing mildewcides.
- Rocks and wood collected outdoors from areas with pesticide/herbicide application or road runoff.
- Painted, glazed, or metal-containing ornaments not intended for aquarium use — glazes may contain lead or cadmium; galvanized metal releases zinc.
- Plastic items that are not food-grade.
- Aquarium sealant repairs made with the wrong product, or not fully cured.
- **New equipment.** New silicone tubing, new plastics, and new filter housings can leach plasticizers initially; rinsing new equipment before use is standard practice.

8.11 Accidental chemical interactions

[PLAUSIBLE, POORLY DOCUMENTED at hobby scale]

- Dechlorinator (a reducing agent) + strong oxidizer (potassium permanganate, hydrogen peroxide, chlorine-based treatment) → mutual neutralization, unpredictable dosing, and oxygen consumption.
- Ammonia-binding conditioner + certain ammonia test chemistries → misleading readings.
- pH-adjusting products + low-KH water → overshoot, then rebound. **“pH down”/“pH up” products used in unbuffered water are a well-known cause of pH instability**, and the resulting oscillation is more harmful than the original off-target pH would have been. **[GENERALLY SUPPORTED]**
- Phosphate-based pH buffers + calcium-rich water → precipitation, cloudiness.
- Medications + carbon left in the filter → underdose, then overdose when carbon saturates.
- Two “beneficial bacteria” products of different formulations dosed together — mainly a waste of money rather than a hazard.

8.12 Contaminants summary table

Contaminant class	Detected by standard 5 tests?	Typical concentration of concern	Realistic home source	Speed of effect
Free chlorine	No	0.003 mg/L chronic; 0.2–0.3 mg/L rapidly toxic	Tap water without conditioner	Minutes-hours
Chloramine	No	LC50 ~0.3–0.35 mg/L (patent data)	Tap water without conditioner	Minutes-hours
Copper	No	17 µg/L (soft) to ~100+ µg/L (hard) for sensitive taxa	Plumbing, meds, fertilizers	Hours-days
Lead / zinc / cadmium	No	Varies; more toxic in soft water	Plumbing, ornaments	Days
Pyrethroids	No	Nanomolar (sub-µg/L)	Household bug spray, flea products	Minutes-hours
Fipronil	No	ng/L for sensitive invertebrates	Flea spot-ons, ant baits	Hours
Surfactants (soap)	No	Highly variable	Buckets, hands, sponges	Minutes-hours
Solvent/paint vapor	No	Unquantified in this context	Renovation	Hours-days
Medication overdose	No	Product-specific	Aquarist	Hours
Glutaraldehyde “liquid carbon”	No	~2–5x label dose	Aquarist	Hours
Salt overdose	No (raises TDS, not GH)	Species-dependent	Aquarist	Hours-days

SECTION 9. TEMPERATURE

9.1 Why temperature is simultaneously the most-measured and most-misunderstood parameter

Temperature is one of the standard five, and yet it is routinely mismeasured and misinterpreted. It is not a single number; it is a distribution across space and time.

9.2 Acute temperature shock

Covered mechanistically in 2.2. Key restatements:

- The physiological response is measurable (heat-shock protein induction, cortisol elevation).
- The magnitude required to produce measurable HSP response in laboratory studies was substantially larger than hobby folklore suggests — on the order of 14–27°F sudden increases in the cited studies, without mortality [AQSCI-SHIFTS].
- **However**, laboratory subjects are healthy, unstressed, well-fed fish. A recently shipped, immunosuppressed, starved fish is a different animal.
- **Directional asymmetry**: rapid warming raises oxygen demand and lowers oxygen supply simultaneously; rapid cooling depresses immune function and metabolism. Both matter; the mechanisms differ.

9.3 Chronic inappropriate temperature

[GENERALLY SUPPORTED — a major cause of “slow mystery decline”]

The classic examples in the hobby:

Situation	Consequence
Goldfish or White Cloud minnows kept at 78–80°F	Chronically elevated metabolism, shortened lifespan, reduced oxygen availability. Extremely common because they are sold in “tropical” displays
Corydoras kept at 80–82°F for a discus or angelfish tank	Many <i>Corydoras</i> species are stressed above ~78°F; chronic decline over months
Kept too cool (72°F) for species requiring 78–82°F	Suppressed immune function; classic setup for chronic <i>Ichthyophthirius</i> susceptibility
Hillstream loaches at tropical temperatures	Require cool, high-oxygen, high-flow water; chronically stressed in a standard warm community tank
Bettas in unheated rooms	Frequently kept at 68–72°F; lethargy, reduced immunity, susceptibility to fin rot and columnaris

Diagnostic significance: chronic temperature mismatch produces slow attrition over months with excellent water parameters throughout. It is one of the highest-yield questions to ask when the pattern is “one fish dies every few weeks, water always perfect.”

9.4 Heater failure modes

Enumerated in 3.7. Additional detail:

- **Stuck-on thermostat** is the most lethal single equipment failure. Small tanks reach lethal temperatures fastest. Symptom: all fish dead or dying, water hot to the touch, everything else normal.
- **Stuck-off thermostat** produces slow decline; often first noticed when fish develop Ich.
- **Undersized heater in a cold room** cannot maintain setpoint; temperature oscillates with room temperature.
- **Oversized heater in a small tank** overshoots between thermostat cycles, producing a larger diurnal swing than a correctly sized unit.
- **Heater set correctly but calibration drifted**. Heater dial settings are notoriously inaccurate; the dial is not a thermometer.
- **Heater in a low-flow area** heats a local pocket while the thermostat reads that pocket, leaving the rest of the tank cold.
- **Two heaters, one failed on, one failed off** — a redundancy strategy that requires an independent thermometer to be useful.

9.5 Temperature gradients

[GENERALLY SUPPORTED]

- Large tanks, low-flow tanks, and heavily planted tanks develop measurable gradients — commonly 2–4°F between the heater end and the far end, more in tanks over 4 feet.
- Substrate surface can be cooler than the upper water column in a low-flow tank — relevant for bottom-dwelling species.
- A tank against an exterior wall or under a window has a cold face.
- A tank under strong lighting has surface heating.
- **A single thermometer reports one point**. Checking both ends of a long tank is a five-second diagnostic that is almost never performed.

9.6 Thermometer inaccuracy

- **Stick-on LCD strip thermometers** measure the glass, which is influenced by room air on the outside face. Errors of several degrees are common. These are the most commonly used and least accurate type.
- **Floating glass alcohol thermometers** are reasonably accurate but are read at whatever depth they float.
- **Digital probe thermometers** are generally the most reliable and can be positioned away from the heater.
- **Calibration**: an ice-water slurry should read 32°F/0°C. Almost no hobbyist checks this.
- **The heater’s own dial is not a measurement**.

- **Infrared thermometers measure the surface**, not the water column, and are affected by evaporative cooling.

9.7 Temperature and dissolved oxygen

See 4.2. The key table point bears repeating: **oxygen solubility falls as temperature rises while metabolic oxygen demand rises**. Elevated temperature is therefore a compounding stressor in any tank that is near its oxygen limit.

9.8 Temperature and metabolism

- Metabolic rate roughly doubles for each 10°C rise ($Q_{10} \approx 2$) over the physiological range.
- Consequences: faster ammonia excretion, faster food processing, faster growth, faster aging, higher oxygen demand, faster drug uptake and toxicity.
- **This is why medication toxicity increases with temperature** and why treatment protocols that combine “raise the temperature” with “medicate” require careful attention to aeration.

9.9 Temperature and pathogen development

[ESTABLISHED]

Pathogen	Temperature relationship
<i>Ichthyophthirius multifiliis</i>	Life cycle accelerates with warmth; the classic treatment strategy raises temperature (within host tolerance) to shorten the vulnerable free-swimming window. Cold water lengthens the cycle dramatically
<i>Flavobacterium columnare</i>	Severity and occurrence generally greater above 20°C [FHS-COLUMNARIS]; experimental work found significantly higher mortality at higher temperature [PMC-GOLDENSHINER]
<i>Flavobacterium psychrophilum</i>	Most severe at 4–10°C; signs not expected above 18°C [MERCK-AQUA]
<i>Saprolegnia</i> (water mold)	Favored by cooler water
Velvet (<i>Piscinoodinium</i>)	Accelerated by warmth
Koi herpesvirus	Active in a specific window, roughly 18–28°C

Practical implication: raising temperature to treat Ich simultaneously increases columnaris risk and reduces available oxygen. This is a genuine trade-off, not a free action, and it is a plausible mechanism for “I raised the temperature for Ich and lost more fish.”

9.10 Species-specific temperature tolerance — reference table

Species/group	Recommended range	Notes
Goldfish (single-tail)	60–72°F	Coldwater; frequently mis-housed
Goldfish (fancy)	68–74°F	Less cold-tolerant than single-tails
White Cloud Mountain minnow	64–72°F	Sold as tropical; is not
Hillstream loach	68–75°F	High oxygen, high flow required
Zebra danio	65–77°F	Very wide tolerance; often kept too warm
Neon tetra	70–78°F	Often kept too warm alongside discus
Cardinal tetra	75–84°F	Warmer than neons — a common confusion
Most <i>Corydoras</i>	72–78°F	Some species (<i>C. sterbai</i>) tolerate 82°F+
Guppy / platy / molly	74–82°F	
Betta splendens	78–82°F	
Angelfish	78–84°F	
Discus	82–86°F	High temp = low oxygen; requires strong aeration
Rift Lake cichlids	76–82°F	
Neocaridina shrimp	65–78°F	Breed better cooler; short-lived when warm
Caridina shrimp	68–74°F	Genuinely intolerant of warmth

A frequently overlooked scenario: a tank stocked with species whose ranges do not overlap. There is no single temperature that suits both discus and *Corydoras aeneus*, or both a betta and a White Cloud minnow. The result is that one group is chronically stressed while parameters read perfectly.

SECTION 10. BEHAVIORAL AND STOCKING PROBLEMS

10.1 Why this category is systematically underweighted

Behavioral causes of death share three properties that make them easy to miss:

1. They leave no chemical trace.
2. Much of the relevant behavior occurs **when the lights are off or when the keeper is not watching**.
3. The visible endpoint (a dead or injured fish) often resembles disease, because the proximate cause of death is frequently a **secondary bacterial infection of an aggression wound**.

10.2 Aggression, territoriality, and bullying

[GENERALLY SUPPORTED]

Common mechanisms: - **Territorial defense** around a cave, plant, or corner. Intensifies dramatically during spawning. - **Dominance hierarchy formation**, particularly in cichlids, gouramis, and some barbs. The lowest-ranked individual is subjected to continuous harassment. - **Chronic low-grade harassment** without visible wounds. The victim is prevented from resting, from reaching food, and from using preferred space. Death is from cumulative stress and starvation, and there are **no marks on the body**. - **Mate aggression** in pair-bonding species when one animal is not receptive. - **Species that become aggressive with maturity** but are sold as juveniles — the single most common stocking failure. Examples: many gouramis, most “community” cichlids, tiger barbs in small groups, Chinese algae eaters (which become large, aggressive, and may rasp on the flanks of other fish).

Key point: many aggression-driven deaths occur in tanks that are objectively fine chemically, and the aggressor is often the fish that looks healthiest.

10.3 Nocturnal aggression and predation

[GENERALLY SUPPORTED]

- Many catfish (*Synodontis*, some plecos), loaches, and eels are crepuscular or nocturnal. Aggression occurs in the dark.
- Sleeping fish are vulnerable; a fish found in the morning with damaged fins or a missing eye has usually been attacked at night.
- **Plecos and some *Ancistrus*** are documented to rasp at the mucus and flanks of slow-moving, flat-sided fish (angelfish, discus, goldfish) at night, particularly if underfed. **[GENERALLY SUPPORTED]**
- **Straightforward predation** is often the answer to “the fish just disappeared.” An adult angelfish will eat neon tetras. A large gourami will eat shrimp. A betta will eat shrimplets. There is no mystery and no body — small fish are consumed completely.
- **Scavenging after death** confuses the record: snails, shrimp, and catfish rapidly consume a body, so absence of a corpse does not indicate predation as the cause of death.

10.4 Fin nipping

- Chronic fin damage from nippers (tiger barbs, serpaes tetras, some danios, black skirt tetras) creates open wounds that become entry points for *Aeromonas*, *Flavobacterium*, and *Saprolegnia*.
- Long-finned varieties (fancy guppies, veiltail bettas, angelfish) are disproportionately targeted.
- **The keeper observes “fin rot” and treats for bacteria.** The bacteria are real; the cause is behavioral. Treatment without removing the nippers fails.
- Fin nipping is strongly reduced by keeping nippy species in **larger groups**, which redirects the behavior intraspecifically.

10.5 Competition for food

[GENERALLY SUPPORTED — a leading cause of slow, unexplained loss]

- Fast midwater feeders (danios, barbs, tetras) consume flake food before it reaches slower or bottom-dwelling species.
- **Bottom dwellers are the most common victims:** *Corydoras*, loaches, plecos, and shrimp in a tank with fast surface feeders.
- **Plecos in particular** are frequently assumed to “eat algae” and are never fed directly. A common pleco in a clean tank with no wood and no supplemental feeding is slowly starving.
- Nocturnal species that are only fed during the day.
- Shy or newly added fish that will not compete during the acclimation period.
- Weight loss is difficult to see on a fish viewed from the side; the diagnostic view is **from above**, looking for a narrowing body and a “razor back” or pinched dorsal profile.

10.6 Incompatible species and sex ratios

- **Size incompatibility:** “if it fits in the mouth, it is food.”
- **Temperament incompatibility:** peaceful and boisterous species in one tank.
- **Environmental incompatibility:** high-flow and low-flow species; hard-water and soft-water species.
- **Sex ratio problems:**
 - Livebearers (guppies, platies, mollies, swordtails): males harass females continuously. **Recommended ratio is at least 2-3 females per male.** A single female with multiple males is harassed to exhaustion and death — a very common and entirely behavioral cause of death.

- Multiple male bettas, multiple male gouramis, or two males of a territorial cichlid species in inadequate space.
- Some species require an excess of one sex; some require single pairs only.
- **Continuous reproduction in livebearers** exhausts females; a female guppy producing broods every 4 weeks for a year has a substantially shortened lifespan.

10.7 Stress from insufficient cover, and open-tank stress

[GENERALLY SUPPORTED]

- Most small aquarium fish are prey animals. A bare tank with bright light and no visual breaks produces chronic vigilance behavior and elevated cortisol.
- **Floating plants** measurably reduce stress behavior in many surface-oriented and blackwater species.
- **Dark substrate and background** reduce stress and improve coloration in many species.
- A tank in a high-traffic area, next to a television, or with people passing constantly imposes repeated startle responses.
- Vibration transmitted through a stand or floor.
- Lighting that switches from full darkness to full intensity instantly (no dawn/dusk ramp) causes a startle response daily.
- **Insufficient hiding places for the number of territorial individuals** guarantees conflict.

10.8 Schooling and shoaling requirements

[GENERALLY SUPPORTED]

- Obligate shoaling species kept in inadequate numbers exhibit chronic stress: hiding, reduced feeding, faded coloration, increased aggression toward each other and toward other species.
- Commonly cited minimums: **6 as an absolute floor, 10-15 for good behavior** in most tetras, rasboras, danios, and *Corydoras*.
- Species where inadequate group size converts into aggression: tiger barbs, serpaes tetras, black skirt tetras, and many danios — **the fin-nipping problem in 10.4 is frequently a group-size problem**.
- *Corydoras* in groups of 2-3 show markedly less foraging activity than groups of 6+.
- **A group that has attrited from 8 to 3 over a year becomes progressively more stressed**, which accelerates the remaining losses. This produces a self-reinforcing decline that looks like a mysterious ongoing problem.

10.9 Jumping

[GENERALLY SUPPORTED — a leading cause of “the fish vanished”]

- Many species are habitual jumpers: killifish, hatchetfish, danios, rainbowfish, bettas, loaches, some rasboras, and eels/knifefish.
- Triggers: startle response, aggression, poor water quality, spawning behavior, insufficient territory, and **maintenance disturbance with the lid off**.
- Gaps around filter housings, heater cables, and feeding hatches are sufficient.
- A fish that jumps out overnight and is found dry in the morning is often assumed to have “died and been thrown out by the filter” or is simply never found (behind the stand, under furniture, taken by a pet).
- **Cross-reference to water quality:** jumping is also a documented escape response to acutely irritating water (chlorine, ammonia, low pH). A fish found on the floor after a water change is diagnostically significant, not merely unlucky.

10.10 Entrapment

- Behind filter intakes and inside intake strainers (small fish, fry, shrimp).
- Inside filter housings and impeller chambers.
- Inside hollow ornaments with narrow openings — a fish grows too large to exit.
- Between rock and glass.
- Under wood or rock that shifts.
- Inside plant thickets in low-oxygen corners.
- In an overflow box or sump.
- Wedged under a heater bracket or suction cup.
- **Beneath an undergravel filter plate.**

10.11 Injuries, including non-obvious internal injury

[PLAUSIBLE, POORLY DOCUMENTED at hobby level, but mechanistically clear]

- **Netting trauma** — internal bruising, spinal injury in small fish, jaw injury in fish that strike the net.
- **Being dropped** during transfer.
- **Collision injury** during a startle response, particularly in a small tank.
- **Barotrauma** in fish that surface rapidly (rare in home aquaria).
- **Swim bladder injury** from impact, producing buoyancy problems that are usually misattributed to “constipation” or “swim bladder”

- disease” (which is a symptom description, not a diagnosis).
- **Spinal injury** from a fall or a collision, producing progressive loss of coordination over days.
 - **Internal injury from aggression**, particularly ramming behavior in cichlids.
 - **Egg binding / dystocia** in livebearers and egg-layers — presents as a distended, non-feeding female that dies over days with perfect water.

10.12 Overstocking as a multi-mechanism stressor

Overstocking does not cause harm by a single route; it acts through at least six:

1. Higher ammonia production per unit volume, requiring more biofilter capacity.
2. Higher oxygen consumption and higher CO2 production.
3. Higher organic load, increasing bacterial oxygen demand.
4. Faster nitrate and dissolved-organic accumulation between water changes.
5. Greater social stress, competition, and aggression.
6. Faster disease transmission once a pathogen enters.

A key consequence: an overstocked tank can show acceptable ammonia, nitrite, and nitrate readings because the biofilter and water-change schedule are keeping up chemically — while the tank remains oxygen-limited and socially stressful. **This is a canonical “perfect parameters, dying fish” configuration.**

SECTION 11. NUTRITION

11.1 Why nutrition is a leading cause of slow unexplained loss

Nutritional problems develop over **weeks to months** and produce no water-chemistry signature at any point. They are therefore invisible to the entire diagnostic framework most hobbyists use.

11.2 Starvation

[GENERALLY SUPPORTED]

Scenarios: - **Newly acquired fish** that have gone through a long supply-chain fast (Section 6.6) and do not resume feeding. - **Bottom dwellers and nocturnal species** outcompeted at feeding time (10.5). - **Algae eaters assumed to be self-sufficient.** Plecos, otocinclus, and Siamese algae eaters all require supplemental feeding. **Otocinclus in particular have a very high mortality rate in the first weeks, largely from starvation** — they arrive already emaciated from the supply chain and require a mature biofilm-covered tank plus supplemental vegetables. [GENERALLY SUPPORTED — widely reported] - **Wild-caught fish** that do not recognize prepared food. - **Specialist feeders** in a general community: some *Corydoras*, some catfish, and pipefish/gobies with live-food requirements. - **Fish that appear to eat but do not swallow.** Common with large-particle food and small-mouthed fish; the fish takes food in and spits it out repeatedly. - **Long absences without an auto-feeder**, or an auto-feeder that jammed. - **A fish that stopped eating for another reason** (disease, aggression, water quality) and then starves as a secondary consequence.

Recognition: view fish from **above**. A healthy fish has a rounded or at least straight dorsal profile; a starving fish shows a narrowed body, prominent dorsal ridge, sunken belly, and enlarged-looking head.

11.3 Incorrect and species-inappropriate diets

[GENERALLY SUPPORTED]

Group	Dietary requirement	Common error
Herbivores (many plecos, mollies, silver dollars, some cichlids)	High plant matter, low protein, high fiber	Fed high-protein flake exclusively → digestive disorders, “Malawi bloat” analogues
Carnivores (many cichlids, predatory catfish, arowana)	High protein, animal-based	Fed plant-heavy diets → poor growth
Omnivores (most community fish)	Mixed	Usually fine on quality flake
Wood-eating plecos (<i>Panaque</i>)	Require lignin/driftwood	Kept without wood → digestive failure
<i>Tropheus</i> and some Lake Tanganyika herbivores	Very high fiber, very low protein	High-protein food → documented severe enteritis (“bloat”), frequently fatal
Goldfish	Low protein, high fiber, sinking food preferred	Floating flake → air ingestion and buoyancy problems
Bettas	Carnivorous	Fed generic flake → poor condition
Shrimp	Biofilm, algae, detritus, plus supplemental	Overfed protein → water quality problems and molting issues

11.4 Vitamin and micronutrient deficiency

[GENERALLY SUPPORTED]

- **Vitamin C deficiency** → skeletal deformity (scoliosis/lordosis), impaired wound healing, reduced immune function. Documented extensively in aquaculture.
- **Thiamine (B1) deficiency** → neurological signs; classically associated with diets high in raw fish containing **thiaminase** (some feeder fish, some frozen foods). Feeding live feeder goldfish or rosy reds long-term is a documented thiaminase risk.
- **Vitamin E and selenium deficiency** → muscular degeneration.
- **Vitamin A deficiency** → eye and skin problems.
- **Fatty acid imbalance** → poor condition, reduced reproduction.
- **Practical driver:** deficiencies arise from feeding a single food type for a long time, or from feeding **old food**. Water-soluble vitamins in dry food degrade substantially after the container is opened; six months is a common practical shelf-life recommendation once opened.

11.5 Overfeeding

[ESTABLISHED — but the mechanism is usually indirect]

Overfeeding rarely kills fish directly. It kills through: 1. **Uneaten food decomposition** → ammonia production, oxygen consumption, bacterial bloom. 2. **Increased fish waste** → higher ammonia load, faster nitrate accumulation. 3. **Organic accumulation in the substrate** → the deferred hazard released during the next deep gravel vacuum (3.2). 4. **Digestive overload in individual fish** → constipation, buoyancy problems, particularly in fancy goldfish and long-finned varieties. 5. **Obesity and fatty liver disease (hepatic lipidosis)** in long-lived, overfed fish — a documented condition in ornamental fish and a plausible contributor to “the old fish just died.”

Diagnostic note: a tank that is chronically overfed but water-changed adequately can show clean nitrogen parameters while accumulating a substantial substrate organic reservoir. This is the mechanism that links overfeeding to the water-change/gravel-vacuum incident pattern.

11.6 Constipation versus actual disease

[GENERALLY SUPPORTED, with important caveats]

- Buoyancy problems and a swollen abdomen in a fish that is otherwise alert are commonly attributed to constipation, and dietary management (fasting 2–3 days, then feeding blanched deshelled pea or daphnia) is a widely used, low-risk intervention.
- **However, “swim bladder disease” is a description, not a diagnosis.** The same presentation can result from: bacterial infection of the swim bladder, physical injury, a tumor or cyst pressing on the swim bladder, egg binding, organ enlargement, congenital deformity (especially in balloon-bodied and short-bodied morphs), or *Mycobacterium granulomas*.
- **Dropsy** — the pinecone appearance of raised scales from generalized edema — is likewise a **sign, not a disease**. It indicates fluid accumulation, usually from renal or hepatic failure, frequently secondary to systemic bacterial infection. Prognosis is poor. It is routinely and incorrectly described as a discrete disease with a cure.
- The pea remedy is harmless and sometimes effective; the risk is that it delays recognition of a more serious condition.

11.7 Food contamination and spoilage

[PLAUSIBLE, POORLY DOCUMENTED at hobby level]

- **Oxidized fats.** Dry foods high in fish oil become rancid with exposure to air, heat, and light. Rancid fats are cytotoxic and reduce vitamin content. Storing food in a warm room above the aquarium hood accelerates this.
- **Mold and mycotoxins.** Fish food stored damp — including food scooped with wet fingers or a wet spoon — can develop mold. **Aflatoxin contamination of fish feed is a documented and serious problem in aquaculture**, causing liver damage and immunosuppression.
- **Live foods as disease vectors.** Live blackworms, tubifex, feeder fish, and pond-collected foods can introduce parasites and bacteria. Feeder goldfish in particular are a documented route for introducing pathogens.
- **Frozen food thawed and refrozen.**
- **Very old food** — a container purchased three years ago and used sparingly.

11.8 Nutritional problems that develop slowly despite excellent parameters — summary

The complete diagnostic profile of a nutritional problem:

- Onset over weeks to months
- Progressive weight loss, or progressive condition decline without weight loss (vitamin deficiency)
- Water parameters normal throughout
- No visible lesions until late
- Frequently affects one species or one size class (the ones losing the food competition)
- Does not respond to water changes, medication, or “fixing” parameters
- **The keeper’s own feeding regime is the last thing examined**

SECTION 12. BIOLOGICAL FILTRATION AND TRANSIENT EVENTS

12.1 What “cycled” actually means and does not mean

[ESTABLISHED]

A “cycled” aquarium has a microbial population capable of oxidizing the current ammonia production rate to nitrate at a rate that keeps ammonia and nitrite undetectable. Three implications follow that are frequently missed:

1. **Capacity is matched to the current load.** A tank cycled for three small fish is not cycled for twelve. Adding livestock faster than the population can grow produces a spike.
2. **The population takes days to weeks to grow.** Nitrifying organisms are slow-growing autotrophs. Doubling times are measured in hours to days under ideal conditions and longer in practice.
3. **The population can shrink.** Reduced feeding, a long absence, or a period of low stocking reduces the population. Returning to normal feeding after a two-week vacation can outrun a diminished biofilter.

12.2 Brief ammonia and nitrite spikes, and why testing afterward shows nothing

[ESTABLISHED — this is the central mechanism of the entire “transient event” category]

An established biofilter can process a moderate ammonia pulse within hours. The sequence is:

1. Event occurs (dead fish, overfeeding, filter disruption, media replacement).
2. Ammonia rises over 2–12 hours. If the pH is high, the toxic NH₃ fraction is substantial.
3. Fish are exposed. Gill damage and cortisol elevation occur. Some fish die.
4. Nitrifiers respond; ammonia falls. Nitrite may rise transiently and then fall.
5. The keeper notices dead fish at hour 24 or 36 and tests.
6. **Ammonia reads 0. Nitrite reads 0. Nitrate reads normal.**
7. Conclusion: “the water is fine, it must be a disease.”

This sequence is entirely consistent with the observations and is probably the single most common explanation for the core question of this document.

Aggravating factor: if the ammonia exposure damaged gills, fish can continue dying for days *after* the ammonia is gone. UF/IFAS’s statement that fish may become sick even weeks after a water-quality problem has been corrected applies directly [UF-FA16].

12.3 Sudden increases in bioload

- Adding several fish at once.
- Adding a single large fish (a 10-inch pleco produces far more waste than six tetras).
- A large livebearer brood surviving.
- Increased feeding.
- Fish growing (a common slow-onset version — the tank was cycled for juveniles).
- A new keeper taking over and feeding more generously.

12.4 Dead organisms decomposing unnoticed

[GENERALLY SUPPORTED — the most common single cause of an unexplained ammonia spike]

- A fish that died behind the filter, under a rock, inside an ornament, or in a plant thicket.
- A dead snail. **Snails are a particular hazard** because a large mystery/apple snail represents a substantial mass and because a dead snail inside its shell is not visually obvious. The characteristic smell is diagnostic.
- Dead shrimp (less mass, but shrimp deaths often indicate the underlying cause too).
- A fish that jumped into the filter compartment.
- A dead fish consumed by the tank’s own cleanup crew, leaving no evidence — the ammonia pulse occurs without a visible cause.
- **Cross-reference:** one technical source states the most common cause of an ammonia spike is a dead fish decaying in an unseen corner of the tank [AQSCI-OTS].

12.5 Filter disruption

Covered in Section 3.5–3.6. Cross-referenced here for retrieval: media replacement, over-cleaning, drying, extended shutdown, restart of a hypoxic canister, impeller failure, and flow reduction all reduce nitrification capacity.

12.6 Medication effects on nitrifying organisms

[GENERALLY SUPPORTED]

- **Antibiotics** (erythromycin, kanamycin, nitrofurazone, tetracyclines, sulfa drugs) are the primary concern; nitrifying bacteria are bacteria. The magnitude of the effect varies by drug and dose. Broad-spectrum antibiotics used at full dose for a full course in a display tank can measurably impair nitrification.
- **The clinical trap:** the keeper treats a sick fish with an antibiotic; the biofilter is suppressed; ammonia rises; more fish become sick; the keeper concludes the disease is spreading and increases treatment.
- **Antiparasitics** (formalin, malachite green, praziquantel, copper) have less effect on nitrifiers but formalin and potassium permanganate exert an oxygen demand and are directly biocidal at higher doses.
- **This is a primary argument for treating in a hospital tank** rather than the display.
- Ammonia-binding conditioners used during treatment can mask a rising ammonia problem on some test chemistries.

12.7 Chlorine exposure to the biofilter

Nitrifiers are as susceptible to chlorine as other bacteria in principle. See 3.5.1 for the disputed magnitude of the brief tap-rinse scenario. The undisputed cases are: refilling a substantial fraction of the tank with unconditioned chlorinated water, and soaking media in tap water.

12.8 Power loss

[GENERALLY SUPPORTED]

Consequences, in order of onset: 1. **Immediately:** surface agitation and circulation stop; gas exchange drops sharply. 2. **Within 1-3 hours:** oxygen falls in a stocked tank; media inside a canister goes hypoxic. 3. **Within 3-12 hours:** aerobic nitrifiers in dense media begin dying; temperature drifts. 4. **On restoration:** the canister injects its accumulated hypoxic water (3.6); the biofilter has reduced capacity; an ammonia rise follows over the next 24-72 hours.

Correct practice during an outage: battery air pump; manual water agitation; insulate the tank; do not feed; on restoration, open and rinse any canister that was off more than a few hours.

12.9 Bacterial blooms

[GENERALLY SUPPORTED]

- A bacterial bloom is a rapid proliferation of **heterotrophic** bacteria feeding on dissolved organic carbon. It appears as milky/white cloudiness (distinct from green algal cloudiness).
- Triggers: new tank setup, overfeeding, a death, substrate disturbance, media replacement, adding a large quantity of organic matter.
- **The bloom itself consumes oxygen.** In a warm, stocked tank, a heavy bloom can produce meaningful oxygen depletion, which is the actual harm.
- Heterotrophic bacteria also compete with nitrifiers for oxygen and surface area.
- Blooms are usually self-limiting once the dissolved organic carbon is consumed.
- **The bloom is a symptom of an organic input event.** Its presence following a water change is evidence that something organic was released — a useful diagnostic marker for substrate disturbance.

12.10 Decaying plant matter

- A large melt event (crypts, or newly added tissue-culture plants adjusting) releases a substantial organic load.
- Removing a large mass of plants leaves cut root systems to decompose.
- Dying plants also stop consuming ammonia and nitrate.
- A plant “melt” following a water change is common and usually reflects a chemistry change (CO₂, temperature, or nutrients), not the plant dying — but the decomposing tissue still adds load.

12.11 What is scientifically plausible versus speculative in this category

Claim	Classification
A transient ammonia spike can kill fish and be undetectable 24 hours later	[ESTABLISHED]
An unnoticed dead animal is a common cause of ammonia spikes	[GENERALLY SUPPORTED]
Wholesale media replacement causes a measurable cycle disruption	[ESTABLISHED]
Antibiotics can suppress nitrification	[GENERALLY SUPPORTED]
Restarting a long-stopped canister introduces harmful water	[GENERALLY SUPPORTED]
Bacterial blooms consume oxygen	[ESTABLISHED]
“The cycle crashed” as an explanation for deaths with 0 ammonia and 0 nitrite at the time of testing	[SPECULATIVE unless a spike was actually measured, or unless pH/KH data support the OTS pattern]
Water changes crash the cycle	[UNSUPPORTED — see 17.7]
Beneficial bacteria live primarily in the water column	[UNSUPPORTED — the great majority are attached in biofilm]

SECTION 13. pH, GH, KH, TDS AND CONDUCTIVITY — A TECHNICAL REFERENCE

13.1 pH

What it measures. The negative base-10 logarithm of hydrogen ion activity. pH 6 is ten times more acidic than pH 7; pH 5 is one hundred times more acidic.

What it does not measure. - The **quantity** of acid or base present (that is alkalinity/acidity, i.e., buffering). - Mineral content of any kind. - Which ions are present. - How stable the value is.

Key relationships. - pH governs the **ammonia/ammonium equilibrium** (Section 1.2) — the single most important consequence of pH in an aquarium. - pH affects the **speciation and toxicity of metals** (copper more toxic at low pH; aluminum more bioavailable at low pH). - pH affects **nitrogen sulfide and hydrogen sulfide speciation**. - pH affects **CO₂/bicarbonate/carbonate equilibrium** and therefore plant carbon availability. - pH affects **nitrifying bacteria activity**, which declines substantially below roughly 6.5 for the classical species [AQSCI-OTS].

Why identical pH does not mean identical water. Two waters at pH 7.6: one could be RO water with a pinch of sodium bicarbonate (TDS 40, GH 0, KH 2); the other could be hard well water (TDS 500, GH 18, KH 12). These are physiologically completely different environments for a fish, and moving between them is a major osmoregulatory transition despite an identical pH reading. **This point is the single most useful correction to hobbyist reasoning about water chemistry.**

13.2 KH (carbonate hardness / alkalinity / buffering capacity)

What it measures. The concentration of bicarbonate (HCO₃⁻) and carbonate (CO₃²⁻) ions — operationally, the water’s capacity to neutralize added acid. Reported in dKH (German degrees) or ppm CaCO₃. **1 dKH ≈ 17.86 ppm CaCO₃.**

What it does not measure. - Calcium or magnesium content (that is GH). - pH. - Total dissolved solids.

Why it matters. - **It determines pH stability.** A tank with KH 8 resists acidification for a long time; a tank with KH 1 can experience large pH swings from small inputs. - **It is consumed by nitrification.** Ammonia oxidation produces hydrogen ions that neutralize bicarbonate. Alkalinity therefore declines over time in a stocked tank without water changes — the mechanism underlying Old Tank Syndrome (2.12). - **It is consumed by CO₂ injection systems indirectly** and by acidic substrates, driftwood tannins, peat, and botanicals. - **It supports nitrifying organisms** as an inorganic carbon source. - **It buffers against the diurnal photosynthesis/respiration pH swing.**

Critical warning: KH below about **3 dKH** should be treated as a monitoring priority. Practical Fishkeeping states that unless keeping genuinely soft-water fish, a reading below 5 dKH is a warning that cannot be ignored [PFK-OTS]. At KH 0, pH becomes essentially unbuffered.

13.3 GH (general hardness / total hardness)

What it measures. The combined concentration of divalent cations, principally calcium (Ca²⁺) and magnesium (Mg²⁺). Reported in dGH or ppm CaCO₃. **1 dGH ≈ 17.86 ppm CaCO₃.**

What it does not measure. - Sodium, potassium, chloride, or sulfate — **softened water can read low GH while having high TDS.** - pH or buffering. - The Ca:Mg ratio, which matters for plants and some invertebrates but is not resolved by a standard GH test.

Why it matters. - **Osmoregulation.** Environmental calcium reduces gill permeability and lowers the metabolic cost of ion regulation (7.1). - **Invertebrate molting.** Shrimp and snails require calcium (and magnesium) to build exoskeleton and shell. Chronically low GH produces failed molts (“white ring of death”) and shell erosion. - **Metal toxicity modification.** Copper (and zinc, cadmium, lead) toxicity decreases with increasing hardness [ANZG-COPPER]. - **Plant nutrition.** - **Species suitability.** Rift Lake cichlids and livebearers do poorly in very soft water; many wild blackwater species do poorly in hard water.

Conversion note: 1 dGH ≈ 17.86 ppm as CaCO₃ ≈ 7.14 ppm Ca (if all hardness were calcium).

13.4 The GH/KH independence point

GH and KH are **independent** parameters that happen to correlate in most natural waters because calcium carbonate dissolution raises both simultaneously.

Configuration	GH	KH	Real-world example	Consequence
Typical hard water	High	High	Limestone aquifer, most tap water in hard-water regions	Stable pH, good for hard-water species
Typical soft water	Low	Low	Granite catchment, rainwater-fed reservoir, RO	Unstable pH; needs buffering attention
Low GH, high KH	Low	High	Ion-exchange softened	Poor for invertebrates and

			water ; RO + sodium bicarbonate only	hard-water fish; pH stable but minerals absent
High GH, low KH	High	Low	RO + GH-only remineralizer; water with calcium sulfate but little bicarbonate	Minerals present but pH unstable

Practical diagnostic value: measuring GH and KH separately, and comparing them to the source water, identifies softeners, RO errors, buffer depletion, and evaporation-driven mineral concentration — none of which is visible from pH alone.

13.5 Conductivity and TDS

Conductivity (specific conductance). Measures the water’s ability to conduct electrical current, which is proportional to the concentration and mobility of dissolved ions. Units: microsiemens per centimeter (µS/cm). Temperature-compensated meters normalize to 25°C.

TDS (total dissolved solids). In hobby practice, TDS meters do **not** measure dissolved solids directly. They measure conductivity and multiply by a conversion factor (commonly 0.5, sometimes 0.64 or 0.7, varying by manufacturer and by assumed ion composition). **A “TDS reading” is therefore a conductivity reading in disguise, converted with an assumption that may not hold for a given water.**

Implications: - Two meters with different conversion factors will report different TDS for the same water. Neither is wrong; they use different assumptions. - Comparing TDS numbers between hobbyists is less meaningful than comparing each hobbyist’s own readings over time. - Trends within one system with one meter are highly informative; absolute cross-system comparisons are not.

13.6 Why TDS alone cannot identify what is dissolved

[ESTABLISHED — a core conceptual point]

A TDS reading of 300 ppm can be produced by any of the following, which are physiologically and chemically dissimilar:

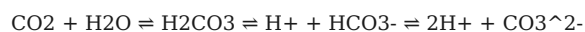
Composition	GH	KH	Physiological character
Calcium bicarbonate (hard water)	High	High	Well-buffered hard water
Sodium chloride (salted or softened water, or brackish contamination)	~0	~0	High osmotic load, no hardness, no buffering
Sodium bicarbonate	~0	High	Buffered but mineral-poor
Potassium nitrate (very high nitrate)	~0	~0	High nitrate; possible chronic toxicity
Magnesium sulfate (Epsom salt)	High	~0	Hardness without buffering
Dissolved organic carbon (tannins, humics)	~0	Variable	Low ionic contribution; TDS meters read these poorly
A copper sulfate contamination event	~0	~0	Potentially lethal to invertebrates at TDS-invisible concentrations

Additional limitation: TDS meters do not detect non-ionic dissolved substances. Sugars, most organic molecules, dissolved gases, and many pesticides contribute little or nothing to conductivity. **A tank contaminated with a lethal concentration of a pyrethroid shows no TDS change.**

Correct use of TDS in the hobby: as a **change detector** within a single system. A shrimp keeper who knows their tank runs at 180 ppm and reads 260 ppm has learned that something changed and should investigate with GH, KH, and nitrate tests. TDS tells you *that* something changed, never *what*.

13.7 Buffering explained

Buffering is the resistance of a solution to pH change on addition of acid or base. In freshwater aquaria the dominant buffer system is the **carbonate system**:



- Adding acid (H⁺) drives the equilibrium left; bicarbonate absorbs the hydrogen ions and pH barely moves — until the bicarbonate is exhausted.
- **When KH approaches zero, the buffer is gone and pH becomes highly sensitive.** Small acid inputs from nitrification, decomposing organics, driftwood tannins, or CO₂ can then produce large swings.
- **Phosphate buffers** (in commercial “pH 7.0” products) work on a different equilibrium and can interact badly with calcium-rich water and with planted-tank chemistry.
- **Substrate buffering:** crushed coral, aragonite, and limestone dissolve to raise KH and GH (pH-dependent dissolution, self-limiting around pH 8.2–8.3). Peat, some botanicals, and cation-exchange “active” aquasoils lower KH and pH — **active substrates deplete over time**, meaning a tank’s KH and pH can drift upward after a year or two as the substrate’s exchange capacity exhausts.

13.8 Sudden versus gradual change

[GENERALLY SUPPORTED]

The general principle across all these parameters: **fish acclimate to a wide range of stable conditions and tolerate gradual change far better than abrupt change**. The physiological basis is that acclimation involves gene expression changes, gill ionocyte remodeling, and enzyme adjustment — processes that take days.

However, this principle is frequently over-applied. Important qualifications:

- 1. The **direction and mechanism** matter more than the numeric magnitude (see 7.2). A pH change caused by CO2 degassing is not equivalent to a pH change caused by a mineral shift.
- 2. The **starting condition** matters. A fish in an already-harmful environment may benefit from rapid correction of an acutely toxic condition even at the cost of a chemistry step-change. **When ammonia is acutely toxic, a large water change is correct even though it changes chemistry rapidly**.
- 3. **Some changes are protective**. Adding aeration, adding chloride against nitrite toxicity, and lowering pH to reduce ammonia toxicity are all rapid interventions that reduce harm.

13.9 Species adaptation and the “stability over ideality” principle

The most defensible general statement available from the sources reviewed:

For most captive-bred ornamental freshwater fish, **a stable value somewhere in a broad tolerable range outperforms an unstable value that is on average closer to the species’ natural ideal**. The main exceptions are (a) parameters that change the toxicity of something else (pH’s effect on ammonia and metals; hardness’s effect on metals), (b) genuine physiological requirements (calcium for invertebrate molting; carbonate for buffering), and (c) wild-caught specimens with narrow adaptations.

This is why chasing pH with buffering chemicals is generally counterproductive: it converts a stable but non-ideal parameter into an unstable one.

13.10 Source water versus aquarium water

The gap between the two is one of the most informative measurements available, and one of the least frequently taken.

Parameter	If aquarium > source	If aquarium < source
pH	CO2 degassing from source water (2.20); alkaline substrate/rock; low CO2; heavy photosynthesis	Nitrification acidification; CO2 injection; tannins/peat; buffer depletion (OTS)
KH	Alkaline substrate or rock dissolving; buffer product added	Nitrification consuming alkalinity — the OTS signature
GH	Evaporation concentration; calcareous substrate/rock; remineralizer added	Active/cation-exchange substrate; RO dilution; plant/invertebrate uptake
TDS	Evaporation concentration; nitrate accumulation; ferts and additives	RO dilution
Nitrate	Normal — nitrification product	Heavy plant uptake, or source water already high in nitrate

Protocol note for correct source-water testing: draw a sample, **let it stand or aerate for 12-24 hours**, then test pH. Testing straight from the tap gives a CO2-depressed value that does not represent what the water will do in the aquarium (2.20).

13.11 Evaporation and mineral accumulation

Covered in 2.13. Restated as chemistry: evaporation removes pure water and leaves all non-volatile solutes. Over months of top-offs with tap water, GH, KH, TDS, sodium, nitrate, and any trace contaminant all concentrate. The magnitude depends on evaporation rate (open-top, warm, high-flow tanks evaporate faster) and on how much of the tank volume is replaced by evaporation over time.

Rough illustration: a 20-gallon open tank in a dry warm room losing 1 gallon per week and topped off with tap water at 200 ppm TDS adds roughly 200 ppm × 1 gallon of solute per week to a fixed volume. Over three months, without a single water change, that is a substantial accumulation. **Water changes remove solutes; top-offs add them**.

13.12 Quick conversion and reference table

Conversion	Value
1 dGH	17.86 ppm (mg/L) as CaCO3
1 dKH	17.86 ppm (mg/L) as CaCO3
NO3- to NO3-N	divide by 4.43
NO3-N to NO3-	multiply by 4.43
NH3/NH4+ (as N) to (as NH3)	multiply by 1.216
NO2- to NO2-N	divide by 3.28

TDS (ppm) from conductivity (µS/cm)	multiply by 0.5–0.7 depending on meter
°C to °F	(°C × 9/5) + 32

Retrieval warning: the nitrate unit ambiguity is a genuine source of error. A drinking-water nitrate limit of 10 mg/L NO₃-N equals about 44 mg/L NO₃⁻, which on an aquarium test kit reading in NO₃⁻ looks like a moderate aquarium value. Well-water users in agricultural regions may be adding meaningful nitrate with every water change.

SECTION 14. SIGNS AND TIMING AS DIAGNOSTIC CLUES

14.0 Mandatory framing

These are probability shifters, not diagnostic rules. Every pattern below has exceptions. Multiple causes frequently operate simultaneously. The purpose of this section is to narrow an investigation, never to conclude one.

14.1 Immediate distress during the water change (seconds to minutes)

Investigate first: 1. **Chlorine or chloramine** — conditioner forgotten, wrong dose, wrong product, or added after the water. Highest prior probability. 2. **Temperature shock** — check the temperature of the incoming water at the tap and in the tank. 3. **Direct exposure to the inflow stream** — fish sitting in undiluted tap water. 4. **Gas supersaturation** — visible fine bubble coating on glass, decor, and fish. 5. **Contaminated bucket, hose, or hands** — soap, cleaner, lotion, insecticide residue. 6. **pH/ionic step change** in a neglected tank receiving a large change (2.11). 7. **Physical disturbance** — fish trapped, crushed, or netted.

Evidence that would support each: - Chlorine: municipality is chlorinated/chloraminated; a chlorine test on the tap water; reproducibility on subsequent changes; gasping and reddened gills; smaller fish worst affected. - Temperature: a thermometer in the incoming stream; a large delta measured. - Gas supersaturation: bubbles on every surface, on fish, and in fins; the water was drawn hot or through a hose. - Bucket contamination: recent household use of the bucket; suds when agitated; a distinctive smell. - pH/ionic: measured pre- and post-change pH, KH, GH, TDS on a tank that had not been changed for a long time.

Evidence that would weaken: if the same procedure has been performed identically for months without incident, chlorine and technique become less likely and something that *changed* becomes more likely — the municipality's chlorine conversion, a new bottle of conditioner, a different bucket, a new plumbing fixture.

14.2 Distress within minutes to hours

More likely: 1. Chlorine/chloramine (slightly delayed presentations occur). 2. **Oxygen depletion** — from substrate disturbance, a stopped filter, reduced surface agitation, or a bacterial bloom starting. 3. **Ammonia released by substrate disturbance** or by a disturbed dead animal. 4. **A restarted canister** injecting hypoxic, organically loaded water. 5. **Chemical contamination** — aerosol, insecticide, cleaner. 6. **Medication or dechlorinator overdose** (oxygen consumption). 7. **CO₂ system malfunction**. 8. **Temperature climbing** from a heater fault or a stuck-on heater.

Differentiators: does vigorous aeration produce visible improvement within 15 minutes? If yes, oxygen. Does the ammonia test now read positive? If yes, ammonia. Are invertebrates dying while fish are merely lethargic? If yes, suspect copper or an insecticide.

14.3 Fish dead the following morning

[This is the single most common presentation reported by hobbyists.]

Consider, roughly in order: 1. **Overnight oxygen minimum**. The pre-dawn hours are the daily oxygen low point in any planted or heavily stocked tank. Any additional oxygen demand (substrate disturbance yesterday, bacterial bloom, warm water, a stopped filter) tips it over. 2. **Equipment left off**. Filter, air pump, or heater unplugged during maintenance and not restarted. **Check every plug and every device before anything else.** 3. **Heater failure — stuck on**. Check the water temperature immediately, before anything else changes it. 4. **CO₂ solenoid stuck open** in an injected tank. 5. **Nocturnal aggression or predation**. 6. **Jumping** — search the floor and behind the stand. 7. **A delayed ammonia spike** from yesterday's disturbance. 8. **Chemical exposure overnight** — an air freshener, a plug-in insect repellent, a cleaner used in the evening. 9. **Acute columnaris** (can kill in 24–72 hours). 10. **Progression of an insult that occurred during yesterday's maintenance** — gill damage from chlorine or ammonia manifesting as respiratory failure hours later.

High-value immediate observations: water temperature; whether every piece of equipment is running; whether the surface is agitated; the position of the dead fish (surface = often respiratory; wedged in decor = possible entrapment; near the substrate with fins clamped = various); the appearance of gills (brown = nitrite; red = irritant; pale = anemia/chronic).

14.4 Deaths beginning several days after introducing new fish

More plausible: 1. **Introduced pathogen with an incubation period** — Ich (3–7 days), velvet (2–7 days), flukes (days to weeks), columnaris (1–5 days after stress). **This is the highest-probability category.** 2. **The new fish were already compromised** and are now dying from supply-chain damage (Section 6). 3. **Bioload increase** exceeding biofilter capacity, producing a modest ammonia excursion. 4. **Aggression** as territories are contested — either against the new fish or triggered among the residents. 5. **Old Tank**

Syndrome pattern — the residents are acclimated to conditions the newcomers cannot tolerate (2.12). Strong indicator: **only the new fish die and the residents are unaffected**. 6. **Stress-mediated flare of a latent infection** in the existing population.

Differentiating question: are the deaths confined to the new fish, confined to the residents, or spread across both? - New fish only → supply-chain damage, OTS pattern, aggression against newcomers, or acclimation failure. - Residents only or residents first → introduced pathogen to which the new fish are resistant carriers. - Both → introduced pathogen, or a shared environmental cause coincidental with the addition.

14.5 One species affected while others remain healthy

Potential indications: 1. **Species-specific pathogen**. Many pathogens have host preferences (e.g., *Pleistophora* in characins; *Camallanus* in livebearers; dwarf gourami iridovirus). 2. **Species-specific parameter intolerance** — temperature (9.10), hardness, oxygen requirement, flow requirement. 3. **Species-specific size/sensitivity** — small-bodied fish affected first by a toxicant; large-bodied fish affected first by hypoxia. 4. **Aggression targeting one species**. 5. **Food competition** disadvantaging one species (bottom dwellers, slow feeders). 6. **Scaleless species** (loaches, some catfish) affected first by medications, salt, and irritants. 7. **All the affected fish came from the same source/batch** — this is a purchasing cohort, not a species effect. **Always check whether “the species” is actually “the batch.”** 8. **Age cohort** — all the fish of that species were purchased together two years ago and are reaching end of life together.

14.6 All fish suddenly affected at once

Strongly shifts toward an environmental/whole-tank cause and away from infectious disease: 1. **Toxin introduced to the water** (chlorine, insecticide, cleaner, medication overdose). 2. **Acute oxygen depletion**. 3. **Temperature excursion**. 4. **Massive ammonia or nitrite excursion**. 5. **Gas supersaturation**. 6. **Electrical fault** — a failing heater or pump leaking current into the water. [PLAUSIBLE, POORLY DOCUMENTED but reported; symptoms include all fish clustered at one end or acting erratically, and a tingling sensation when the keeper touches the water. A ground-fault interrupter is the correct mitigation and is rarely used in the hobby.] 7. **pH collapse** in an unbuffered tank.

Infectious disease essentially never affects every fish simultaneously within hours. Simultaneity is the strongest single argument for an environmental cause.

14.7 Only bottom-dwellers affected

Investigate: 1. **Substrate-associated exposure** — something released from the substrate, or a contaminant that has settled. 2. **Localized low oxygen** at the substrate in a low-flow tank. 3. **Medication and salt sensitivity** — most bottom dwellers in home aquaria are scaleless or nearly so (*Corydoras*, loaches, plecos) and are the classic victims of copper, formalin, malachite green, and salt at doses tolerated by scaled fish. 4. **Sharp substrate** damaging *Corydoras* barbels, leading to infection. 5. **Starvation** — bottom dwellers outcompeted at feeding (10.5, 11.2). 6. **Temperature** — many *Corydoras* are stressed at temperatures suiting the rest of the tank. 7. **Physical trapping or crushing** during maintenance. 8. **Cold layer** near the substrate in a low-flow tank. 9. **Nocturnal aggression**, since many bottom dwellers are active when the aggressors are.

14.8 Only newly introduced fish affected

Suggests, in order: 1. **Old Tank Syndrome / large chemistry gap** between the store’s water and the (drifted) tank water — the residents are acclimated, the newcomers are not. 2. **Supply-chain damage** manifesting on schedule. 3. **Acclimation failure** (Section 7). 4. **Aggression from established residents** against newcomers. 5. **Inability to compete for food** during the first days. 6. **A pathogen carried by the residents** to which the residents are immune and the newcomers are naive. **This possibility is systematically overlooked** — the assumption is always that the new fish brought the disease.

Highest-value diagnostic: compare tank water and store water directly on pH, KH, GH, TDS, and nitrate. A large gap on any of these — particularly a large nitrate and TDS gap combined with a lower tank pH — is the OTS signature.

14.9 Only invertebrates affected (shrimp, snails)

Not in the original outline; added because it is one of the highest-information patterns available.

Shrimp and snails dying while fish are unaffected points strongly to: 1. **Copper** — from medication, plumbing, or trace fertilizers. 2. **Insecticide/pesticide** — especially from treated aquatic plants, houseplant systemics, or flea products. 3. **Ammonia and nitrite** — invertebrates are more sensitive than most fish. 4. **Low GH** — molting failure. 5. **Rapid TDS change**. 6. **Fish medications** at fish doses. 7. **Predation** by fish that have grown or become bolder.

Conversely, fish dying while shrimp are unaffected argues *against* metals and insecticides and toward pathogens, oxygen (shrimp tolerate lower DO than many fish and can leave the water), aggression, or fish-specific parameters.

14.10 Deaths that follow a regular schedule

- **Weekly, after the water change** → strongly suggests something in the water-change procedure (Sections 2 and 3).
- **Monthly, after filter maintenance** → suggests the filter procedure.
- **Every few weeks with no pattern** → chronic condition: mycobacteriosis, nutrition, chronic parameter mismatch, chronic aggression, age cohort.

- **Seasonal** → room temperature, municipal water changes, seasonal insecticide use, seasonal source-water switching, heating/cooling system operation.
- **After every absence/vacation** → automatic feeder malfunction, sitter overfeeding, equipment failure unnoticed, temperature drift.

14.11 Behavioral signs and their possible interpretations

Behavior	Possible interpretations
Gasping at the surface	Low oxygen; ammonia; nitrite; gill parasites; gill bacterial disease; chlorine; high CO ₂ ; gill damage from any prior insult
Hanging at the filter outflow	Seeking oxygen or seeking flow (species-normal for some)
Rapid gill movement	Same list as gasping
Clamped fins	General malaise: parasites, poor water quality, temperature, stress, early bacterial infection
Flashing / scratching on objects	Ectoparasites (Ich, velvet, flukes, <i>Chilodonella</i> , <i>Trichodina</i>); chemical irritants; ammonia; extreme pH
Shimmying (rocking in place without moving forward)	Classic in mollies and livebearers: low mineral content, temperature stress, or general nervous/osmotic distress
Erratic darting, spiraling, twitching	Neurotoxin (insecticide); severe ammonia; neurological disease; extreme pH
Sitting on the bottom, upright, not moving	Weakness from many causes; cold; late-stage disease
Lying on the side	Advanced illness; buoyancy disorder; near death
Loss of equilibrium / rolling	Swim bladder problem; neurological; buoyancy disorder
Head-down or tail-down hovering	Buoyancy disorder; swim bladder
Hiding constantly	New arrival; aggression; insufficient cover; illness; inadequate group size
Not eating	Almost any problem; also normal for 1–3 days after introduction
Eating and spitting	Food size; mouth injury; internal disease
Faded or darkened color	Stress (either direction depending on species); illness; substrate/background
White stringy feces	Internal parasites (flagellates, nematodes); enteritis; also seen with a fasted fish shedding gut lining
Gulping at the surface (labyrinth fish)	May be normal; excessive frequency indicates low oxygen
Increased aggression suddenly	Spawning; territory disruption after rearranging; overcrowding; a fish reaching maturity
Rubbing gills / opercular flaring	Gill parasites
Yawning/coughing motion	Gill irritation
Swimming into the glass repeatedly	New tank; reflection; neurological; extreme stress

SECTION 15. COMMON MISDIAGNOSES

15.1 “It was ammonia”

When this is over-invoked: ammonia is the default explanation for any gasping fish, because it is the parameter beginners are taught to fear. It is invoked even when the ammonia test reads zero, on the theory that “it must have spiked and come down.”

Why it is often wrong: - The identical presentation is produced by hypoxia, nitrite, gill parasites, gill bacterial disease, chlorine, and high CO₂ (4.7). - **A TAN reading without pH is uninterpretable** (1.2). At pH 6.5, a TAN of 1.0 mg/L is a very different situation than at pH 8.3. - Some ammonia-binding conditioners interfere with certain test chemistries, producing readings that are not what they appear to be.

When it genuinely is ammonia: - New tank, no cycling, fish added immediately — the most common true case. - A measured, positive ammonia reading with a pH that makes the un-ionized fraction significant. - A documented ammonia source: dead animal, media replacement, medication suppressing the biofilter, massive overfeeding, filter off for a long period. - Old Tank Syndrome with high TAN at low pH, followed by a pH-raising water change (2.11) — **the most dangerous true ammonia scenario, and the one most likely to be missed** because the pre-change ammonia was relatively harmless.

15.2 “It was the water change”

When this is over-invoked: almost always, because the water change is the label for a bundle of simultaneous actions (3.1) and because it is the most salient recent event.

Why it is often wrong: the actual cause is frequently a co-occurring maintenance action — filter cleaning, substrate disturbance, equipment left off, a heater exposed to air, decor moved, a fish injured. Sometimes it is an unrelated event that merely coincided.

When it genuinely is the water change (the new water itself): - Chlorine or chloramine not neutralized. - A large temperature delta. - A large chemistry delta into a neglected tank. - A change in the municipal supply. - Contaminated replacement water (bucket, hose, plumbing work). - Gas supersaturation.

Diagnostic test: perform a small water change (10–15%) with carefully pre-treated, temperature-matched water and observe. If the fish are undisturbed, the replacement water is probably not the problem, and the maintenance actions are.

15.3 “The cycle crashed”

When this is over-invoked: whenever ammonia is detected at all, and frequently when no ammonia is detected at all. One technical source observes directly that many hobbyists label any ammonia spike a “crash” and attribute it to Old Tank Syndrome, when the vast majority of ammonia spikes are not accompanied by a pH drop and are not OTS [AQSCI-OTS].

Why it is often wrong: - A single ammonia excursion from a discrete input (a dead fish, a heavy feeding) is a **load event**, not a loss of biofilter capacity. The distinction matters because the responses differ: find and remove the source, versus rebuild the biofilter. - If the tank had truly lost its nitrifying population, ammonia and nitrite would remain elevated for days to weeks. A spike that resolves within 24 hours demonstrates the biofilter is working.

When it genuinely is a biofilter failure: - Wholesale media replacement or thorough cleaning of all media. - Filter off for an extended period. - A full course of a broad-spectrum antibiotic in the display tank. - Substantial chlorine exposure to the media. - pH collapse below ~6.0 suppressing nitrification. - Ammonia and nitrite **both** persistently elevated over multiple days.

15.4 “It was the pH”

When this is over-invoked: whenever a pH reading differs from a species’ published “ideal,” or whenever a pH change is noted around the time of a problem.

Why it is often wrong: see 2.8 and 7.2. pH is a log of hydrogen ion activity, not a substance and not a measure of water character. The evidence for magnitude-driven “pH shock” as an independent killer is weak.

When pH genuinely matters: - **As the controller of ammonia toxicity.** This is the dominant real pH mechanism. - **As the controller of metal speciation and toxicity.** - **As an indicator of buffer depletion** — a falling pH with a falling KH is diagnostically important. - **At physiological extremes** — sustained pH below ~5.0 or above ~9.0 is directly harmful to most community species. - **As a marker of a CO2 problem** — a sudden pH drop in an injected tank means CO2 is accumulating.

A frequent secondary error: chasing pH with buffering products in unbuffered water, converting a stable non-ideal pH into an unstable one, which is worse (13.9).

15.5 “It was the pet store’s fault”

When this is over-invoked: as a reflexive attribution whenever a new fish dies, often without any specific evidence.

When it is genuinely justified: the evidence in Section 6 is substantial. Supply-chain damage, latent infection, starvation, and centralized-system pathogen sharing are real and common. A fish purchased on the day of delivery, from a system where other tanks show sick fish, is a genuinely elevated risk.

Why the attribution is nonetheless often unhelpful: it is not actionable, it is usually unverifiable, and it forecloses examination of the home tank. The productive version of this hypothesis is **specific:** “the fish likely arrived carrying flukes” is testable (by microscopy or by response to praziquantel); “the store’s fault” is not.

15.6 “It was a disease”

When this is over-invoked: whenever water parameters read fine, because disease becomes the residual category. It is particularly over-invoked when a fish shows any visible mark, which is then treated with a broad-spectrum antibiotic.

Why it is often wrong: - Visible lesions are frequently **secondary** to injury, aggression, or an environmental insult (5.11). - Buoyancy problems, “dropsy,” and “swim bladder disease” are **signs, not diseases** (11.6). - Treating with medication in the display tank imposes real costs: biofilter suppression, oxygen demand, invertebrate mortality, and plant damage — all of which can worsen the actual problem. - Merck’s explicit warning against “shotgunning” fish with multiple drugs applies directly [MERCK-OWNER].

When disease is genuinely the answer: a recognizable presentation (Ich’s discrete white dots, velvet’s gold dusting, columnaris’ saddleback), progressive fish-to-fish spread over days, correlation with a recent introduction, or a positive microscopic finding.

15.7 “It was the substrate”

When this is over-invoked: the substrate hydrogen sulfide claim (3.4) is the canonical example — invoked confidently, mechanistically dubious under aquarium conditions, and repeated across social media.

When the substrate is genuinely implicated: - **Organic accumulation released by disturbance**, creating an oxygen-demand pulse and an ammonia/nutrient release (3.2). This is the real, well-supported version of the “substrate killed my fish” claim. -

Active/buffering substrates altering KH, GH, and pH — either intentionally (aquasoil lowering pH) or unintentionally (crushed coral, calcareous sand, or certain rocks raising them). - **Sharp substrate** injuring *Corydoras* barbels and other bottom dwellers. - **A substrate that trapped and killed something**, which then decomposed. - **A newly added substrate leaching** — some soils release ammonia for days to weeks after setup. - **Under-gravel filter plenum sludge** released when flow stops or resumes (3.4, Position C).

15.8 “It was a chemical” (unspecified)

When this is over-invoked: as an untestable catch-all whenever nothing else fits.

When it is genuinely justified: the evidence pattern is fairly specific. - **Invertebrates dead, fish affected less or later** → strong evidence for copper or an insecticide. - **A plausible exposure event identified in the timeline** — spraying, painting, a new bucket, a pet treatment, plumbing work. - **All fish affected simultaneously with normal nitrogen parameters and adequate oxygen.** - **Rapid neurological signs** — spiraling, twitching, violent darting — which are characteristic of pyrethroid and organophosphate exposure and are not typical of ammonia. - **Improvement following heavy activated carbon plus large water changes**, which is weak evidence (it is also the correct response to several other causes) but is consistent.

15.9 The general structure of misdiagnosis

Every entry above shares a pattern:

1. A **salient, familiar, testable** cause is available (ammonia, pH, the water change).
2. A **non-salient, unfamiliar, untestable** cause is actually operating (oxygen, a transient spike already resolved, supply-chain damage, aggression, starvation, a contaminant).
3. The familiar cause is selected because it can be investigated, not because it is more likely.
4. The intervention aimed at the familiar cause fails, and often adds harm (medication in the display tank, chasing pH, another large water change).

The corrective discipline is to enumerate what has *not* been measured before concluding from what has.

SECTION 16. EMERGENCY RESPONSE

16.0 Principle

The two dominant errors in an aquarium emergency are **doing nothing** and **doing everything at once**. The second is more common and more damaging: a keeper who simultaneously performs a 90% water change, adds three medications, raises the temperature, adds salt, and replaces the filter media has destroyed both the fish’s environment and all diagnostic information.

16.1 What to check first — in order

This ordering reflects speed of harm and cost of checking.

1. **Is every piece of equipment running?** Filter, heater, air pump, powerhead. Look at the water surface: is it moving? This costs five seconds and resolves a substantial fraction of emergencies.
2. **What is the temperature, right now, measured with a thermometer you trust?** Before adding anything or changing anything.
3. **Is the surface agitated? Is there a surface film?**
4. **Count the livestock.** Is anything missing? A dead animal is both a cause and an ongoing source.
5. **Smell the water.** Hydrogen sulfide, decay, chemical odors, and chlorine are all detectable.
6. **Look at the fish, specifically:** gill color and movement rate; body condition; lesions; position in the tank; whether they are all affected or only some.
7. **Test ammonia, nitrite, nitrate, and pH** — and **test the tap water too**.
8. **Reconstruct the last 72 hours.** Every maintenance action, every addition, every household event (see 16.6).

16.2 When a water change is appropriate

[GENERALLY SUPPORTED] A water change is the correct first intervention when the problem is, or may be, a dissolved toxicant.

Indicated when: - Ammonia or nitrite is measurably elevated. - A contaminant is suspected (chemical, medication overdose, insecticide). - Nitrate is extremely high **and** the tank has been maintained (i.e., not the OTS scenario). - The tank is new and uncycled with fish present. - You do not know what is wrong and the fish are acutely distressed — **a partial change with properly conditioned, temperature-matched water is a low-risk default action** in most circumstances.

Execution requirements when doing an emergency change: - Match temperature. - Fully dechlorinate. - Add the new water gently, not as a stream onto the fish. - 25–50% is usually sufficient; a 50% change removes half of any dissolved toxicant. - Repeating a 50% change twice removes 75%; three times removes 87.5%. **Serial moderate changes are usually better than one enormous one.**

16.3 Aeration — the near-universal first action

[GENERALLY SUPPORTED — highest benefit-to-risk ratio of any emergency intervention]

Adding an air stone or increasing surface agitation: - Directly treats hypoxia, the most under-diagnosed acute killer (Section 4). - Helps with elevated CO₂. - Helps drive off volatile contaminants. - Assists during medication use. - Assists during and after gravel vacuuming or any organic-load event. - **Is harmful in essentially no realistic aquarium scenario** except a CO₂-injected planted tank (where it will drive off CO₂ — but in an emergency, fish outrank plants) and the rare case of gas supersaturation (where the aeration actually

helps degas the water to equilibrium).

Because it is nearly free of downside and treats a common invisible cause, aeration should be the default first action in almost any aquarium emergency.

16.4 When another water change might make things worse

[GENERALLY SUPPORTED]

1. **The Old Tank Syndrome scenario.** A neglected tank with low pH and accumulated ammonia: a large water change with higher-pH water can convert relatively harmless ammonium into toxic un-ionized ammonia (2.11). **Test ammonia before changing water in a long-neglected tank.** If ammonia is present and pH is low, change small volumes and consider addressing ammonia first.
2. **Untreated or improperly treated replacement water.** If the last water change is the suspected cause, repeating it with the same water and the same procedure repeats the insult. **Identify the water first.**
3. **When the source water is itself contaminated** — during a municipal chlorine conversion, after a main break, or with a known plumbing problem.
4. **When temperature cannot be matched** and the delta would be large.
5. **Mid-medication**, where a large change removes the therapeutic dose and creates a dosing problem; unless the medication itself is the suspected cause, in which case removing it is correct.
6. **When it would destroy diagnostic information** without a specific reason — see 16.7.

Note on a common hobby claim: “never do a water change during an emergency” is not a defensible general rule (17.11). The defensible version is: **understand what you are changing before you change it, and match temperature and dechlorinate properly.**

16.5 Other evidence-based emergency actions

Action	When appropriate	Notes
Stop feeding	Almost always, for 24–72 hours	Reduces ammonia production and organic load; healthy fish tolerate short fasts easily
Increase aeration	Almost always	16.3
Check and correct temperature	Always	Correct slowly unless the current temperature is acutely lethal
Add fresh activated carbon	Suspected chemical contamination, or ending a medication course	Adsorbs many organics, medications, and some contaminants. Not effective for ammonia, nitrite, nitrate, or most metals
Add a dechlorinator dose	Suspected chlorine exposure	Immediate and harmless
Add chloride/salt	Measured nitrite present	~10 mg/L chloride per 1 mg/L nitrite [THEFISHSITE-AMMONIA]; check species tolerance
Test the source water	Always, in any water-change-associated event	The most-skipped high-value test
Isolate affected fish	Suspected infectious disease, or aggression	A hospital tank allows treatment without dosing the display
Isolate an aggressor	Aggression identified	Often more effective than removing the victim
Remove the suspected contaminant source	Identified exposure	Insecticide, aerosol, ornament, medication
Remove carbon	Before dosing most medications	Or the dose is adsorbed
Water change	16.2	With the cautions in 16.4
Do nothing further and observe	When nothing specific is indicated	Legitimate and underused

16.6 Preserving evidence and recording the timeline

[STRONGLY RECOMMENDED — this section is arguably the highest-value practical content in the document]

Before changing everything:

1. **Test and record** ammonia, nitrite, nitrate, pH, KH, GH, TDS, and temperature — **of the tank and of the source water.** Write the numbers down with the time.
2. **Photograph and video the fish.** Video captures gill rate, swimming pattern, and buoyancy that a still photo and a text description cannot. Photograph gills, lesions, and the whole fish including a view from above.
3. **Photograph the tank as it currently is**, including the equipment and water level.
4. **Do not discard the dead fish immediately.** Refrigerate (do not freeze — freezing destroys tissue architecture for histopathology) if there is any chance of veterinary examination. Freshly dead is far more informative than long-dead; decomposition begins within an hour and post-mortem bacterial overgrowth quickly makes bacteriology uninterpretable.
5. **Write the 72-hour timeline** with times: every maintenance action, every addition, every product dosed, every household event (cleaning, spraying, cooking, painting, guests, pets treated), power interruptions, and weather/room temperature changes.

6. **Note what changed** — new bottle of conditioner, new food, new bag of substrate, new plant, new fish, new equipment, new plumbing.
7. **Check the water utility's website** for notices.
8. **Keep the current filter media** rather than replacing it.

Rationale: the diagnostic value of an aquarium emergency degrades within hours. A keeper who performs a full reset before recording anything has usually eliminated any possibility of identifying the cause, guaranteeing a repeat.

16.7 The specific danger of the “reset everything” response

Doing all of the following at once — 90% water change, all new media, deep gravel vacuum, three medications, raised temperature, added salt — produces: - Loss of biofilter capacity at the moment ammonia handling matters most. - Multiple simultaneous chemistry changes to already-compromised fish. - Oxygen demand from medications, at elevated temperature. - Complete destruction of the causal record. - Inability to attribute any subsequent improvement or deterioration to any specific action.

If more than one intervention is required, separate them by hours and observe between them.

16.8 On medications

[ESTABLISHED — a strong consensus point across veterinary sources]

- Medications are **not** a general-purpose response to sick fish. They are targeted treatments for identified conditions.
- Merck warns explicitly that rapidly changing antibiotics or “shotgunning” fish with many different drugs is dangerous and can create resistant bacterial strains, and that laboratory testing is needed to determine which antibiotics will be effective [MERCK-OWNER].
- Costs of speculative medication: biofilter suppression (12.6), oxygen demand (4.4), invertebrate mortality, plant damage, resistance development, delayed correct diagnosis, and additional stress on already-compromised fish.
- **Water quality correction and supportive care resolve a substantial fraction of presentations without any medication.**
- Where a medication is used, the correct approach is: identify the target, treat in a hospital tank where practical, complete the full course, remove carbon during and add it after, provide extra aeration, and do not stack agents.

16.9 When to seek veterinary or aquatic-animal-health help

[RECOMMENDED]

- Repeated unexplained mortality events.
- High-value fish or a large collection.
- Suspected mycobacteriosis (which is also a human health concern).
- Any presentation where microscopy would be decisive — which is most parasitic and bacterial presentations.
- When antibiotic selection matters (culture and sensitivity testing).
- **Aquatic veterinarians exist**, and the profession has a recognized specialty. The World Aquatic Veterinary Medical Association and the American Association of Fish Veterinarians maintain practitioner directories. State agricultural extension services and university fish diagnostic laboratories (e.g., UF/IFAS Tropical Aquaculture Laboratory) accept diagnostic submissions in many regions.
- **A skin scrape and gill clip examined under a microscope by someone competent is the single highest-yield diagnostic step available**, and it is inaccessible to the overwhelming majority of hobbyists.

SECTION 17. MYTHS AND CONTROVERSIES

Classification key: Strongly supported / Generally supported but context-dependent / Plausible but poorly documented / Primarily anecdotal / Misleading / Unsupported (myth).

17.1 “Never change more than 25% of aquarium water.”

Classification: MISLEADING.

The rule captures a genuine risk in the wrong variable. The hazard is the **magnitude of the chemistry difference** between old and new water, not the percentage replaced (2.10). In a well-maintained tank where tank water and treated tap water are nearly identical, a 90% change moves almost nothing. In a neglected tank with a large chemistry gap, even a 40% change can be a large step.

The rule's origin appears to be defensive: it protects a beginner who may not dechlorinate correctly, may not match temperature, and may have a neglected tank — by limiting the size of any error. As a safety margin for the inexperienced it is reasonable; as a statement about aquarium biology it is not.

Better formulation: *the safe size of a water change is determined by how similar the new water is to the old water, and by whether the new water is properly conditioned and temperature-matched.*

17.2 “Large water changes kill beneficial bacteria.”

Classification: UNSUPPORTED (myth).

Nitrifying organisms live in **biofilm attached to surfaces** — filter media, substrate, glass, decor, plants. The free-floating population in the water column is a small fraction of the total. Removing 50%, 75%, or 90% of the water removes very little of the biofilter.

The kernel of truth: if the replacement water contains chlorine or chloramine, that is harmful to bacteria (and much more harmful to fish). The mechanism is the disinfectant, not the volume of water.

Secondary kernel: a very large water change temporarily reduces the ammonia concentration available to the nitrifiers, which over long periods could reduce population size if the fish load also fell. This is not a “crash” and is not relevant on the timescale of a single water change.

17.3 “Never rinse filter media in tap water.”

Classification: DISPUTED / probably overstated — see 3.5.1.

Traceable evidence suggests that disinfection is a concentration × contact-time process and that a brief rinse at typical tap residuals is unlikely to meaningfully harm an established biofilm, with Cunliffe (1991) cited for chloramine requiring roughly ninety minutes to kill even small quantities of nitrifying bacteria at typical tap concentrations [AQSCI-TAPRINSE]. The dramatic figures circulating in consumer articles (70% kill in 30 seconds, irreversible destruction in 10 seconds) could not be traced to any primary source and should be classified **UNSUPPORTED**.

However: the conservative practice costs nothing, and the risky variants (extended soaking, hot water, soap, bleach, vinegar) are genuinely harmful. The advice is therefore harmless even if the stated mechanism is overstated.

17.4 “Perfect parameters mean the fish cannot be dying from the aquarium.”

Classification: UNSUPPORTED (myth) — and it is the central error this document exists to address.

Five measured parameters do not describe an aquarium. Not measured: dissolved oxygen, CO₂, chlorine, chloramine, metals, pesticides, surfactants, organics, pathogens, gas saturation, flow, gradients, or the fish’s nutritional and social condition (1.3). Further, a test describes the moment it was taken and not the conditions that killed a fish six hours earlier (1.8).

17.5 “A pH swing automatically kills fish.”

Classification: MISLEADING.

See 2.8 and 15.4. The evidence for magnitude-driven pH shock as an independent killer is weak, and there is old field evidence of fish tolerating very large pH transfers in both directions without mortality [AQSCI-SHIFTS citing Wiebe 1931]. pH matters enormously as the controller of ammonia and metal toxicity, as an indicator of buffer depletion, and at physiological extremes. It is much less potent as a stand-alone stressor at moderate magnitudes than the hobby believes.

17.6 “Old tank water is safer than fresh water.”

Classification: UNSUPPORTED as a general claim; contains a narrow kernel of truth.

The general claim is backwards: old water accumulates nitrate, dissolved organics, and possibly growth-inhibiting substances, and loses buffering capacity.

The narrow kernel: for a fish that has been living in that specific water, the *familiar* chemistry is less of an acute transition than a large volume of very different water. This is the Old Tank Syndrome mechanism (2.11), and it justifies *gradual* correction rather than continued neglect. It does not make old water good.

Related genuine practice: using removed tank water to rinse filter media is reasonable — not because the water is beneficial but because it is chemistry-matched and disinfectant-free.

17.7 “Water changes can crash your cycle.”

Classification: UNSUPPORTED (myth), with one specific exception.

See 17.2. The exception is a water change performed with **unconditioned chlorinated or chloraminated water** at a large volume — which is a disinfectant exposure, not a water-change effect.

A second, softer version has some validity: **a water change during the initial fishless cycling process can slow the cycle** by removing the ammonia the bacteria are growing on. This is a delay, not a crash, and is a different situation entirely.

17.8 “If one fish is healthy, the water must be safe.”

Classification: UNSUPPORTED (myth).

Species, size, age, and prior condition all modify vulnerability. Specific well-established asymmetries: - Labyrinth fish (bettas, gouramis) breathe atmospheric air and can appear unaffected in water with dangerously low dissolved oxygen. - Scaleless fish are far more sensitive to medications, salt, and irritants than scaled fish. - Invertebrates are far more sensitive to copper and insecticides than fish. - Large fish are affected by hypoxia before small fish; small fish are affected by many toxicants before large fish. - A fish adapted over months to a condition tolerates it while a newcomer does not (the OTS pattern).

17.9 “If ammonia and nitrite are zero, the tank is healthy.”

Classification: UNSUPPORTED (myth).

Identical in structure to 17.4, and worth stating separately because it is the most compressed version of the error. Zero ammonia and zero nitrite establish that nitrification is currently keeping pace with the current load. They establish nothing about oxygen, contaminants, pathogens, nutrition, temperature suitability, social conditions, or what the values were six hours ago.

17.10 “TDS shock kills fish.”

Classification: GENERALLY SUPPORTED BUT CONTEXT-DEPENDENT, and frequently over-applied.

- **Well supported** for invertebrates, especially Caridina shrimp, where TDS/GH changes affect molting directly.
- **Well supported** for very large ionic transitions — hard water to unrem mineralized RO, or the reverse.
- **Weakly supported** for routine water changes with the keeper’s own consistent tap water in robust community fish.
- **Confounded** by the fact that the situations where “TDS shock” is invoked (a neglected tank receiving a big change) usually also involve a pH rise that mobilizes un-ionized ammonia — a far better-documented killer.

Additional limitation: TDS is a proxy for ionic strength that cannot identify composition (13.6), so “TDS shock” as a diagnosis is inherently imprecise.

17.11 “Never do a water change during an emergency.”

Classification: MISLEADING.

When the emergency is a dissolved toxicant — ammonia, nitrite, a contaminant, a medication overdose — a water change is the correct and often the only effective intervention. Withholding it can be fatal.

The defensible version: do not perform a *large, unmatched, unconditioned* change into an unknown situation, and specifically be cautious in the neglected-tank/low-pH/ammonia-present scenario (16.4). The correct emphasis is on *how*, not *whether*.

17.12 Additional claims classified

Claim	Classification	Note
“Hydrogen sulfide from deep substrate kills fish”	Primarily myth as usually stated	See 3.4; specific stopped-filter scenarios excepted
“Deep sand beds perform anoxic denitrification in aquaria”	Disputed / probably unsupported	Nitrate reduction observed is more likely assimilatory uptake by plants, bacteria, and algae [AQSCI-ANAEROBIC]
“Letting tap water stand overnight removes chlorine”	Generally supported for free chlorine; UNSUPPORTED for chloramine	Utilities state chloramine does not dissipate by standing [ACWD-CHLORAMINE]
“Ammonia-binding conditioners detoxify ammonia”	Disputed	Manufacturers assert it; critics argue the released ammonia from chloramine is trivially small at normal pH and that the claims are marketing [AQSCI-CHLORAMINE] vs. [FRITZ-CHLORAMINE]
“Nitrate above 40 ppm is dangerous to fish”	Disputed — see 17.13	
“Aquarium salt cures most fish diseases”	Misleading	Salt has genuine but limited uses; it is not broadly antimicrobial at tolerable concentrations and harms many species and plants
“Bettas can live in a bowl because they breathe air”	Unsupported	Air-breathing addresses oxygen, not ammonia, temperature, or space
“Fish grow to the size of their tank”	Misleading	Growth is stunted by poor conditions; internal organs continue growing, producing deformity and shortened life
“You must remove chlorine before adding water or bacteria die instantly”	Misleading	The urgent reason to dechlorinate is the fish, not the bacteria (3.5.1)
“Cloudy water means the tank is dirty”	Misleading	Bacterial blooms indicate a dissolved organic input event; the oxygen consumption is the real concern
“Surface agitation drives CO2 into the tank”	Unsupported — reversed	Agitation moves dissolved gases toward atmospheric equilibrium; in a CO2-rich tank that means CO2 leaves
“You can cycle a tank in 24 hours with a bottled bacteria product”	Disputed	Some products demonstrably seed nitrifiers; instant full capacity for a full stocking load is not supported
“A UV sterilizer prevents disease”	Misleading	UV kills free-floating organisms passing through it; it does not affect parasites on fish, in substrate, or in biofilm
“Activated carbon causes lateral line erosion (HLE)”	Disputed / poorly documented	Proposed in the marine hobby; evidence is limited and confounded with nutrition and water quality

"Snails cause ammonia spikes / are pests that harm fish"	Misleading	Snail populations track available food; a bloom indicates overfeeding. A dead large snail is a genuine ammonia source
"Plants oxygenate the tank at night"	Unsupported — reversed	Plants respire continuously and photosynthesize only under light; the pre-dawn hours are the daily oxygen minimum
"Airstones oxygenate water via the bubbles themselves"	Misleading	Most oxygen transfer from an airstone occurs through the surface disturbance the rising bubbles create, not through the bubble walls

17.13 The nitrate controversy — a fully worked example of a genuine disagreement

Classification: DISPUTED. This case is included in full because it illustrates how a widely repeated hobby number relates to the underlying literature.

The hobby standard: keep nitrate below 20–40 ppm; above 40 ppm causes chronic stress and immune suppression; above 80 ppm risks organ damage.

What environmental toxicology says. Camargo, Alonso & Salamanca (2005), a widely cited review in *Chemosphere*, concluded that **a maximum of 2 mg NO₃-N/L would be appropriate for protecting the most sensitive freshwater species**, and that 10 mg NO₃-N/L (the US federal drinking-water maximum) can adversely affect, during long-term exposures, certain freshwater invertebrates, fishes (rainbow trout, Chinook salmon, cutthroat trout), and amphibians [CAMARGO2005]. In NO₃- units, 2 mg NO₃-N/L is roughly **9 ppm nitrate** — far below the hobby's target.

What aquaculture and ornamental-relevant work says. UF/IFAS's ammonia publication states nitrate is considered harmless to fish in natural systems and ponds, but in closed systems with little water exchange it accumulates and **may be harmful above 250 mg/L** [THEFISHSITE-AMMONIA, summarizing UF/IFAS]. A study of post-smolt Atlantic salmon in freshwater recirculating systems evaluated chronic nitrate effects at levels far above hobby thresholds [DAVIDSON2017]. A technical hobby review compiling the LC50 literature reports an average of sixteen freshwater LC50 values of approximately **4,918 ppm nitrate (1,117 ppm NO₃-N)**, with the lowest observed being an LC50 of 191 mg NO₃-N/L (~840 ppm nitrate) for guppy fry, and cites studies finding no long-term effects on adult fish at 440 ppm nitrate [AQSCI-NITRATE].

Where the disagreement actually lies:

- Species and life stage.** The protective guidelines are driven by highly sensitive salmonids, amphibians, and stream invertebrates — not by tropical ornamental fish. Juveniles and fry are far more sensitive than adults; the aquarium literature is largely about adults.
- Endpoint definition.** Environmental guidelines protect against any measurable *effect* (including sublethal, reproductive, and community-level effects) in wild ecosystems. Aquarium guidance is concerned with health and longevity of individual captive animals. As one technical reviewer notes, "effect" is not synonymous with "harmful" [AQSCI-NITRATE].
- Regulatory conservatism.** Environmental guideline-setting deliberately applies safety factors and precautionary framing. A technical hobby review criticizes some of these documents as reflecting a policy bias toward maximal environmental protection [AQSCI-NITRATE] — a criticism that is itself a value judgment rather than a scientific finding, and should be flagged as such.
- Confounding in aquarium practice.** Nitrate is a **marker** for accumulated nitrification, accumulated dissolved organics, accumulated TDS, and accumulated everything else. A tank at 160 ppm nitrate is not simply a tank with nitrate; it is a tank that has not been water-changed. **Attributing observed harm specifically to nitrate rather than to what nitrate co-varies with is not supported by the evidence.**
- Known physiological mechanisms.** Reviews document methemoglobinemia, decreased immune response and osmoregulatory function, and endocrine and reproductive effects from nitrate exposure in various species [DAVIDSON2017, citing Camargo et al. 2005; Hrubec et al. 1996/1997; Guillet & Edwards 2005]. The mechanisms are real; the concentration at which they matter for a captive-bred adult ornamental fish is what is in dispute.

Defensible synthesis: - The commonly repeated "40 ppm is the danger line" is **not derived from a specific study of ornamental fish** and should not be presented as an established threshold. - Acute nitrate toxicity to adult fish requires concentrations far above anything a maintained aquarium reaches. - Chronic effects at moderate concentrations are plausible and are supported for some species and life stages, but the ornamental-specific evidence is thin. - **Low nitrate remains a good husbandry target** — not primarily because nitrate itself is dangerous at 40 ppm, but because keeping nitrate low requires the water changes that also remove dissolved organics, replenish minerals and buffering, and dilute unmeasured contaminants. **The water change is the intervention; nitrate is the convenient indicator.**

This reframing is more defensible than either extreme position and preserves the practical recommendation without misrepresenting the evidence.

SECTION 18. SCENARIO LIBRARY

18.0 How to use this section

Each scenario provides: observable symptoms, timeline, recent events, most plausible possibilities, less obvious possibilities, differentiating tests and questions, and what evidence would support or weaken each hypothesis.

No scenario has a single guaranteed answer. Several are deliberately constructed with multiple simultaneous contributing factors, because that is how real aquarium problems usually present. Where a scenario has a “most likely” answer, it is stated as a probability, not a conclusion.

SCENARIO 1 — Gasping five minutes after a 70% water change

Symptoms. All fish at the surface within minutes, gaping, rapid opercular movement. Two small tetras dead within twenty minutes. Larger fish still alive but distressed.

Timeline. Distress began during the refill.

Recent events. 70% water change using a python-style hose directly from the tap. Conditioner added to the tank before refilling, dosed for the tank volume. Temperature “felt about right.”

Most plausible: 1. **Chlorine/chloramine exposure.** Conditioner added to the tank but new water introduced faster than mixing, exposing fish near the inflow to essentially undiluted tap water. Small fish affected first, consistent with gill surface area to body mass. **Strongest hypothesis.** 2. **Temperature shock.** “Felt about right” is not a measurement; a 10–15°F delta is common with hand-mixed tap water. 3. **Gas supersaturation.** Cold water heated in-line or by mixing, delivered under pressure and released to atmosphere.

Less obvious: 4. **Insufficient conditioner** — dosed for the tank but the municipality is in a free-chlorine conversion period with elevated residual (2.7.1). 5. **Contaminated hose** — the python was left with stagnant water in it for weeks; or the outdoor spigot was used and the hose had garden chemistry in it. 6. **pH/ammonia interaction** if the tank had been neglected (2.11). 7. **Plumbing work or a main break** in the neighborhood mobilizing metals.

Differentiating tests and questions: - What is the temperature of the tank now, and what was the tap water temperature? (Direct measurement; a delta of 10°F+ supports #2.) - Does the municipality use chlorine or chloramine, and is there a current notice? (Supports #1/#4.) - Is there a fine bubble coating on the glass and on the fish? (Supports #3.) - Test the tap water for chlorine, ammonia, and pH. - Has this exact procedure been done many times before without incident? (If yes, weakens #1 and #2 and strengthens “something changed” — #4, #5, #7.) - Test tank ammonia now and pH before/after (if pre-change values exist).

Supporting/weakening evidence: - Reddened gills → supports chlorine. - Bubbles on fish and glass → supports supersaturation. - Positive ammonia → shifts toward #6 or chloramine residual. - Fish improve dramatically within 30 minutes of adding an air stone → supports an oxygen component, possibly supersaturation resolution or simply improved gas exchange.

Immediate action regardless of hypothesis: add vigorous aeration; add a full dose of conditioner; do not perform another large change with the same water until the tap water is tested.

SCENARIO 2 — Everything perfect for eight months, then all fish dead one morning

Symptoms. Six fish dead on the substrate, one alive and lying on its side. No lesions. Water clear.

Timeline. Fine at 10pm; dead at 7am.

Recent events. None reported. Routine 25% water change three days earlier.

Most plausible: 1. **Heater failure, stuck on.** Check temperature immediately. A small tank can reach lethal temperature in hours. 2. **Equipment stopped.** Filter or air pump failed or was unplugged; overnight oxygen depletion. 3. **Power interruption** overnight that reset or failed to restart equipment. 4. **Airborne contaminant** — evening use of an aerosol, plug-in repellent, cleaner, or paint in the room.

Less obvious: 5. **CO2 solenoid failure** in an injected tank, or a CO2 timer failure. 6. **Dead animal decomposing** since the water change, producing an overnight ammonia and oxygen event. 7. **Electrical fault** leaking current into the water from a failing heater or pump. 8. **A pest control visit** or a neighbor’s treatment in a shared building. 9. **Something introduced during the water change three days ago** that has been building — a bacterial bloom consuming oxygen, or organics released by substrate disturbance.

Differentiating tests and questions: - **Temperature right now**, before touching anything. - Is every device running? Is the surface moving? - Does the room smell of anything? Was anything sprayed, cleaned, painted, or plugged in yesterday evening? - Ammonia, nitrite, pH now. - Is the water cloudy? (Supports #9.) - Any missing livestock? (Supports #6.) - Was there a power blip? (Check clocks, the filter’s state.)

Supporting/weakening evidence: - Water at 95°F → #1, essentially confirmed. - Water at normal temperature, filter running, water clear, all parameters zero → shifts strongly toward #4/#7/#8, none of which is measurable. - Cloudy water with a slight smell → #9 and #6. - Survivor recovers rapidly in a bucket of aged, aerated water → supports a water-borne cause.

Note: this scenario is deliberately constructed to have no measurable answer for several of its most likely causes. That is realistic and important.

SCENARIO 3 — New fish added; five days later the established fish start dying

Symptoms. Two established rasboras dead. Others flashing against decor. The three new fish look fine.

Timeline. New fish added Sunday. First flashing Wednesday. First death Friday.

Recent events. Three fish purchased from a chain store the day a shipment arrived. Bag water was poured into the tank. No quarantine.

Most plausible: 1. **Introduced ectoparasite** — Ich, velvet, or monogeneans, with the new fish acting as resistant carriers. Flashing is a strong ectoparasite indicator. **Strongest hypothesis.** 2. **Introduced bacterial pathogen** with the established fish naive. 3. **Bioload increase** producing a small ammonia excursion — but this would not explain flashing, and would generally not spare the new fish.

Less obvious: 4. **A chemical carried in the bag water** — copper or another medication from the store's system, poured directly into the tank. Would affect invertebrates first if present. 5. **Stress-triggered flare of a latent pathogen** already present in the established population, triggered by the disturbance of adding new fish. 6. **Aggression** from the new fish.

Differentiating tests and questions: - Examine the fish closely under a bright light at an angle: discrete white dots (Ich), fine gold dust (velvet), or nothing visible (flukes/gill parasites)? - Are any invertebrates present, and are they affected? (Affected invertebrates → #4.) - Ammonia and nitrite now. - Are the new fish genuinely unaffected, or merely not yet symptomatic? - Which species is dying — is it one species or across species?

Supporting/weakening evidence: - Visible white spots → Ich; treat and note the timeline fits a 3–7 day incubation exactly. - No visible spots but heavy flashing and rapid gill movement → gill flukes or gill-only Ich; requires microscopy for confidence. - Invertebrates dead → shifts to #4. - Ammonia positive → #3 is contributing, though probably not alone.

Key learning point: this is the canonical argument for quarantine (5.12) and for not adding bag water (6.10).

SCENARIO 4 — Neglected tank receives its first water change in eight months

Symptoms. Within two hours of an 80% water change, fish are lethargic, gasping, and hanging near the bottom. Two dead by evening.

Timeline. Change performed at noon.

Recent events. Tank not water-changed for eight months; topped off with tap water throughout. Conditioner used correctly. Temperature matched.

Most plausible: 1. **pH-driven ammonia conversion.** The tank had accumulated ammonia at low pH (as harmless ammonium); the fresh higher-pH water converted a large fraction to toxic NH₃ essentially instantly. **Strongest hypothesis and the most dangerous one, because the pre-change ammonia was not harming the fish.** 2. **Large ionic/osmotic transition** from high-TDS old water to source water (2.9, 2.13). 3. **Substrate disturbance** during the change, releasing organics and creating oxygen demand.

Less obvious: 4. **KH restoration causing a rapid pH rise** with the same ammonia consequence. 5. **Loss of accumulated dissolved organics** that had been complexing metals. 6. **Fish that were already in poor condition** and were pushed over by any disturbance.

Differentiating tests and questions: - **Test ammonia now.** A positive result with a pH now well above the old pH essentially confirms #1. - What was the pH before, and what is it now? (If pre-change data does not exist, test the tap water and the substrate interstitial water.) - TDS before and after, if a meter is available. - Was the substrate vacuumed, and how deeply? - Was the water cloudy afterward? (Supports #3.)

Supporting/weakening evidence: - Ammonia positive and pH risen by 1+ unit → #1. - Ammonia zero, pH change modest, water cloudy → #3. - Ammonia zero, pH stable, TDS dropped by 300+ ppm → #2.

Correct handling had it been anticipated: test ammonia first; if present, use serial 15–20% changes over 1–3 weeks (2.11).

SCENARIO 5 — Shrimp all die; fish are completely unaffected

Symptoms. All twelve Neocaridina shrimp dead over 48 hours. Snails also dead or sealed shut. Eight fish behaving entirely normally.

Timeline. First shrimp deaths noticed a day after a routine water change and the addition of new plants.

Recent events. Water change; three new plants added from a big-box store; the dog received a flea treatment two days prior.

Most plausible: 1. **Insecticide on the new plants.** Systemic insecticides used at plant nurseries are a documented and specific hazard to invertebrates, with fish unaffected (8.6.3). **Strongest hypothesis.** 2. **Copper exposure** — from a fertilizer, a medication, plumbing, or the new plants' packaging. 3. **Fipronil or permethrin transfer** from handling the treated dog and then entering the tank.

Less obvious: 4. **A GH crash** — but this would be visible on a GH test and would produce failed molts rather than rapid mass death. 5. **A rapid TDS swing** from the water change. 6. **Ammonia or nitrite excursion** — invertebrates are more sensitive, but a level that killed all shrimp would usually visibly affect fish. 7. **Temperature spike** — shrimp are less heat-tolerant than most community fish.

Differentiating tests and questions: - Where did the plants come from, and were they labeled shrimp-safe? - Timeline: did deaths begin before or after the plants went in? - Was the flea treatment handled without washing hands before tank work? - GH, KH, TDS now versus the usual values. - Ammonia, nitrite. - Temperature. - Copper test if available.

Supporting/weakening evidence: - Deaths beginning within hours of plant addition → #1 strongly. - Snails also affected → consistent with #1, #2, #3 (all invertebrate-selective). - Fish entirely normal → strongly argues **against** ammonia, nitrite, oxygen, and temperature as primary causes, and **for** an invertebrate-selective toxicant.

Action: remove the plants; run heavy fresh activated carbon; perform serial 50% water changes; do not restock invertebrates for several weeks.

SCENARIO 6 — One fish dies every three weeks for six months; parameters always perfect

Symptoms. Individual fish gradually lose weight, become reclusive, sometimes develop a slight spinal curve, then die. No consistent external lesions.

Timeline. Ongoing for six months. No pattern relative to maintenance.

Recent events. None.

Most plausible: 1. **Mycobacteriosis.** Chronic, progressive, emaciation with occasional spinal deformity, low-level ongoing mortality, unresponsive to any intervention, normal water parameters throughout. **This presentation is close to textbook.** 2. **Internal parasites** (nematodes, *Camallanus*, flagellates) — wasting with a normal appetite. 3. **Nutritional deficiency** — a single food type fed for a long period, or old food.

Less obvious: 4. **Chronic starvation from food competition** affecting specific individuals (10.5). 5. **Chronic parameter mismatch** — a temperature or hardness that suits most of the tank but not this species. 6. **An age cohort** — all the fish were purchased together and are reaching end of life together (6.9). 7. **Chronic low-grade aggression** producing a rotating victim. 8. **A slow contaminant leach** — an ornament, a rock, or degrading equipment.

Differentiating tests and questions: - Is it always the same species, or across species? (Same species → #1, #2, #5, #6; across species → #3, #5, #8.) - View the fish from above: are they thin? - Are the fish eating? (Wasting despite eating → #2 strongly.) - White stringy feces? (→ #2.) - How old are the fish and when were they purchased? (→ #6.) - What exactly is fed, how often, and how old is the container? (→ #3.) - Is there any observed chasing, especially after lights-out? (→ #7.) - Would veterinary histopathology on the next fatality be feasible? (This is the only way to confirm #1.)

Supporting/weakening evidence: - Spinal curvature plus emaciation plus chronic course → #1. - Response to a course of levamisole or fenbendazole → #2. - Improvement after diversifying the diet → #3. - All losses in the same size/age cohort → #6.

Important note: #1 is not treatable, is potentially zoonotic (*M. marinum*), and warrants gloves during maintenance.

SCENARIO 7 — Fish fine for three days after the water change, then start dying

Symptoms. Progressive; fish develop white patches on the body and around the mouth. Two dead by day four.

Timeline. Water change Saturday. Nothing unusual Sunday–Monday. Patches Tuesday. Deaths Wednesday.

Recent events. Water change including a full filter clean; the fish were netted into a bucket during the change because the keeper wanted to move a large rock.

Most plausible: 1. **Columnaris following handling injury.** Netting damages the mucus layer and epithelium; *Flavobacterium* is an opportunist present in most systems; acute columnaris can kill in 24–72 hours after establishing. The 3-day delay fits precisely.

Strongest hypothesis. 2. **Saprolegnia (true fungus) on handling wounds** — the differential to #1. 3. **Ammonia excursion from the filter clean**, with the white patches secondary. Would need a positive ammonia test.

Less obvious: 4. **Ich** — but Ich presents as discrete dots, not patches. 5. **Epistylis** — associated with wounds and high organic load; fuzzier and more tufted than columnaris. 6. **A pre-existing infection** unmasked by stress.

Differentiating tests and questions: - Are the patches flat and grey-white with a saddleback or mouth distribution (columnaris), or three-dimensional and cottony at discrete wound sites (fungus), or fine and tufted at fin rays (Epistylis)? - Ammonia and nitrite now. - Water temperature? (Higher temperature strongly favors columnaris [FHS-COLUMNARIS].) - How were the fish handled, and were any visibly damaged? - Are the deaths rapid (columnaris) or slow (fungus)?

Supporting/weakening evidence: - Rapid progression at warm temperature → #1. - Localized cottony tufts at obvious wounds, slower course → #2. - Positive ammonia → #3 contributing.

Key learning point: this is the archetypal example of Section 3's thesis — the water change is blamed, but the causal action was the **netting**, and the fatal agent is a bacterium.

SCENARIO 8 — Fish gasping every morning, normal by afternoon

Symptoms. Fish at the surface at 6am; behaving normally by noon. Repeats daily. No deaths yet.

Timeline. Ongoing for two weeks.

Recent events. New LED light installed a month ago; plants growing well.

Most plausible: 1. **Overnight oxygen minimum in a planted tank.** Plants respire all night; the pre-dawn hours are the daily oxygen low. Improved plant growth from the new light has increased both the plant biomass and the nightly respiratory demand. **Strongest hypothesis, and this presentation is close to diagnostic.** 2. **Overnight CO₂ accumulation** driving a pH drop and impairing respiration — related, and often concurrent with #1. 3. **CO₂ injection running overnight** if present.

Less obvious: 4. **A heater cycling** and driving a nightly temperature excursion. 5. **Surface film** reducing gas exchange, most limiting when demand peaks. 6. **A timer-controlled device** turning off overnight. 7. **Fish being fed heavily in the evening**, increasing overnight oxygen demand from both fish and decomposition.

Differentiating tests and questions: - Test pH at 6am and again at 6pm. A significant diurnal swing supports #1/#2. - Is there a

surface film? (→ #5.) - Does adding an air stone on a night timer eliminate the symptom? **This is both the diagnostic test and the fix.** - Is CO2 injected, and does the solenoid shut off with the lights? (→ #3.) - Temperature logged overnight? (→ #4.)

Supporting/weakening evidence: - Symptom disappears with overnight aeration → #1/#2 confirmed practically. - pH at 6am is 0.4–1.0 units below the 6pm value → confirms nightly CO2 accumulation. - Largest fish affected first → consistent with hypoxia (4.6).

SCENARIO 9 — Betta in a 5-gallon tank; fine for months; dies two days after a filter cartridge change

Symptoms. Lethargy, clamped fins, resting on the bottom, then death. Ammonia at 1.0 ppm on testing.

Timeline. Cartridge replaced Sunday; symptoms Monday; death Tuesday.

Recent events. Cartridge replaced per the manufacturer's monthly schedule; the old cartridge was thrown away.

Most plausible: 1. **Biofilter loss from wholesale media replacement** in a small tank with limited alternative biofilm surface.

Strongest hypothesis and essentially confirmed by the ammonia reading.

Contributing factors: 2. **Small volume** — 5 gallons has minimal buffering against any load change. 3. **pH** — if the tank runs at pH 8.0+, 1.0 ppm TAN represents a substantially higher un-ionized ammonia fraction than at pH 7.0 (1.2). Check pH before concluding the ammonia level was “only” 1.0. 4. **Temperature** — bettas require 78–82°F; a cool tank compounds the stress.

Less obvious: 5. **A contaminant on the new cartridge** — some cartridges contain carbon and other additives; rinsing new media before installation is standard practice. 6. **The old cartridge was the majority of the biofilm**, meaning the “cycle” was effectively contained in a disposable part.

Differentiating tests and questions: - What is the pH? (Determines how toxic 1.0 ppm TAN actually was.) - Was the old cartridge kept? (Reinstalling it alongside the new one would have preserved the biofilter.) - Temperature? - Was the new cartridge rinsed?

Key learning point: cartridge-based filtration design creates a scheduled biofilter destruction event. The standard mitigation is to run two cartridges and replace only one at a time, or to replace cartridges with permanent sponge/ceramic media.

SCENARIO 10 — Multiple simultaneous factors: the “spring cleaning” event

Symptoms. Cloudy water within 12 hours; fish lethargic at 24 hours; three deaths at 48 hours; two more over the following week with white edges on the fins.

Timeline. Major maintenance session Saturday morning.

Recent events. All of the following on the same day: 60% water change; deep gravel vacuum of the entire substrate for the first time in a year; all filter media rinsed thoroughly under the tap; carbon replaced; two large rocks removed and scrubbed; a large sword plant uprooted and divided; the fish were netted out during the rock work; the heater was unplugged and left unplugged.

This scenario has no single cause. It has at least six.

Contributing factors, each independently plausible: 1. **Deep substrate vacuum** → large organic release → oxygen demand and bacterial bloom (explains the cloudiness at 12 hours). 2. **All media cleaned at once** → reduced nitrification capacity. 3. **Uprooted plant** → further substrate disturbance and decomposing root mass. 4. **Netting the fish** → mucus layer damage → columnaris/fungus (explains the white fin edges at day 5–7). 5. **Heater left unplugged** → declining temperature → immune suppression and stress. 6. **60% water change** → chemistry step-change, possibly minor. 7. **Bacterial bloom** → additional oxygen consumption.

Differentiating tests and questions: - Temperature now. (#5 is trivially checkable and often the largest single factor.) - Ammonia, nitrite. (#2 and #1.) - Is the water still cloudy? (#7.) - Which fish are showing fin lesions — the ones that were netted? (#4.) - Was the substrate ever vacuumed before? (#1's magnitude scales with the interval.)

The instructive point: attempting to identify “the” cause here is the wrong analysis. Each factor contributed. **The corrective lesson is procedural, not diagnostic: separate maintenance actions in time.**

SCENARIO 11 — Only the Corydoras die; everything else thrives

Symptoms. Six *Corydoras* declining over two months; three dead. Barbels eroded. The tetras, gouramis, and plants are all doing well.

Timeline. Slow, over two months.

Recent events. Tank set up four months ago with a new sharp-edged gravel; keeper raised the temperature to 82°F for the gouramis.

Most plausible: 1. **Barbel erosion from sharp substrate**, leading to bacterial infection. *Corydoras* forage by probing the substrate with their barbels. **Strongly supported by the specific finding of eroded barbels.** 2. **Chronic temperature stress** — many common *Corydoras* species are stressed above ~78°F (9.10). 3. **Starvation** — *Corydoras* outcompeted by faster mid-water feeders (10.5, 11.2).

Less obvious: 4. **Insufficient group size** if fewer than 6 remain, reducing foraging behavior. 5. **Sensitivity to a medication or salt** used previously — *Corydoras* are among the most medication-sensitive common aquarium fish. 6. **Organic accumulation in the substrate** affecting a species that lives at the substrate interface. 7. **Nitrate or hardness levels** tolerated by the other species.

Differentiating tests and questions: - Substrate: is it sharp-edged gravel or smooth sand? Run it between the fingers. - Temperature, and which species is it set for? - Is sinking food specifically provided, and do the *Corydoras* actually get it? - Are the barbels eroded, and were they on arrival? - How many *Corydoras* remain? - Any medications used?

Supporting/weakening evidence: - Eroded barbels plus sharp gravel → #1. - Temperature at 82°F with a species that prefers 74–78°F → #2 contributing. - Thin bodies viewed from above → #3.

Key learning point: an incompatible stocking combination (gouramis wanting warmth, *Corydoras* wanting cooler) with mismatched substrate produces species-selective attrition with permanently perfect water parameters. This is a **Section 9 and Section 10 problem masquerading as a Section 1 mystery**.

SCENARIO 12 — The tank is fine; the fish die within a week of every purchase

Symptoms. Each newly purchased fish declines and dies within 3–7 days. The existing four fish have been healthy for two years.

Timeline. Repeated across four separate purchases over six months.

Recent events. Tank has not had a water change in over a year; topped off only.

Most plausible: 1. **Old Tank Syndrome pattern.** The residents are acclimated to the drifted chemistry; newcomers coming from very different store water are not (2.12). **This presentation is the classic OTS signature.** 2. **Nitrate and TDS accumulation** producing a large gap between store water and tank water. 3. **A persistent pathogen** in the tank to which the residents are immune carriers and newcomers are naive (14.8, point 6).

Less obvious: 4. **Aggression** from established residents against every newcomer. 5. **The source** — all fish came from the same store with the same underlying problem. 6. **Acclimation method** — no acclimation at all, or excessive drip acclimation from long bag times. 7. **Insufficient cover** for a newcomer to establish.

Differentiating tests and questions: - **Test pH, KH, GH, TDS, and nitrate in the tank and compare directly to the tap water and, if possible, to the store's water.** A large gap on nitrate and TDS with a lower tank pH essentially confirms #1/#2. - Do the new fish show physical damage, or do they simply fade? (Damage → #4.) - Are the residents ever seen chasing? (Especially after lights-out.) - Have all fish come from the same store? (→ #5.) - Do the new fish show any parasite signs? (→ #3.)

Supporting/weakening evidence: - Nitrate 160+ ppm, KH near zero, pH 6.2, tap water pH 7.6 → #1 confirmed. - Correcting via serial small water changes over several weeks, then a successful introduction → confirms. - Nitrate and KH normal → shifts to #3, #4, #5.

SCENARIO 13 — Deaths every spring, for three consecutive years

Symptoms. Losses concentrated in a three-week window each March–April. Water always tests normal.

Most plausible: 1. **Annual municipal free chlorine conversion** (2.7.1). Many utilities run these in late winter or spring, lasting 30–60 days [EPA-R6-FCC]. 2. **Seasonal source switching** by the utility, changing hardness, alkalinity, and TDS. 3. **Spring runoff** increasing organic load and therefore disinfectant dosing. 4. **Annual hydrant flushing** mobilizing sediment and metals.

Less obvious: 5. **Seasonal household insecticide use** — spring pest control, ant treatments, lawn treatments. 6. **Room temperature swings** in a transitional season, with heating turned off before consistent warmth arrives. 7. **Spring cleaning** — aerosols, paint, carpet cleaning. 8. **Pet flea treatment** resuming for the season. 9. **Window opening** and pollen/outdoor air exposure — weak.

Differentiating tests and questions: - **Check the water utility's website and notices for a scheduled conversion or flushing program.** This is the single most informative action and takes minutes. - Test tap water chlorine, ammonia, pH, KH, GH, and TDS in March and compare to the same tests in September. - What household products are used seasonally, and when? - Is the tank near a window or an HVAC vent? - Log room temperature.

Key learning point: a seasonal pattern is one of the strongest signals available, and it points outside the aquarium. **Very few hobbyists ever contact their water utility.**

SCENARIO 14 — Fish gasping; ammonia 0, nitrite 0, nitrate 20, pH 7.4, temperature 78°F

Symptoms. All fish near the surface, breathing rapidly. Larger fish worst affected. Water clear.

Timeline. Onset over several hours; no recent maintenance.

Most plausible: 1. **Low dissolved oxygen.** Every measured parameter is normal, the presentation is respiratory, and the largest fish are worst affected — the signature of hypoxia (4.6, 4.7). **Strongest hypothesis.** 2. **Gill pathology** — parasites, bacterial gill disease, or gill damage from an earlier insult, producing functional hypoxia in oxygen-adequate water. 3. **Elevated dissolved CO₂** — impairs respiration independently of oxygen.

Less obvious: 4. **A contaminant with a respiratory mechanism** — chlorine (already dissipated), a surfactant, or a metal. 5. **A transient ammonia spike already resolved** (12.2), with the fish suffering the aftermath of gill damage. 6. **Temperature having recently been higher** and now recovered. 7. **Gas supersaturation.**

Differentiating tests and questions: - **Add vigorous aeration and observe for 15–30 minutes.** Rapid improvement → #1 or #3. This is the highest-value single test. - Is the surface moving? Is there a film? Is the filter flow normal? - Examine gills: brown (nitrite —

but nitrite is zero), bright red (irritant), pale (chronic), with visible mucus (parasites)? - Any flashing behavior? (→ #2.) - Is the tank overstocked? - Bubbles on surfaces? (→ #7.) - Was a large water change or gravel vacuum done in the last 48 hours? (→ oxygen demand.)

Key learning point: this is the canonical “perfect parameters, dying fish” scenario, and the answer is usually the parameter that was not measured.

SCENARIO 15 — Angelfish develops popeye and abnormal buoyancy after a large water change; three other fish die suddenly

Symptoms. Sudden mortality of three fish; survivors show swollen eyes and floating abnormally.

Recent events. 80% water change using a hose connected to a mixing tap, refilled quickly.

Most plausible: 1. **Gas bubble disease from supersaturation.** The presentation — acute mortality plus swollen eyes plus abnormal buoyancy in survivors — matches a documented veterinary account almost exactly [TOLEDO-GBD]. **Strongest hypothesis given this specific symptom cluster.** 2. **Bacterial septicemia** — popeye is also a classic septicemia sign, but the sudden simultaneous mortality argues against an infectious process. 3. **Chlorine exposure** — would explain the mortality but not the buoyancy and eye signs in survivors.

Less obvious: 4. **Temperature excursion.** 5. **Coincidence** — an unrelated septicemia that happened to coincide with the water change.

Differentiating tests and questions: - Were fine bubbles visible on the glass, decor, and fish after the refill? - Was the water drawn hot, or through a hose, or quickly under pressure? - Look for gas bubbles in the fins between the rays (“silver lines”), around the mouth, and under the skin — described as characteristic of acute supersaturation [TOLEDO-GBD]. - Is there any external lesion or reddening consistent with septicemia? - Do the survivors improve over 24–48 hours with the water at atmospheric equilibrium?

Supporting/weakening evidence: - Visible emboli in fins → #1 essentially confirmed. - Reddened, ulcerated, or hemorrhagic areas → shifts toward #2. - Improvement over 24–48 hours without intervention → #1 (supersaturation self-resolves).

Prevention: pre-mix and aerate replacement water rather than refilling directly from a pressurized line.

SCENARIO 16 — Everything dies after a home renovation

Symptoms. All fish and invertebrates dead over 24 hours. No maintenance performed. Water clear, all tested parameters normal.

Recent events. Interior painting in an adjacent room over two days; windows closed.

Most plausible: 1. **Airborne solvent/VOC exposure** partitioning into the water through the surface and being actively drawn in by the air pump (8.5). 2. **Aerosol or spray drift.** 3. **Dust and debris** — less likely to be lethal.

Less obvious: 4. **Concurrent pest control** performed as part of the renovation. 5. **Temperature excursion** from HVAC changes or open doors. 6. **Power interruption** during electrical work. 7. **Plumbing work** mobilizing metals if any water was added.

Differentiating tests and questions: - Was an air pump running, and where does it draw air from? (An air pump in the painted room is a very efficient contaminant delivery system.) - What products were used, and were they solvent-based or water-based? - Was anything else done during the renovation — spraying, caulking, sanding, electrical work? - Was the tank covered? - Temperature log? - Are invertebrates also dead? (Yes → consistent with a toxicant.)

Key learning point: none of this is measurable after the fact, and all standard parameters will read normally. **The timeline is the only evidence.** This scenario belongs in the document specifically to establish that a fully normal test panel is consistent with a lethal chemical exposure.

SCENARIO 17 — Fancy guppies dying at about one year, one after another

Symptoms. Fish become inactive, thin, sometimes with a slight spinal curve, and die. Water parameters excellent throughout. No lesions.

Most plausible: 1. **Normal lifespan.** Fancy guppies commonly live 1–2 years, and are typically sold at 3–6 months of age (6.9). **This may not be a problem at all.** 2. **Genetic fragility of commercial fancy strains** (6.8) — widely reported, poorly documented. 3. **Female exhaustion from continuous reproduction** if females and males are housed together.

Less obvious: 4. **Mycobacteriosis**, which is reported to be common in guppy lines. 5. **Internal parasites** (*Camallanus* is common in livebearers). 6. **Chronic soft water** — guppies do poorly in very soft, mineral-poor water. 7. **Insufficient female-to-male ratio** producing harassment stress (10.6).

Differentiating tests and questions: - How old were the fish when purchased, and how long have they been kept? - Are females producing broods continuously? - What is the GH? (Guppies prefer moderately hard water.) - Sex ratio? - Any red protrusions from the vent? (→ *Camallanus*.) - Is the spinal curve present in multiple individuals? (→ #4 or a vitamin C deficiency.)

Key learning point: “how old is this fish?” is a question almost never asked in troubleshooting, and for short-lived species it is frequently the answer.

SCENARIO 18 — Fish die two days after a medication was added; the original problem was never confirmed

Symptoms. Fish were flashing; a broad-spectrum antibiotic and an antiparasitic were added together. Two days later, multiple fish dead. Ammonia now reads 0.5 ppm.

Most plausible: 1. **Biofilter suppression by the antibiotic** producing an ammonia excursion (12.6). Supported by the ammonia reading. 2. **Oxygen depletion** from the medications’ oxygen demand plus the die-off of suppressed bacteria (4.4). 3. **Direct medication toxicity** from stacking two agents (8.7) — Merck’s warning against shotgunning applies [MERCK-OWNER].

Less obvious: 4. **The original condition progressing** untreated because the medications were not appropriate for it. 5. **Species sensitivity** — scaleless fish and invertebrates affected disproportionately. 6. **Carbon left in the filter** producing an unpredictable dose curve, or removed mid-course producing a spike. 7. **Overdose** from dosing by the tank’s nominal rating rather than actual volume.

Differentiating tests and questions: - Which medications, at what doses, and calculated against what volume? - Was carbon removed? Was it re-added? - Was aeration increased? - Which species died — scaleless first? - What was the original flashing actually caused by? (It was never determined — which is the root problem.)

Key learning point: this scenario illustrates the cost of speculative medication (16.8). The fish may have died from the treatment for a condition that was never identified.

SCENARIO 19 — Tank fine; fish die whenever the keeper goes on vacation

Symptoms. Losses discovered on return from trips of a week or more. Water cloudy; ammonia elevated.

Most plausible: 1. **Sitter overfeeding.** The most common vacation failure by a wide margin. Uneaten food decomposes; ammonia rises; oxygen falls; a bacterial bloom develops. 2. **Automatic feeder malfunction** — dumping the full hopper, or jamming and delivering nothing. 3. **Equipment failure unnoticed** — filter, heater, or air pump stopping with no one to observe it. 4. **Power interruption “helpful” maintenance** — a water change without conditioner, or cleaning the filter.

Less obvious: 5. **Temperature drift** from HVAC being turned down or off. 6. **Evaporation** — no top-offs for two weeks concentrates everything and can expose a heater or filter intake. 7. **A death early in the absence**, producing a compounding cascade. 8. **The sitter performing “helpful” maintenance** — a water change without conditioner, or cleaning the filter.

Differentiating tests and questions: - How much food was left, and how much is gone? - Is the water cloudy and is ammonia elevated? (→ #1/#2/#7.) - Was the water level significantly low? (→ #6.) - Is all equipment running now, and does it appear to have been running throughout? - Was the thermostat changed? - What exactly did the sitter do?

Standard mitigation: healthy adult fish tolerate a week without food far better than they tolerate overfeeding; pre-portioned daily containers; a written “do not clean anything” instruction; a large water change and full maintenance the day before departure.

SCENARIO 20 — Everything perfect; fish stop eating and slowly waste; a new food was introduced two months ago

Symptoms. Gradual condition loss across multiple species. No lesions. Parameters excellent.

Most plausible: 1. **Food quality problem** — rancid, expired, moldy, or nutritionally inadequate (11.7). 2. **Palatability** — the fish are not eating the new food, and the keeper does not realize it because the food disappears (it may be settling into the substrate). 3. **Nutritional inadequacy** for the species mix (11.3).

Less obvious: 4. **A contaminant introduced with the food** — mycotoxins are documented in aquaculture feed. 5. **Coincidental timing** with an unrelated cause (temperature drift, chronic parasitism). 6. **The food being fed in a form the fish cannot use** — pellets too large, floating food for bottom feeders.

Differentiating tests and questions: - Watch a feeding closely for five minutes. Does food actually get eaten, or does it settle? - Smell the food. Rancid oil is detectable. - How old is the container, and how is it stored? - Do the fish eat a different food eagerly? (The decisive test.) - Is the substrate accumulating uneaten food? (This would also explain rising organic load.)

Key learning point: food is a variable that changes, degrades, and is almost never suspected, and it produces slow decline with perfect parameters throughout.

SECTION 19. STRUCTURED TROUBLESHOOTING REFERENCE TABLES

19.1 Symptom → possible causes

Observed symptom	Possible causes (not ranked as certainties)
Gasping at surface	Low dissolved oxygen; ammonia; nitrite; high CO2; chlorine/chloramine; gill

	flukes; gill Ich; velvet; bacterial gill disease; gill damage from a prior insult; gas supersaturation; temperature too high; overstocking; surface film; filter stopped
Rapid gill movement without surfacing	Early hypoxia; gill parasites; irritant; elevated CO2; fever-analogue stress response; recent handling
Flashing / scratching on objects	Ich; velvet; monogenean flukes; <i>Chilodonella</i> ; <i>Trichodina</i> ; chemical irritant; ammonia; extreme pH; very low or very high hardness; new-water irritation
Clamped fins	Nonspecific malaise; parasites; poor water quality; cold; early bacterial infection; stress from aggression or inadequate cover
White discrete dots, uniform size	Ich (most likely); early velvet; spawning tubercles (males of some species); <i>Epistylis</i> (less uniform)
Fine gold or rust dusting	Velvet (<i>Piscinoodinium</i>)
Fuzzy tufted growths, often at wounds	<i>Epistylis</i> ; <i>Saprolegnia</i> ; columnaris
Flat grey-white patches, saddleback or mouth	Columnaris
Cottony 3D tufts at a discrete wound	<i>Saprolegnia</i> (true fungus)
Red streaks in fins or body	Septicemia (<i>Aeromonas</i> / <i>Pseudomonas</i>); ammonia burn; nitrite; severe irritant
Ulcers	Bacterial (motile <i>Aeromonas</i>); mycobacteriosis; injury with secondary infection
Popeye (exophthalmia), one eye	Injury; localized infection
Popeye, both eyes	Systemic bacterial infection; gas bubble disease; severe water quality
Raised scales (pinecone / dropsy)	Generalized edema from renal or hepatic failure; systemic bacterial infection; sign, not a disease
Abnormal buoyancy	Swim bladder infection or injury; constipation/diet; congenital deformity; tumor; egg binding; gas bubble disease; genetics of balloon/short-body morphs
Brown or chocolate gills	Nitrite (methemoglobinemia)
Pale gills	Chronic anemia; heavy parasitism; chronic disease
Bright red or inflamed gills	Ammonia; chlorine; irritant; acute parasitism
Excess mucus / slime coat sloughing	Ectoparasites; chemical irritant; extreme pH; ammonia
Progressive weight loss with appetite	Internal parasites; mycobacteriosis; malabsorption
Progressive weight loss without appetite	Starvation from competition; disease; food refusal; internal obstruction
White stringy feces	Flagellates; nematodes; enteritis; also seen in fasted fish
Spinal curvature	Mycobacteriosis; vitamin C deficiency; congenital; injury; electric shock; <i>Pleistophora</i>
Erratic darting, spiraling, twitching	Neurotoxin (pyrethroid, organophosphate); severe ammonia; neurological disease; extreme pH; electrical fault
Shimmying in place	Low mineral content (classic in mollies); cold; osmotic stress; general nervous distress
Sitting on the bottom upright	Weakness from many causes; cold; late-stage illness
Sudden death without prior signs	Toxin; acute columnaris; heater failure; oxygen crash; entrapment; internal rupture; aggression
Disappearance with no body	Jumping; predation; complete scavenging; entrapment in equipment
Eroded barbels (<i>Corydoras</i>)	Sharp substrate; organic-loaded substrate; secondary bacterial infection
Fin edges frayed or white	Fin nipping with secondary infection; columnaris; poor water quality
Fish clustered at one end of the tank	Temperature gradient; flow preference; aggression; electrical fault; localized contaminant

19.2 Timing → possible causes

Timing of onset	What becomes more likely
During the water change	Chlorine/chloramine; temperature; contaminated container; direct inflow exposure; physical injury; gas supersaturation
Minutes to 2 hours after	Chlorine; oxygen crash from substrate disturbance; ammonia release; restarted canister; chemical contamination; medication or conditioner overdose
2-12 hours after	Ammonia excursion; oxygen decline; bacterial bloom starting; temperature drift; CO2 accumulation
Next morning	Overnight oxygen minimum; equipment left off; heater failure; CO2 overnight;

	nocturnal aggression; jumping; delayed progression of an earlier insult
1-3 days after maintenance	Biofilter disruption; bacterial bloom oxygen demand; handling-injury infection beginning
3-7 days after new fish	Ich; velvet; columnaris; supply-chain damage manifesting; acclimation failure; OTS pattern
1-3 weeks after new fish	Monogeneans; bacterial spread; slower pathogens; starvation of newcomers
Weeks to months, gradual	Nutrition; chronic parameter mismatch; internal parasites; mycobacteriosis; chronic aggression; age; slow contaminant leach
Weekly, tracking maintenance	Something in the maintenance routine
Seasonal	Municipal water changes; room temperature; seasonal chemical use; source switching
After every absence	Feeding failures; equipment failure; temperature drift

19.3 Parameter → what it tells you → what it cannot tell you

Parameter	Tells you	Cannot tell you
Ammonia (TAN)	Total NH ₃ + NH ₄ ⁺ at this moment	Toxic NH ₃ fraction without pH and temperature; what it was earlier; whether an excursion occurred and resolved
Nitrite	Nitrite now	Prior spikes; whether chloride is protecting the fish
Nitrate	Accumulated nitrification and time since dilution	Organic load; oxygen; anything about toxicity in isolation; which unit convention the kit reports
pH	Hydrogen ion activity now	Mineral content; buffering capacity; stability; ionic composition; how it got there
Temperature	Water temperature at one point, now	Gradients; overnight excursions; heater reliability; species suitability
KH	Buffering capacity	GH; pH; TDS
GH	Calcium + magnesium	Sodium, potassium, chloride, sulfate; buffering; pH
TDS / conductivity	Total ionic strength approximation	Which ions; non-ionic contaminants; pesticides; dissolved organics (poorly)
Chlorine test	Free/total chlorine now	Whether an exposure occurred earlier and dissipated
Copper test	Copper now	Bioavailable fraction; other metals
Dissolved oxygen	Oxygen now, at that point	Overnight minimum unless measured then
All standard tests combined	A small subset of water chemistry at one instant	Pathogens; oxygen; gases; contaminants; nutrition; behavior; history

19.4 Maintenance action → potential risk → mechanism → evidence strength

Action	Potential risk	Mechanism	Evidence strength
Adding tap water without conditioner	Acute mortality	Gill oxidation and necrosis by chlorine/chloramine	[ESTABLISHED]
Large change in a neglected tank	Acute mortality	pH rise converts ammonium to toxic NH ₃ ; ionic step change	[GENERALLY SUPPORTED]
Unmatched temperature	Stress, mortality if large	Thermal shock; oxygen demand/supply mismatch	[ESTABLISHED] mechanism, [DISPUTED] thresholds
Deep gravel vacuum after long interval	Delayed hypoxia; bloom; ammonia	Organic release creates biological oxygen demand	[GENERALLY SUPPORTED]
Replacing all filter media	Ammonia/nitrite spike	Loss of nitrifying biomass	[ESTABLISHED]
Cleaning all media at once	Partial biofilter reduction	Biofilm removal	[GENERALLY SUPPORTED]
Brief tap rinse of media	Claimed biofilter loss	Disinfection requires concentration × contact time	[DISPUTED — probably overstated]
Letting media dry	Biofilter loss	Desiccation of aerobic bacteria	[GENERALLY SUPPORTED]
Filter off for many hours	Biofilter loss; hypoxic media	Aerobic organisms consume available oxygen	[GENERALLY SUPPORTED]
Restarting a long-stopped canister	Acute event	Injection of hypoxic, organically loaded water	[GENERALLY SUPPORTED]
Heater left on out of water	Equipment failure, thermal shock	Rapid heating with no thermal sink	[ESTABLISHED]
Heater left unplugged	Slow decline, immune suppression	Temperature drop	[ESTABLISHED]

Netting/handling fish	Infection days later	Mucus and epithelium damage; opportunistic colonization	[GENERALLY SUPPORTED]
Moving hardscape	Injury; aggression reset; organic release	Physical and behavioral	[GENERALLY SUPPORTED]
Uprooting large plants	Substrate disturbance; nutrient sink loss	As above plus decomposition	[PLAUSIBLE, POORLY DOCUMENTED]
Reduced surface agitation after refill	Hypoxia	Gas exchange occurs at the surface	[GENERALLY SUPPORTED]
Refilling from a pressurized hose	Gas supersaturation	Dissolved gas released to atmosphere	[ESTABLISHED] phenomenon, [DISPUTED] home frequency
Shared/contaminated bucket	Acute mortality	Surfactant membrane disruption	[GENERALLY SUPPORTED]
Aerosol use in the room	Acute mortality	Surface partitioning; air pump entrainment	[ESTABLISHED] for insecticides
Dosing medication in the display	Biofilter suppression; oxygen demand	Antibacterial and oxidative action	[GENERALLY SUPPORTED]
Heavy dechlorinator overdose	Oxygen depletion	Thiosulfate consumes dissolved oxygen	[GENERALLY SUPPORTED]
Top-offs instead of water changes	Mineral/TDS accumulation	Evaporation concentrates solutes	[ESTABLISHED]

19.5 Fish behavior → possible interpretations

Covered in full at 14.11. Cross-referenced here for retrieval.

19.6 Water-change problem → questions to ask

1. What is the exact temperature of the tank now, and what was the replacement water?
2. What percentage was changed?
3. Was conditioner used? Which product, what dose, dosed for what volume, added when relative to the water?
4. Does the municipality use chlorine or chloramine? Any current notices?
5. What is the tap water's ammonia, chlorine, pH (after 24h standing), KH, GH, TDS, and nitrate?
6. Was the water drawn from hot, cold, or mixed taps?
7. Was a hose or python used, and had it been sitting?
8. What container was used, and what else has it been used for?
9. How long since the previous water change?
10. What are the tank's current pH, KH, GH, TDS, and nitrate, and how do they compare to the tap water?
11. Were fine bubbles visible on glass, decor, or fish afterward?
12. What else was done during the same session? (Full list — see 19.7.)
13. Has this exact procedure been performed before without incident? If yes, what changed this time?
14. Was anything new used — new conditioner bottle, new bucket, new hose, new plumbing?
15. Which fish were affected first, and where in the tank were they?

19.7 Maintenance session checklist — what else was done?

- Substrate vacuumed? How deeply, how much of it, first time in how long?
- Filter opened? Media rinsed, replaced, or discarded?
- Carbon changed?
- Any equipment unplugged? Was everything plugged back in?
- Heater exposed to air while powered?
- Glass scraped?
- Decor or rocks moved, removed, or scrubbed?
- Plants trimmed, uprooted, added, or removed?
- Fish netted or moved?
- Anything dosed — fertilizer, buffer, salt, medication, bacteria product?
- Fed before, during, or after?
- Air stone, spray bar, or outflow repositioned?
- Water level ended higher or lower than usual?
- New livestock, plants, wood, rock, or substrate added?

19.8 New-fish death → questions to ask

1. Which store, and when did the shipment arrive relative to purchase?
2. How long was the fish in the bag in total?

3. Was bag water added to the tank?
4. What acclimation method, and how long?
5. Was the fish quarantined? If not, is a quarantine tank available now?
6. What are the store's pH, KH, GH, and TDS versus the tank's? (Ask the store; many will tell you.)
7. When was the tank last water-changed, and what is its nitrate and KH now?
8. Are the established fish affected?
9. Was the fish eating at the store? Is it eating now?
10. Wild-caught or captive-bred?
11. Are there aggressive residents, and has chasing been observed — including after lights-out?
12. Is there adequate cover?
13. Does the fish show flashing, spots, dusting, or lesions?
14. Is the species' temperature and hardness range compatible with this tank?
15. How many others from the same batch were purchased, and are they all affected?

19.9 One fish dying versus many fish dying — diagnostic implications

Pattern	Shifts toward	Shifts away from
One fish, others fine	Individual disease; injury; aggression victim; age; congenital problem; species-specific intolerance; individual starvation; internal parasite	Whole-tank toxicant; acute oxygen crash; temperature failure
One species, others fine	Species-specific pathogen; species-specific parameter intolerance; targeted aggression; feeding competition; medication sensitivity; purchase-batch effect; age cohort	Whole-tank toxicant
All fish, over days, sequentially	Infectious disease spreading; progressive water-quality decline; ongoing low-level exposure	Single acute event
All fish, within hours, simultaneously	Toxicant; oxygen crash; temperature failure; massive ammonia/nitrite; gas supersaturation; electrical fault	Infectious disease (essentially excluded)
Only new fish	OTS chemistry gap; supply-chain damage; acclimation failure; resident aggression; resident-carried pathogen	Whole-tank toxicant
Only established fish, new fish fine	Pathogen introduced by resistant carriers	Environmental cause affecting all
Only invertebrates	Copper; insecticide; low GH; TDS swing; fish medication; ammonia/nitrite (invertebrates more sensitive)	Most fish-specific pathogens
Only bottom dwellers	Substrate issues; localized low oxygen; medication/salt sensitivity; starvation; temperature; nocturnal aggression	Surface-associated causes
Only largest fish first	Hypoxia	Most toxicants (which usually hit small fish first)
Only smallest fish first	Toxicant; chlorine; thermal shock; predation	Hypoxia

19.10 Rapid triage decision reference

Observation	First action	Second action
All fish gasping	Add vigorous aeration immediately	Check equipment, temperature, then test
Water hot to the touch	Unplug heater; float sealed bags of cool water; do not cool rapidly	Test temperature; assess survivors
Ammonia or nitrite positive	Partial water change with matched, conditioned water; stop feeding	Identify the source (dead animal, media change, medication)
Suspected chemical exposure	Large water change plus fresh activated carbon; remove the source	Repeat changes; do not medicate
Fish dying right after a refill	Add conditioner; add aeration; do not change more water with the same tap	Test tap water for chlorine, ammonia, pH
Neglected tank, ammonia present, low pH	Small change only (10-15%); aerate; stop feeding	Serial small changes over weeks
Invertebrates dying, fish fine	Carbon; serial water changes; remove recent additions	Test GH/TDS; review recent chemical exposures
Visible parasites	Isolate if practical; identify before treating	Treat the identified agent only
Nothing identifiable, fish stable	Aerate; stop feeding; record everything	Observe 24-48h before intervening further

SECTION 20. ADDITIONAL CAUSES DISCOVERED DURING RESEARCH

This section collects mechanisms and factors that were not adequately represented in the original research brief but emerged as relevant during the compilation.

20.1 Electrical stray voltage and ground faults

[PLAUSIBLE, POORLY DOCUMENTED — mechanistically coherent, widely reported anecdotally, essentially unstudied in home aquaria]

Failing heaters, pumps, and powerheads can leak current into the water. Reported signs: fish clustered abnormally, erratic behavior, unexplained spinal deformities, chronic stress with no chemical explanation, and a tingling sensation when the keeper's hand contacts the water while touching a grounded surface. Mitigation is a GFCI outlet and a grounding probe. This is a recognized concern in the reef hobby and is almost entirely absent from freshwater beginner literature.

20.2 Cyanobacteria and algal toxins

[PLAUSIBLE, POORLY DOCUMENTED at aquarium scale; ESTABLISHED in ponds and lakes]

Cyanobacteria ("blue-green algae") can produce microcystins, anatoxins, and other toxins. This is thoroughly documented in natural waters and ponds. Whether aquarium cyanobacteria blooms reach toxicologically relevant concentrations is not established. What is established is that a dense cyanobacterial mat consumes oxygen at night, smothers substrate and plant surfaces, and indicates a nutrient/flow problem.

20.3 Allelopathy and growth-inhibiting substances

[DISPUTED / PLAUSIBLE, POORLY DOCUMENTED]

- Some aquatic plants release allelopathic compounds affecting other plants; this is documented in aquatic botany.
- The hobby claim that fish release growth-inhibiting hormones which accumulate and stunt tankmates ("growth inhibiting hormone," GIH) is widely repeated. There is genuine research on density-dependent growth suppression in fish, but the specific hobby framing — an identifiable accumulating hormone removed by water changes — is not well substantiated. Density-dependent growth suppression is real; the proposed mechanism is not established.

20.4 Dissolved organic carbon (DOC) as an independent variable

[GENERALLY SUPPORTED]

DOC is invisible to every hobbyist test, accumulates continuously, and has multiple effects: it complexes metals (reducing their toxicity — one reason very soft, low-DOC water is more dangerous), it feeds heterotrophic bacteria (competing with nitrifiers for oxygen), it contributes to surface films that impede gas exchange, and it is the substrate for bacterial blooms. **Water changes remove DOC; nitrate is merely the convenient proxy for how much has accumulated** (see 17.13).

20.5 Disinfection byproducts (DBPs)

[PLAUSIBLE, POORLY DOCUMENTED for fish]

Trihalomethanes and haloacetic acids form when disinfectants react with organic matter in source water. They are elevated during free chlorine conversions [EPA-R6-FCC]. Standard dechlorinators neutralize chlorine but do not remove DBPs. Their toxicity to aquarium fish at drinking-water concentrations is not documented. Activated carbon removes many of them.

20.6 Nanoparticles, microplastics, and PFAS

[PLAUSIBLE, POORLY DOCUMENTED]

Emerging contaminant classes present in some municipal supplies. Not measurable by any hobbyist method, not removed by dechlorinator, partially removed by RO. Currently speculative as a cause of aquarium mortality but worth naming as an acknowledged unknown rather than omitting.

20.7 Silicate, phosphate, and corrosion-control additives

Utilities dose orthophosphate or silicate for corrosion control. Effects on aquaria: elevated phosphate (algae), and — when a utility *changes* its corrosion-control program — altered metal solubility in household plumbing. The Washington DC and Flint lead events both originated in disinfectant/corrosion-control changes. **A change in a utility's treatment chemistry can mobilize metals from plumbing that had been stable for years.**

20.8 Barometric pressure and weather

[PRIMARILY ANECDOTAL]

Hobbyists and pond keepers frequently report behavioral changes and spawning triggered by weather fronts. There is legitimate

research on barometric response in fish. Whether it contributes to mortality in a stable indoor aquarium is unsupported. Weather-associated events in home aquaria more plausibly reflect room temperature, humidity, and power stability.

20.9 Light: intensity, spectrum, photoperiod

[GENERALLY SUPPORTED]

- Excessive photoperiod drives algae and, in planted tanks, larger diurnal pH and oxygen swings.
- Abrupt full-intensity switching causes a daily startle response; ramped lighting or a room light turned on first mitigates it.
- Some species (many catfish, loaches, blackwater species) are photophobic and are chronically stressed under bright open lighting without cover.
- Very high-intensity lighting on a shallow tank contributes heat.
- Continuous lighting (24h) is genuinely harmful and is a common error in office and children's tanks.

20.10 Sound and vibration

[PLAUSIBLE, POORLY DOCUMENTED]

Fish detect vibration acutely via the lateral line and inner ear. Tanks on a subwoofer, next to a slamming door, on a washing machine, or in a high-traffic corridor experience continuous vibrational stress. Tapping on the glass transmits substantial energy. There is aquaculture research on noise stress; home-aquarium data is absent.

20.11 Flow: too much and too little

[GENERALLY SUPPORTED]

- **Too little** produces dead zones, gradients, poor gas exchange, and localized organic accumulation.
- **Too much** exhausts species adapted to still water (bettas, gouramis, fancy goldfish, long-finned varieties), which must swim continuously to hold position. Chronic exhaustion presents as thin, stressed fish with perfect water parameters.
- **Species mismatch within one tank** — hillstream loaches and bettas cannot both be accommodated.

20.12 Tank size, aspect ratio, and territory geometry

Small tanks amplify every problem: less thermal inertia, less dilution of any input, less oxygen reserve, less territory. Aspect ratio matters independently of volume — a tall narrow tank has less surface area per unit volume (worse gas exchange) and less horizontal swimming space than a long shallow tank of identical capacity. Territory disputes are geometric, not volumetric: a long tank permits more territories than a tall one of the same volume.

20.13 New tank syndrome distinct from cycling

Beyond the nitrogen cycle, a genuinely new aquarium lacks: mature heterotrophic microbial diversity, biofilm on all surfaces, established grazing surfaces for algae eaters and shrimp, and microfauna. Several species — otocinclus, many shrimp, some plecos — do poorly in tanks under several months old for reasons unrelated to ammonia. **“Cycled” and “mature” are different states.**

20.14 Leaching from new hardscape, substrate, and equipment

- New aquasoils release ammonia for days to weeks.
- Some rocks alter KH/GH significantly (limestone, coral, some slates and shales).
- Untreated or unsuitable driftwood releases large quantities of tannins and can drop pH substantially in low-KH water.
- Botanicals (leaves, pods, cones) add tannins and organic load; a large addition at once is a meaningful organic input.
- New silicone, tubing, and plastics may leach initially.
- Painted, glazed, or metal ornaments not rated for aquarium use.

20.15 Temperature stratification and heater placement

Distinct from 9.5: in a very low-flow tank, thermal stratification can produce a genuinely warm upper layer and cool lower layer. Bottom dwellers then live in cooler water than the thermometer reports.

20.16 The “everything is fine so I stopped testing” failure mode

[ANECDOTAL but extremely common]

A tank that has been stable for months produces months of identical test results, and the keeper stops testing. The parameters the keeper reports during a later incident are therefore weeks or months old. **“My parameters are perfect” frequently means “my parameters were perfect the last time I checked, some time ago.”** This is an epistemic failure mode, not a chemical one, and it may be the single most common reason the premise of the core question is false in the first place.

20.17 Test kit consistency and cross-checking

Related: a keeper using one kit for two years has no independent check on that kit's degradation. Cross-checking against a second method, a store test, or a known standard occasionally is cheap and rarely done. A kit that has silently drifted produces confidently wrong "perfect parameters."

20.18 Undetected slow leaks and water level changes

A slow leak produces a gradually falling water level, which the keeper compensates for with top-offs — accelerating mineral accumulation (2.13) and eventually exposing heaters, filter intakes, or the surface skimmer.

20.19 Reproductive stress and spawning-related mortality

- Egg-binding/dystocia in livebearers and egg-layers.
- Post-spawning exhaustion, particularly in females spawning repeatedly.
- Male aggression during breeding condition.
- Parental defense converting a peaceful fish into a lethal aggressor within a 30-gallon tank.
- Continuous livebearer reproduction shortening female lifespan (10.6).

20.20 Cumulative and interactive stressors

[ESTABLISHED as a principle]

The most important synthesizing point in this section. Individual stressors that are each survivable become lethal in combination:

- High temperature + high stocking + reduced agitation = oxygen failure.
- Warm water + high pH + a modest ammonia excursion = a large un-ionized ammonia dose.
- Handling injury + warm water + high organic load = acute columnaris.
- Shipping stress + immediate introduction + resident aggression + no quarantine = predictable loss.
- Neglected tank + large water change + cold tap water = multiple simultaneous insults.

Nothing in a five-parameter test panel captures interaction effects. This is the structural reason the core question of this document exists.

20.21 Human factors

- **Multiple caretakers** — a household where two people each feed the fish, each unaware of the other.
- **Children** adding items to the tank.
- **Well-meaning "help"** from guests, sitters, or family.
- **Product-driven intervention** — solving problems that do not exist because a product exists for them.
- **Recall error** — hobbyists reconstructing timelines from memory, days later, reliably misremember the order of events. This is why 16.6 (write it down at the time) has disproportionate value.
- **Confirmation bias in symptom reading** — once a hypothesis is formed, ambiguous colors on a test card and ambiguous fish behavior are read to fit it.

SECTION 21. CONSOLIDATED ANSWER TO THE CORE QUESTION

Restating the core question:

If ammonia, nitrite, nitrate, pH, and temperature appear acceptable, what else could cause aquarium fish to become sick, stressed, behave abnormally, or die — and under what circumstances could a water change or associated maintenance contribute to the problem?

21.1 Master list of causes compatible with a normal five-parameter test panel

A. The measurement itself is wrong or incomplete 1. Test performed incorrectly (unshaken nitrate reagent, expired strips, contaminated vial, wrong timing) 2. Test read incorrectly (color matching, card resolution, unit convention) 3. TAN interpreted without pH/temperature context 4. Test taken long after the event (transient already resolved) 5. Test taken from an unrepresentative location 6. Test result is old ("perfect" means "last month") 7. Kit degraded silently

B. Parameters not on the panel 8. Low dissolved oxygen 9. Elevated dissolved CO₂ 10. Gas supersaturation 11. Chlorine 12. Chloramine and its residual ammonia 13. Copper and other metals 14. Insecticides and pesticides 15. Surfactants and household chemicals 16. Solvent and aerosol exposure 17. Medication overdose or interaction 18. Dissolved organic carbon load 19. GH, KH, TDS extremes and swings 20. Disinfection byproducts and emerging contaminants

C. Biological causes 21. Ectoparasites (Ich, velvet, flukes, sessile and motile ciliates) 22. Internal parasites 23. Bacterial infection (columnaris, *Aeromonas*, *Flavobacterium* gill disease, mycobacteriosis) 24. Fungal/oomycete infection secondary to injury 25. Viral

disease 26. Asymptomatic carriers and incubating infections in new arrivals

D. Fish-condition causes 27. Supply-chain damage (transport, handling, starvation, ammonia and pH exposure in transit) 28. Genetic fragility and congenital defects 29. Age 30. Prior irreversible gill or organ damage 31. Nutritional deficiency, starvation, or inappropriate diet 32. Reproductive stress and egg binding

E. Environmental and physical causes not on the panel 33. Temperature gradients, overnight excursions, chronic mismatch 34. Flow too high or too low 35. Light intensity, spectrum, photoperiod 36. Sound and vibration 37. Insufficient cover, inappropriate tank geometry 38. Stray voltage 39. Physical injury, entrapment, jumping

F. Social and stocking causes 40. Aggression, territoriality, nocturnal attack 41. Fin nipping 42. Food competition 43. Inadequate group size for shoaling species 44. Sex ratio problems 45. Predation 46. Overstocking (social, not chemical)

G. Systemic and interaction causes 47. Cumulative stressor interaction 48. Transient excursions that resolve before testing 49. Human factors (multiple caretakers, recall error, well-meant intervention)

21.2 Circumstances under which a water change or associated maintenance contributes

Directly attributable to the replacement water: 1. Chlorine/chloramine not neutralized or under-neutralized 2. Municipal treatment change (free chlorine conversion, flushing, source switch, corrosion-control change) 3. Large temperature delta 4. Large chemistry delta into a neglected tank (the pH/ammonia mechanism) 5. Large ionic delta (RO, softener, evaporation-concentrated tank) 6. Gas supersaturation from pressurized or heated tap water 7. Metals from plumbing, especially after stagnation, new plumbing, or hot water 8. Contaminated container, hose, or hands 9. Well-water specific issues (CO₂, iron, low oxygen, nitrate, H₂S)

Attributable to co-occurring maintenance actions (Section 3): 10. Substrate disturbance creating oxygen demand and releasing organics 11. Filter media replacement or over-cleaning 12. Filter or heater left off, or restarted incorrectly 13. Restart of a long-stopped canister 14. Heater damaged by exposure during draining 15. Surface agitation reduced by repositioned equipment or water level 16. Fish netted, injured, trapped, or crushed 17. Hardscape moved, resetting territories 18. Large plant removal or uprooting 19. Chemical introduced during the session (bucket, hands, aerosol, dosing error) 20. Multiple actions performed simultaneously, none individually decisive

Attributable to the timing rather than the action: 21. Coincidence — an unrelated cause that happened to occur near the water change 22. A pre-existing trajectory (incubating infection, supply-chain damage, chronic decline) reaching its endpoint

21.3 The three highest-value practical corrections identified in this research

1. **Add aeration first, in any emergency.** It treats the most under-diagnosed acute cause (hypoxia), is essentially never harmful, and is diagnostic — rapid improvement identifies an oxygen component.
2. **Test the source water, not just the tank.** The comparison between source and tank on pH (after standing), KH, GH, TDS, ammonia, and nitrate identifies chlorine problems, softeners, RO errors, buffer depletion, evaporation concentration, and the OTS gap — none of which is visible from tank testing alone.
3. **Write down the 72-hour timeline before changing anything.** The causal record degrades within hours and is destroyed entirely by a “reset everything” response. For the majority of causes in 21.1, the timeline is the only available evidence.

21.4 Explicit statement of what remains uncertain

Areas where this research found genuine, unresolved disagreement or an absence of applicable evidence:

- The practical significance of brief tap-water rinsing of filter media (3.5.1)
- The relative merits of drip acclimation versus rapid transfer for shipped fish (7.6)
- The magnitude at which temperature and pH changes become independently harmful (2.2, 2.8)
- Whether ammonia-binding conditioners provide meaningful benefit at normal municipal chloramine levels (2.4)
- The concentration at which nitrate causes chronic harm to captive-bred adult ornamental fish (17.13)
- The real-world frequency of gas bubble disease in home aquaria (2.19)
- Whether “TDS shock” is a meaningful cause of fish (as opposed to invertebrate) mortality (17.10)
- Whether hobby aquarium cyanobacteria produce toxicologically relevant toxin concentrations (20.2)
- The validity of the growth-inhibiting-hormone hypothesis (20.3)
- The contribution of chronic sound, vibration, and stray voltage to home aquarium mortality (20.1, 20.10)

These should be represented as open questions rather than resolved in either direction.

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[MERCK-BACT] *Bacterial Diseases of Fish*. Merck Veterinary Manual, Exotic and Laboratory Animals — Aquarium Fish. <https://www.merckvetmanual.com/exotic-and-laboratory-animals/aquarium-fish/bacterial-diseases-of-fish> — *Flavobacterium columnare* strain virulence range; unidentified predisposing water-quality conditions; acute mortality.

[MERCK-AQUA] *Bacterial Diseases in Aquaculture*. Merck Veterinary Manual. <https://www.merckvetmanual.com/exotic-and-laboratory-animals/aquaculture/bacterial-diseases-in-aquaculture> — Taxonomic revision into *Flavobacterium*; *F. psychrophilum* temperature range (4–10°C, signs not expected above 18°C); diagnostic methods.

[MERCK-OWNER] *Disorders and Diseases of Fish*. Merck Veterinary Manual, All Other Pets. <https://www.merckvetmanual.com/all-other-pets/fish/disorders-and-diseases-of-fish> — Columnaris; carp pox; *Edwardsiella*; *Pleistophora hyphessobryconis*; the warning against “shotgunning” antibiotics.

[FHS-COLUMNARIS] 1.2.3 *Columnaris Disease*. American Fisheries Society, Fish Health Section Blue Book (2014). <https://units.fisheries.org/fhs/wp-content/uploads/sites/30/2017/08/1.2.3-Columnaris2014.pdf> — Epizootiology; severity greater above 20°C.

[PMC-GOLDENSHINER] *Evaluating Stress-Mediated Microbial Pathogenesis in Golden Shiners, Notemigonus crysoleucas*. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9132093/> — Density and temperature effects on columnaris challenge survival; *F. covae* as an opportunistic pathogen causing heavy mortality in stressed fish.

[CAMARGO2005] Camargo, J. A., Alonso, A., & Salamanca, A. (2005). Nitrate toxicity to aquatic animals: a review with new data for freshwater invertebrates. *Chemosphere*. <https://pubmed.ncbi.nlm.nih.gov/15667845/> — Recommended maximum of 2 mg NO₃-N/L for the most sensitive freshwater species; effects at 10 mg NO₃-N/L. Full text mirror: https://www.waterboards.ca.gov/water_issues/programs/tmdl/records/region_2/2008/ref2426.pdf

[DAVIDSON2017] Davidson, J., et al. (2017). Evaluating the chronic effects of nitrate on the health and performance of post-smolt Atlantic salmon *Salmo salar* in freshwater recirculation aquaculture systems. *Aquacultural Engineering*. <https://www.sciencedirect.com/science/article/pii/S0144860917301231> — Review of nitrate physiological effects: methemoglobinemia, immune and osmoregulatory effects, endocrine disruption; Westin (1974) 96-h LC50 values.

[ABDELTAWWAB2019] Abdel-Tawwab, M., et al. (2019). Fish response to hypoxia stress: growth, physiological, and immunological biomarkers. *Fish Physiology and Biochemistry*, 45(3), 997–1013. <https://pubmed.ncbi.nlm.nih.gov/30715663/> — DO below 1–2 mg/L for a few hours adversely affects growth and can cause death.

[UKY-POND] *Causes of “Water Turnover” and Low Dissolved Oxygen*. University of Kentucky Aquaculture Extension. https://aquaculture.mgcafe.uky.edu/sites/aquaculture.ca.uky.edu/files/pond_turnover.pdf — Warmwater fish require 5 mg/L, coldwater 7 mg/L; prolonged exposure below 3 mg/L lethal to warmwater fish.

[GSA-DO-REQ] *Dissolved oxygen requirements in aquatic animal respiration*. Global Seafood Alliance / Responsible Seafood Advocate. <https://www.globalseafood.org/advocate/dissolved-oxygen-requirements-in-aquatic-animal-respiration/> — Survival minima; the 50%-of-saturation guideline.

[GSA-DO-DYNAMICS] *Dissolved oxygen dynamics*. Global Seafood Alliance. <https://www.globalseafood.org/advocate/dissolved-oxygen-dynamics/> — Below 3 mg/L stressful; 1.0–1.5 mg/L for a few hours can kill warmwater animals.

[GSA-DO-POND] *Dissolved oxygen concentrations in pond aquaculture*. Global Seafood Alliance. <https://www.globalseafood.org/advocate/dissolved-oxygen-concentrations-pond-aquaculture/> — Growth effects below 50% saturation;

saturation concentration examples at 26°C.

[WIKIVET-GBD] *Gas Bubble Disease*. WikiVet. https://en.wikivet.net/Gas_Bubble_Disease — Supersaturation mechanism; causes in aquaria including pump/valve leaks and sudden temperature gradients; references Roberts (2001) *Fish Pathology* and Noga (2010) *Fish Disease: Diagnosis and Treatment*.

[NATURE-GBD] *Establishment of a fish model to study gas-bubble lesions*. Scientific Reports. <https://www.nature.com/articles/s41598-022-10539-8> — Experimental supersaturation methodology and pathology.

[TOLEDO-GBD] Jafarey, Y. (Toledo Zoo). *Fish Health Q&A: Gas Bubble Disease*. Reefs.com. <https://reefs.com/magazine/toledo-zoo-fish-health-q-a-gas-bubble-disease/> — Clinical presentation: swollen eyes, abnormal buoyancy, emboli in gill lamellae and opercula, “silver lines” between fin rays.

[PETMD-GBD] Sanders, J., DVM, DABVP (Fish Practice). *Gas Bubble Disease in Fish*. PetMD. <https://www.petmd.com/fish/conditions/systemic/gas-bubble-disease-fish>

[PETPLACE-SUPERSAT] *Supersaturation*. PetPlace. <https://www.petplace.com/article/fish/general/supersaturation> — Over-aeration at high temperature; cavitating pumps.

[ANZG-COPPER] *Copper in freshwater and marine water*. Australian and New Zealand Guidelines for Fresh and Marine Water Quality. <https://www.waterquality.gov.au/anz-guidelines/guideline-values/default/water-quality-toxicants/toxicants/copper-2000> — Hardness/alkalinity/DOC modification of copper toxicity; USEPA acute range of 17–10,000 µg/L across 41 genera; Gauss et al. (1985) soft vs. hard water EC50 comparison.

[NPIC-FIPRONIL] *Fipronil Fact Sheet*. National Pesticide Information Center, Oregon State University. <https://npic.orst.edu/factsheets/fipronil.html> — Highly toxic to freshwater and marine fish and invertebrates; metabolites more toxic than the parent compound.

[CAWB-FIPRONIL] *The insecticide fipronil and its degradates as contributors to toxicity*. California Water Boards SWAMP workplan. https://www.waterboards.ca.gov/water_issues/programs/swamp/docs/workplans/reg_wrkpln_fipronil_%20final.pdf — Mysid shrimp acute EC50 of 140 ng/L; life-cycle effects below 5 ng/L (EPA 1996).

[BRADBURY-COATS-1989] Bradbury, S. P., & Coats, J. R. (1989). Toxicity of pyrethroid insecticides to fish. *Environmental Toxicology and Chemistry*, 8(5), 381. <https://academic.oup.com/etc/article-abstract/8/5/381/7860964> — 96-hour LC50 values in the nmol/L range; mechanisms involving nervous system, respiratory surfaces, and renal ion regulation.

[EPA-R6-FCC] *Microbial and physicochemical water quality changes within distribution and premise plumbing systems during a chlorine conversion*. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11980844/> — Free chlorine conversion frequency (typically annual), timing, and 30–60 day duration; DBP elevation; explicit mention of inconvenience to fish tank owners.

[ZOOLOGIA-TRANSPORT] *Water acidification causes death of marine ornamental fish (Perciformes: Pomacentridae) during transport*. Zoologia. <https://zoologia.pensoft.net/article/25083/> — pH correlated with transport mortality, especially in the first 24 hours; hypoxia at 24–48 hours; ammonia build-up not problematic even at 72 hours.

[PMC-HYPOXIA-BASS] *Transcriptome and 16S rRNA Analyses Reveal That Hypoxic Stress Affects the Antioxidant Capacity of Largemouth Bass*. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9854696/> — Effects of sustained low DO on oxidative stress and tissue damage.

[PMC-HARDNESS-REPRO] *Water hardness influenced variations in reproductive potential of two freshwater fish species; Poecilia reticulata and Betta splendens*. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7678082/> — Species-divergent hardness effects on growth and reproduction.

[TNC-AMMONIA-REVIEW] *A Literature Review of Effects of Ammonia on Fish* (2010). The Nature Conservancy. <https://www.nature.org/content/dam/tnc/nature/en/documents/literature-review-ammonia.pdf> — EPA acute and chronic ammonia criteria; the 0.020 mg/L un-ionized ammonia reproduction-safety figure; one pH unit increases un-ionized ammonia tenfold.

[CUNLIFFE1991] Cunliffe, D. A. (1991). Bacterial nitrification in chloraminated water supplies. *Applied and Environmental Microbiology*. — Cited via [AQSCI-TAPRINSE] for chloramine contact time required to kill nitrifying bacteria at typical tap concentrations. **Note: this citation was not independently verified against the primary text during this compilation and is reported as cited by a secondary source.**

Tier 5-6: Aquaculture literature and water utility / municipal sources

[THEFISHSITE-AMMONIA] *Ammonia in Aquatic Systems*. The Fish Site (reproducing UF/IFAS content by Francis-Floyd, Watson, Petty, and Poudel). <https://thefishsite.com/articles/ammonia-in-aquatic-systems> — Nitrite toxic at 0.10 mg/L; the 10 mg/L chloride per 1 mg/L nitrite mitigation ratio; nitrate potentially harmful above 250 mg/L in closed systems.

[ACWD-CHLORAMINE] *Chloraminated Water & Your Pet Fish (FAQ)*. Alameda County Water District. <https://www.acwd.org/faq.aspx?TID=19> — Chloramine breakdown releases ammonia; chloramine does not dissipate by letting water stand; koi equally susceptible.

[GREENSBORO-FCC] *Temporary Free-Chlorine Conversion Program*. City of Greensboro, NC. <https://www.greensboro-nc.gov/departments/water-resources/water-system/temporary-free-chlorine-conversion-program>

[GUC-FCC] *Chlorine Conversion FAQs*. Greenville Utilities Commission, NC. <https://www.guc.com/water/annual-temporary-switch-chlorine/chlorine-conversion-faqs>

[GSWSA-FLUSH-FAQ] *Questions for Water Line Flushing and the Conversion to Free Chlorine*. Grand Strand Water & Sewer Authority. https://www.gswwsa.com/gswsa_public_site/userfiles/file/Flushing%20and%20Free%20Chlorine%20Q&A.pdf — All methods for

removing chloramine also remove chlorine, but not the reverse.

[COCOA-FCC] *Free Chlorine Conversion FAQ*. City of Cocoa, FL. <https://www.cocoafl.gov/m/faq?cat=55> — Transition weeks may contain chlorine, ammonia, or a mixture.

[BRUNSWICK-FCC] *Brunswick County water system to temporarily switch to chlorine*. WECT. <https://www.wect.com/2024/09/25/brunswick-county-water-systems-temporary-switch-chlorine-through-november/> — Conversion combined with hydrant flushing; sediment and discoloration.

[LAKEJACKSON-FCC] *Free Chlorine Conversion*. City of Lake Jackson, TX. <https://www.lakejackson-tx.gov/797/Free-Chlorine-Conversion>

[KILGORE-FCC] *Free Chlorine Conversion*. City of Kilgore, TX. <https://www.cityofkilgore.com/513/Free-Chlorine-Conversion>

[PALMBAY-FCC] *Palm Bay Switching to Free Chlorine for Water System Maintenance*. <https://news.thepalmbayer.com/news/palm-bay-free-chlorine-water-maintenance-march-2026/> — Guidance to treat before large water changes and to be especially careful during transition periods.

[ASRS-AERATION] Kirby, C. S., et al. *Aeration to degas CO₂, increase pH, and iron oxidation rates*. American Society of Reclamation Sciences. <https://www.asrs.us/DOI/Proceedings/2007v1/JASMR07010373-Kirby.pdf> — Experimental demonstration that aeration drives off dissolved CO₂, raising pH.

[USPTO-4666610] *Method and product for removal of chloramines, chlorine and ammonia from aquaculture water*. US Patent 4,666,610. <https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/4666610> — Chlorine at 0.2–0.3 ppm rapidly toxic to fishes; EPA 0.003 ppm continuous-exposure recommendation; municipal chlorine up to 2.5 ppm; excess thiosulfate reacting with dissolved oxygen. **Patent literature; motivated source.**

[USPTO-4786434] *Agent for the elimination of active chlorine compounds from water*. US Patent 4,786,434. <https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/4786434> — Chloramine LC₀/LC₅₀/LC₁₀₀ figures; lethal action within 5–30 minutes; the claim that replacing more than 10% of aquarium water with chloraminated water may prove fatal. **Patent literature; motivated source; figures should be treated as indicative rather than authoritative.**

[USPTO-4883789] *Substituted phenyltrialkylsilane insecticides*. US Patent 4,883,789. <https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/4883789> — Cypermethrin 100% fish kill at 0.01 ppm, cited as a comparative reference standard.

[RG-TRANSPORT-DENSITY] *A neglected fish stressor: mechanical disturbance during transportation impacts susceptibility to disease in a globally important ornamental fish / related transport-density research*. <https://www.researchgate.net/publication/331676367> — Density effects on ammonia, temperature, pH, mortality, gill and liver damage, cortisol, ROS, and IgG.

[RG-TAMBAQUI] *Effect of Fish Density During Transportation on Stress and Mortality of Juvenile Tambaqui Colossoma macropomum*. <https://www.researchgate.net/publication/230028908> — Ammonia and nitrite as stressors stimulating corticosteroid release; link between elevated corticosteroids, impaired immune function, and decreased disease resistance.

[RG-TRANSPORT-METHODS] *Methods for reducing stressors and maintaining water quality associated with live fish transport in tanks*. <https://www.researchgate.net/publication/228375829> — Lactate, cortisol, malondialdehyde, enzyme activity; mortality increasing with density and duration.

[RG-TRANSPORT-OVERVIEW] *An overview of stress physiology of fish transport: Changes in water quality as a function of transport duration*. <https://www.researchgate.net/publication/299477002>

[ACADEMIA-ORNAMENTAL-TRANSPORT] *A novel approach to ornamental fish transportation for the aquarium hobbyist*. <https://www.academia.edu/125103395/> — 24–36 hour practical limit for sealed oxygenated bags; cortisol rising within 4 hours and returning to control 2 hours after arrival; low bag pH keeping NH₃ at safe levels during transport.

Tier 7-9: Aquarium publications, manufacturer documentation, and experienced hobbyist technical sources

These sources are cited where they provide mechanism, calculation, or practical detail not available in the tiers above. Several are opinionated and several disagree with each other; that disagreement is documented rather than resolved in the body text.

[AQSCI-AMMONIA] 5.2.1 *Ammonia in Depth*. Aquarium Science. <https://aquariumscience.org/index.php/5-2-1-ammonia-in-depth/> — TAN vs. free-ammonia test chemistry; conversion of Seachem thresholds into API-equivalent readings.

[AQSCI-CHLORAMINE] 5.5.2.1 *Ammonia in Chloramine in the Aquarium* and 5.5.3.1 *Ammonia Detoxifying*. <https://aquariumscience.org/index.php/5-5-2-1-ammonia-in-chloramine/> and <https://aquariumscience.org/5-5-3-1-ammonia-detoxifying/> — The 2 ppm chloramine → 0.62 ppm TAN → 0.0035 ppm NH₃ calculation at pH 7.0/25°C; the argument that ammonia-detoxifier claims are marketing. **Strongly opinionated; presented in the body text as one side of a documented dispute.**

[AQSCI-NITRATE] 5.4 *Safe Nitrate Levels* and 5.4.1 *Nitrate in Depth*. <https://aquariumscience.org/index.php/5-4-safe-nitrate-levels/> and <https://aquariumscience.org/index.php/5-4-1-nitrate-in-depth/> — Compilation of LC₅₀ literature; the 4,918 ppm average of sixteen freshwater LC₅₀ values; the guppy fry LC₅₀ of 191 mg NO₃-N/L; the “effect is not synonymous with harmful” argument; criticism of environmental guideline documents as policy-driven. **Contains explicit editorial position; flagged as such in 17.13.**

[AQSCI-OTS] 18.6 *Old Tank Syndrome*. <https://aquariumscience.org/18-6-old-tank-syndrome/> — OTS defined as high ammonia, high nitrate, low pH, low KH; nitrification failure below pH 6.5; the argument that most ammonia spikes are not OTS and that the most common cause is an unnoticed dead fish.

[AQSCI-ANAEROBIC] 14.2.4 *Anaerobic Myth* and 14.2.4.1 *Anaerobic Science*. <https://aquariumscience.org/14-2-4-anaerobic-myth/> and

<https://aquariumscience.org/14-2-4-1-anaerobic-science/> — The argument that toxic H₂S production in aquarium substrates does not occur, alongside the acknowledgment that hypoxic pockets producing ammonia, nitrite, and bacterial toxins can occur but are rare.

[AQSCI-GBD] *11.19 Gas Bubble Disease*. <https://aquariumscience.org/index.php/11-19-gas-bubble-disease/> — The theoretical tap-water supersaturation pathway; the argument that normal aquarium aeration resolves it before disease develops; the commercial account of an aquatics retailer installing a degassing device.

[AQSCI-TAPRINSE] *6.9 Tap Water Rinsing*. <https://aquariumscience.org/6-9-tap-water-rinsing/> — The concentration × contact-time argument; citation of Cunliffe (1991).

[AQSCI-SHIFTS] *4.8.1 Rapid Thermal and pH Shifts in the Aquarium*. <https://aquainfo.nl/en/4-8-1-rapid-thermal-and-ph-shifts-in-the-aquarium/> — HSP induction thresholds citing Airaksinen et al. (1998), Basu et al. (2001), Currie et al. (2000); Wiebe (1931) lake-transfer observations. **The primary studies were not independently verified during this compilation.**

[AQSCI-CO2PH] *4.4.3 Carbon Dioxide and pH*. <https://aquariumscience.org/4-4-3-carbon-dioxide-and-ph/> — Tap water pH shifting by a full unit or more in either direction as CO₂ equilibrates.

[AQSCI-EPISTYLIS] *10.2.4 Epistylis*. <https://aquariumscience.org/index.php/10-2-4-epistylis/> — Visual differentiation from Ich (fuzzy/translucent/protruding vs. defined/white/slightly raised).

[FISHKEEPCHILL-EPISTYLIS] *Ich versus Epistylis*. Fishkeep & Chill, interview with a veterinarian. <https://fishkeepandchill.com/ich-versus-epistylis/> — Sessile vs. motile ciliate distinction; Ich as a primary pathogen capable of killing healthy fish; *Epistylis* as opportunistic.

[PFK-OTS] *How to avoid 'old tank syndrome'*. Practical Fishkeeping. <https://www.practicalfishkeeping.co.uk/features/how-to-avoid-old-tank-syndrome/> — Carbonate depletion consequences for osmoregulation and pH stability; the below-5°KH warning.

[FRITZ-CHLORAMINE] *About Chloramine*. Fritz Aquatics. <https://fritzaquatics.com/resources/articles/about-chloramine> — Manufacturer technical documentation: standard dechlorinators leave ammonia; pH dependence of the resulting risk. **Manufacturer source with a commercial interest in the claim.**

[AQCOOP-COND] *Water Conditioner: How It Works and How Much to Use in Aquariums and Does dechlorinator remove ammonia?* Aquarium Co-Op. <https://www.aquariumcoop.com/blogs/aquarium/water-conditioner-for-fish> and <https://www.aquariumcoop.com/blogs/faqs/does-dechlorinator-remove-ammonia> — Ammonia binding for up to 24 hours; the recommendation to add an air stone for 3–4 hours after heavy dosing due to oxygen reduction. **Retailer source with a commercial interest.**

[AQUASCAPEGUIDE-TAPRINSE] *Cleaning your filter with chlorinated tap water*. <https://www.aquascapeguide.com/post/cleaning-your-filter-with-chlorinated-tap-water> — Biofilm protection; laboratory-cultured bacteria being less resistant than natural populations.

[FOSSA-GBD] *Gas Bubble Disease: It's Not About the Bubbles*. Fossa Method. <https://fossamethod.ru/en/posts/bubble/> — The distinction between visible bubbles (a consequence of degassing) and the invisible supersaturated condition that causes disease; the observation that conventional aquarium pumps rarely generate sufficient supersaturation alone.

[PAKIS-SUBSTRATE] *Substrate*. Planted Aquarium Keep It Simple. http://www.aquaworldaquarium.com/ebooks/PAKIS/014_Substrate.html — The position that H₂S does develop under some substrate conditions but never reaches harmful concentrations in systems with good surface agitation.

[AQPONDANSWERS-H2S] *Hydrogen Sulfide in Aquariums*. Aquarium Pond Answers. <https://www.aquarium-pond-answers.com/2008/11/hydrogen-sulfides.html> — The position that sulfate reduction requires total deprivation of both oxygen and nitrate; identification of stopped undergravel and canister filters as the realistic risk configuration.

[AQWIKI-PHSHOCK] *pH Shock*. The Aquarium Wiki. https://www.theaquariumwiki.com/wiki/PH_Shock — Transport-bag pH shift within 20 minutes of opening; clinical description; cites van Ginneken et al. (1997), *Journal of Fish Biology* 51:795–806, on tilapia tolerance of long-term low environmental pH.

[GLASSBOX-OTS] *How To Avoid Old Tank Syndrome*. Glass Box Diaries. <https://glassboxdiaries.com/old-tank-syndrome/> — The observation that soft-water systems may be dominated by ammonia-oxidizing archaea, *Nitrospira*, or *Nitrotoga* rather than the classical *Nitrosomonas/Nitrobacter* model.

[VETVERIFIED-COLUMNARIS] *Understanding and Treating Columnaris in Aquarium Fish*. Vet Verified. <https://vetverified.com/articles/understanding-and-treating-columnaris-in-aquarium-fish> — Quarantine recommendation of 2–4 weeks; cleaning nets and tools between tanks; columnaris as bacterial rather than fungal.

[IERE-COPPER] *Is copper in water bad for fish?* and *How toxic is copper to fish?* Institute for Environmental Research and Education. <https://iere.org/is-copper-in-water-bad-for-fish/> and <https://iere.org/how-toxic-is-copper-to-fish/> — New copper pipe leaching; flushing recommendation; boiling concentrating rather than removing copper; pH and hardness effects.

[FISHHANDBOOK-METALS] *Heavy Metals*. Fish Handbook. <https://www.fishhandbook.org/part-ii-guides-and-practical-knowledge/chemistry/heavy-metals> — Invertebrate copper sensitivity figures; the observation that topping off concentrates metals. **Specific numerical figures from this source should be treated as approximate.**

[AQADVICE-DEGAS] *Aeration for true pH (gassing off CO₂)*. Aquarium Advice forum. <https://www.aquariumadvice.com/threads/aeration-for-true-ph-gassing-off-co2.106763/> — A worked example of tap water shifting from pH 7.2 to pH 8.0 on aeration at high KH. **Forum source; the calculation is chemically coherent and is used illustratively.**

Tier 10: Forum and community sources (anecdotal; cited only to document what hobbyists

report and where disagreements lie)

Explicit caveat: the following are cited to identify recurring real-world experiences, common misconceptions, and the shape of active disagreements. **They are not cited as evidence for any biological or chemical claim.** Several contain chemistry that is incorrect.

[FISHLORE-DRIP] *To Drip? Or Not To Drip?* and *Drip Acclimation Is Dangerous?* FishLore. <https://www.fishlore.com/aquariumfishforum/threads/to-drip-or-not-to-drip.269364/> and <https://www.fishlore.com/aquariumfishforum/threads/drip-acclimation-is-dangerous.384619/>

[REEF2REEF-DRIP] *Drip acclimate or not?* Reef2Reef. <https://www.reef2reef.com/threads/drip-acclimate-or-not.1102071/>

[HUMBLEFISH-ACCLIMATION] *Fish acclimation inherently damaging to our fish.* Humble.Fish. <https://humble.fish/community/threads/fish-acclimation-inherently-damaging-to-our-fish.6587/> — **Contains at least one demonstrably incorrect chemistry claim (that an ammonia reducer requires oxygen to decompose ammonia); cited in 7.6 as an illustration of why forum consensus is not evidence.**

[MEDIUM-ACCLIMATION] Kamath, S. *What they don't tell you about acclimating new fish to your freshwater tank.* Medium. <https://medium.com/@kamath.s.suraj/what-they-dont-tell-you-about-acclimating-new-fish-to-your-freshwater-tank-72b971dd4655> — Individual hobbyist opinion piece arguing against drip acclimation.

[UKAPS-OTS] *Old tank syndrome.* UK Aquatic Plant Society. <https://www.ukaps.org/forum/threads/old-tank-syndrome.63108/> — A thread specifically asking whether anyone can categorically attribute fish deaths to OTS; responses are near-misses and second-hand reports.

[R2R-TESTERRORS] *The errors of water testing and more.* Reef2Reef. <https://www.reef2reef.com/threads/the-errors-of-water-testing-and-more.849402/> — Hobbyist documentation of reagent separation causing falsely favorable readings.

[FISHLORE-NITRATE] *Nitrate Toxicity In Fish.* FishLore. <https://www.fishlore.com/aquariumfishforum/threads/nitrate-toxicity-in-fish.385550/> — A thread in which hobbyists trace the 40 ppm guideline back to the primary literature and find it unsupported by studies of ornamental species.

[TANNIN-ACCLIMATION] *The acclimation evolution.* Tannin Aquatics. <https://tanninaquatics.com/blogs/the-tint-1/the-acclimation-evolution> — Retailer account of low-pH acclimation practice to keep ammonia bound as ammonium during transfer.

Sources consulted and deliberately not used

- Several consumer-facing articles and AI-generated content-farm pages asserting specific quantitative claims about chlorine killing filter bacteria (e.g., “0.2 ppm destroys 70% of surface-adhered *Nitrospira* within 30 seconds,” “10 seconds of tap contact destroys biofilm irreversibly”). **These figures could not be traced to any primary source and are classified as unsupported in 3.5.1 and 17.3.**
- Commercial pages presenting nitrate danger thresholds (40 ppm / 80 ppm) as established findings without citation.
- AI-generated aggregator pages on test-kit accuracy that reproduce manufacturer marketing without independent testing.

APPENDIX A. GLOSSARY AND ALTERNATE TERMINOLOGY

Included to improve retrieval matching across the varied vocabulary used by hobbyists, veterinarians, and researchers.

Term	Alternate / related terms	Definition
TAN	Total ammonia nitrogen; “ammonia” on hobby kits	NH3 + NH4+ combined
Un-ionized ammonia	NH3; free ammonia; gaseous ammonia; UIA	The toxic ammonia species
Ammonium	NH4+; ionized ammonia	The comparatively non-toxic species
Nitrite	NO2-	Intermediate nitrification product; causes methemoglobinemia
Methemoglobinemia	Brown blood disease	Nitrite-induced conversion of hemoglobin to a non-oxygen-carrying form
KH	Carbonate hardness; alkalinity; buffering capacity; acid-neutralizing capacity	Bicarbonate + carbonate concentration
GH	General hardness; total hardness; permanent hardness	Calcium + magnesium concentration
TDS	Total dissolved solids; conductivity (related); EC; specific conductance	Ionic strength proxy
DO	Dissolved oxygen	Oxygen concentration in water
BOD	Biological oxygen demand	Oxygen consumed by biological decomposition
DOC	Dissolved organic carbon; “organics”; dissolved organic matter; DOM	Accumulated dissolved organic compounds
Biofilter	Biological filtration; the cycle; nitrifying bacteria; beneficial bacteria; BB	The nitrifying microbial community

Cycling	Nitrogen cycle establishment; new tank syndrome (related)	Establishing nitrification capacity
Old Tank Syndrome	OTS; tank crash (loosely and often incorrectly)	Chronic neglect producing low pH, low KH, high nitrate, accumulated organics
Ich	Ick; white spot disease; <i>Ichthyophthirius multifiliis</i>	Ciliate ectoparasite
Velvet	Rust disease; gold dust disease; <i>Piscinoodinium</i> ; <i>Oodinium</i> ; <i>Amyloodinium</i> (marine)	Parasitic dinoflagellate
Columnaris	Cottonmouth; mouth fungus (misnomer); saddleback disease; <i>Flavobacterium columnare</i> ; formerly <i>Flexibacter columnaris</i>	Bacterial infection, frequently mistaken for fungus
Monogeneans	Gill flukes; skin flukes; <i>Dactylogyrus</i> ; <i>Gyrodactylus</i>	Flatworm ectoparasites
Mycobacteriosis	Fish TB; fish tuberculosis; wasting disease; <i>Mycobacterium marinum/fortuitum</i>	Chronic granulomatous bacterial disease; zoonotic
Dropsy	Pinecone disease; ascites; generalized edema	A sign of fluid accumulation, not a disease
Popeye	Exophthalmia	Protrusion of the eye
Gas bubble disease	GBD; the bends; supersaturation disease	Emboli from dissolved gas supersaturation
Flashing	Scratching; glancing; rubbing	Scraping against surfaces, usually indicating irritation
Shimmying	Shimmies; rocking	Rocking in place without forward motion
Piping	Surface gasping; gulping	Breathing at the air-water interface
Dechlorinator	Water conditioner; declorinator; thiosulfate; Prime/Safe/etc.	Product neutralizing chlorine (and sometimes chloramine and ammonia)
Free chlorine conversion	Chlorine burn; chlorine burnout; temporary chlorine conversion; FCC	Utility practice of switching from chloramine to free chlorine temporarily
Plop and drop	Net and release; float and dump	Acclimation by rapid transfer after temperature matching
Drip acclimation	Slow drip; drip line acclimation	Gradual water exchange over an extended period
Hospital tank	Quarantine tank; QT; Q-tank; treatment tank	Separate system for isolation and treatment
Bioload	Stocking load; biological load	Total waste production relative to system capacity
Melt	Plant melt; die-back	Rapid decay of plant tissue after a change in conditions
Bacterial bloom	Bacterial bloom; white cloudy water; heterotrophic bloom	Rapid heterotrophic bacterial proliferation
Blackwater	Soft acidic water; tannin-stained water	Low-mineral, low-pH, tannin-rich conditions
Remineralization	Reconstitution; GH+/GH+KH+ dosing	Adding minerals back to RO or distilled water

APPENDIX B. DOCUMENT INTEGRITY NOTES

B.1 Known limitations of this compilation

- Several citations are second-hand.** Where a technical source cites primary research (e.g., Cunliffe 1991; Airaksinen et al. 1998; Basu et al. 2001; Currie et al. 2000; Wiebe 1931; Gauss et al. 1985; Russo & Thurston 1991), the primary texts were **not independently verified** during this compilation. These are flagged at the point of use and in the bibliography. Anyone relying on these specific figures for a public claim should verify the primary sources.
- Ornamental-fish-specific data is scarce.** A large fraction of the quantitative toxicology available comes from food-fish aquaculture (salmonids, catfish, tilapia) or from environmental protection literature (wild stream and lake species). Extrapolation to captive-bred tropical ornamentals is an assumption, not a finding, and is noted where it matters most (nitrate, dissolved oxygen, ammonia thresholds).
- No controlled studies were located** for several of the hobby's most-debated practices, notably acclimation method comparison, water-change percentage effects, and filter media rinsing. Those sections document positions rather than resolving them.
- Commercial interest is present in several sources.** Manufacturer and retailer documentation is cited where it provides useful technical detail, and is flagged as commercially interested. Conversely, at least one heavily-cited technical source has an explicit editorial position against manufacturer claims, and is flagged as such.
- Anecdotal sources are labeled but are not evenly distributed.** Forum evidence clusters around exactly the topics where research is thinnest, which is a systematic bias that should be borne in mind.
- Currency.** Municipal water practices, product formulations, and pathogen taxonomy all change. The *Flavobacterium* taxonomic

revisions noted by Merck are a live example. Utility conversion schedules are local and annual.

B.2 Suggested queries this document is structured to answer

- Why would fish die when ammonia, nitrite, nitrate, pH, and temperature all read normal?
- What does a hobbyist test kit not measure?
- How can a water change kill fish?
- What is actually happening during Old Tank Syndrome?
- Why do new fish die within a week?
- Is drip acclimation good or bad?
- Does rinsing filter media in tap water crash the cycle?
- Does hydrogen sulfide from substrate kill fish?
- What causes gasping when parameters are fine?
- Why did the shrimp die but not the fish?
- What is gas bubble disease and does it happen in home aquaria?
- What should be checked first in an aquarium emergency?
- Which common aquarium claims are myths?
- Why do fish die every spring?
- What does the timing of symptoms suggest?

B.3 Compilation metadata

- **Document type:** research compilation / source corpus
- **Intended consumer:** NotebookLM, as source material for later podcast development
- **Explicitly not included, per the research brief:** episode script, outline, narrative structure, talking points, hooks, or recommendations regarding episode framing
- **Evidence labeling:** applied throughout per the key in section 0.4
- **Compiled:** August 2026

END OF DOCUMENT